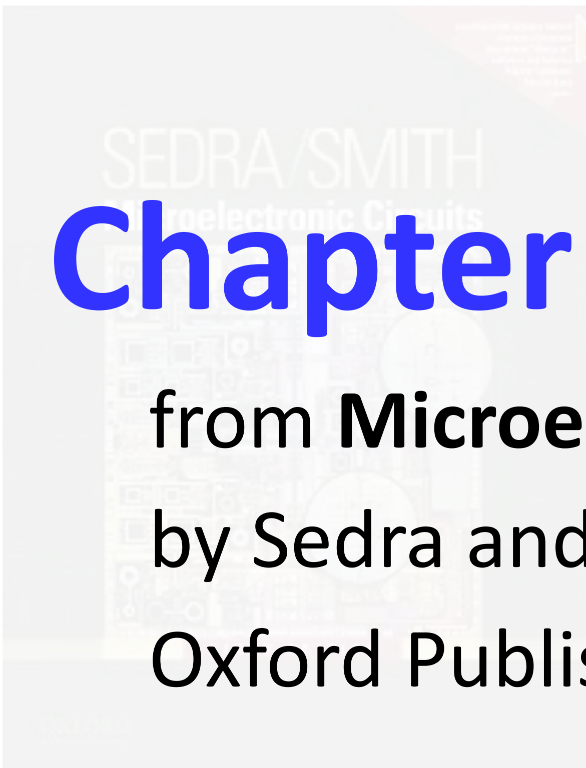



# Chapter #5: MOSFET's

from **Microelectronic Circuits** Text

by Sedra and Smith

Oxford Publishing

# Introduction

- **IN THIS CHAPTER WE WILL LEARN**
  - The physical structure of the MOS transistor and how it works.
  - How the voltage between two terminals of the transistor control the current that flows through the third terminal, and the equations that describe these current-voltage characteristics.
  - How the transistor can be used to make an amplifier, and how it can be used as a switch in digital circuits.

# Introduction

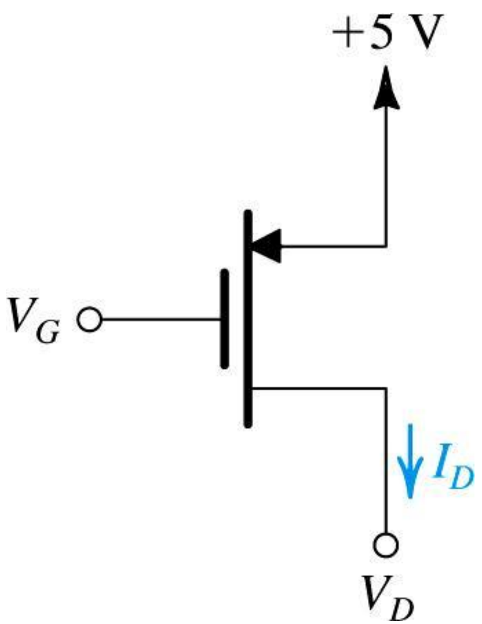
- **IN THIS CHAPTER WE WILL LEARN**
  - How to obtain linear amplification from the fundamentally nonlinear MOS transistor.
  - The three basic ways for connecting a MOSFET to construct amplifiers with different properties.
  - Practical circuits for MOS-transistor amplifiers that can be constructed using discrete components.

# Introduction

- We have studied two-terminal semi-conductor devices (e.g. diode).
- However, now we turn our attention to three-terminal devices.
- They are more useful because they present multitude of applications, e.g:
    - signal amplification, digital logic, memory, etc…

# Introduction

- **Q:** What, in simplest terms, is the desired operation of a three-terminal device?
  - **A:** Employ voltage between two terminals to control current flowing in to the third.



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# Introduction

**note:** MOSFET is more widely used in implementation of modern electronic devices

- **Q:** What are two major types of three-terminal semiconductor devices?
  - metal-oxide-semiconductor field-effect transistor (MOSFET)
  - bipolar junction transistor (BJT)
- **Q:** Why are MOSFET's more widely used?
  - size (smaller)
  - ease of manufacture
  - lesser power utilization

- MOSFET technology
  - It allows placement of approximately 2 billion transistors on a single IC
    - backbone of very large scale integration (VLSI)
  - It is considered preferable to BJT technology for many applications.

# 5.1. **Device Structure and Operation**

- Figure 5.1. shows general structure of the *n*-channel enhancement-type MOSFET

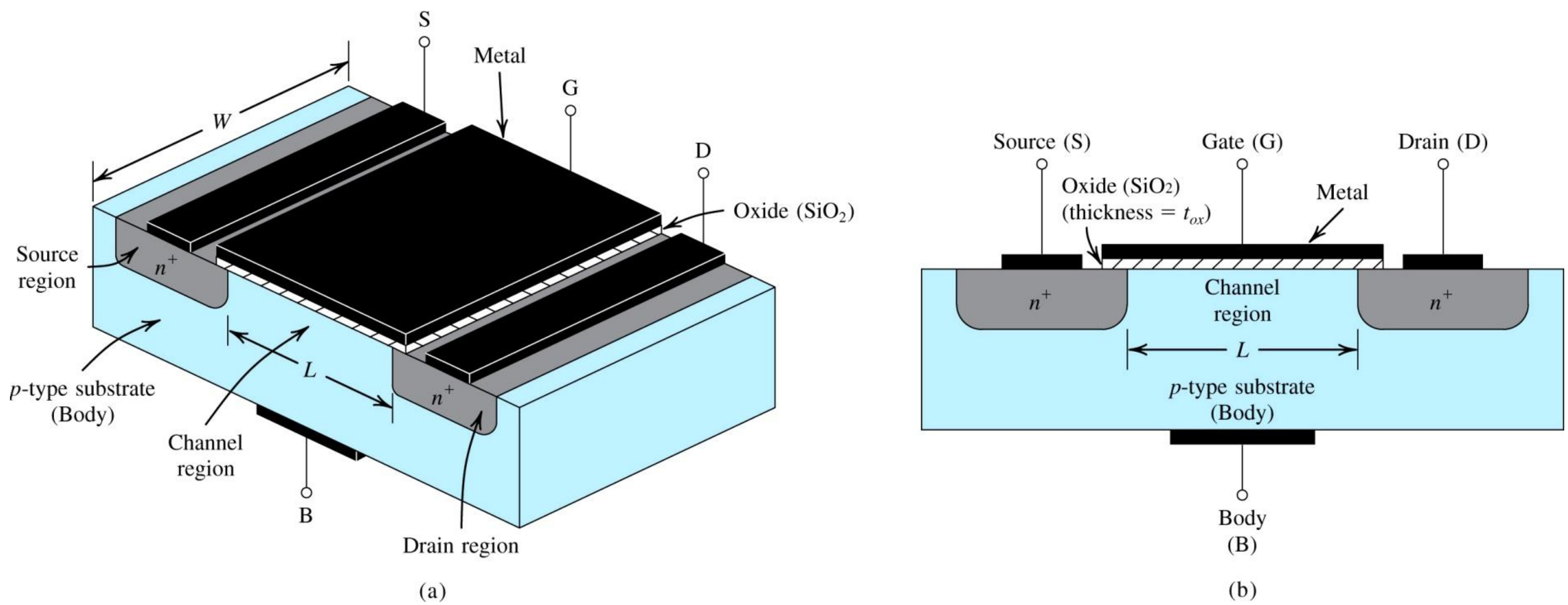


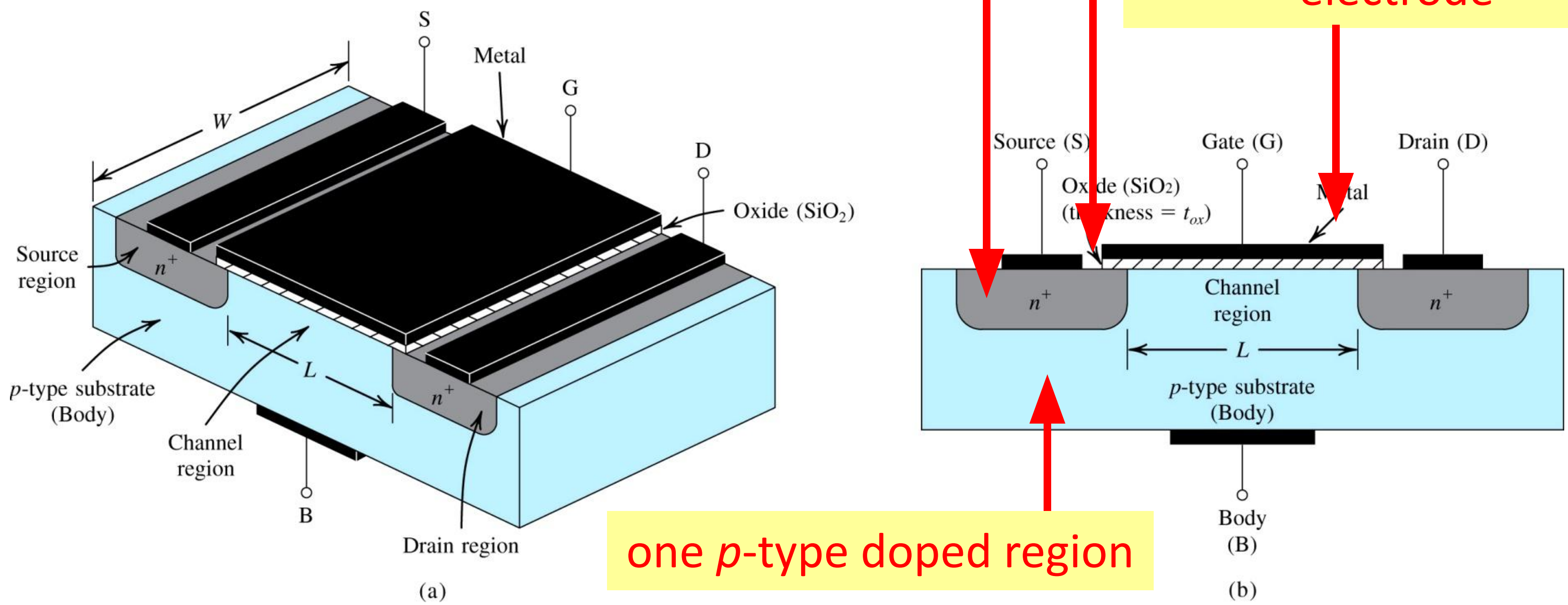

**Figure 5.1:** Physical structure of the enhancement-type NMOS transistor: **(a)** perspective view, **(b)** cross-section.  Note that typically L = 0.03*um* to 1*um*, W = 0.1*um* to 100*um*, and the thickness of the oxide layer ($t_{ox}$) is in the range of 1 to 10*nm*.

# 5.1. **Device Structure and Operation**

two *n*-type doped regions (drain, source)

layer of $SiO_2$ separates source and drain

metal, placed on top of $SiO_2$, forms gate electrode




one *p*-type doped region

**Figure 5.1:** Physical structure of the enhancement-type NMOS transistor: **(a)** perspective view, **(b)** cross-section. Note that typically L = 0.03*um* to 1*um*, W = 0.1*um* to 100*um*, and the thickness of the oxide layer ($t_{ox}$) is in the range of 1 to 10*nm*.

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# 5.1. Device Structure and Operation

- The name MOSFET is derived from its physical structure.
- However, many MOSFET's do not actually use any "metal", polysilicon is used instead.
  - "This" has no effect on modeling / operation as described here.
- Another name for MOSFET is insulated gate FET, or IGFET.
- The device is composed of two pn-junctions, however they maintain reverse biasing at all times.
  - Drain will always be at positive voltage with respect to source.
- We will not consider conduction of current in this manner.

# 5.1.2. **Operation with Zero Gate Voltage**

- With zero voltage applied to gate, two back-to-back diodes exist in series between drain and source.
- "They" prevent current conduction from drain to source when a voltage $v_{DS}$ is applied.
  - yielding very high resistance ($10^{12}$ *ohms*)

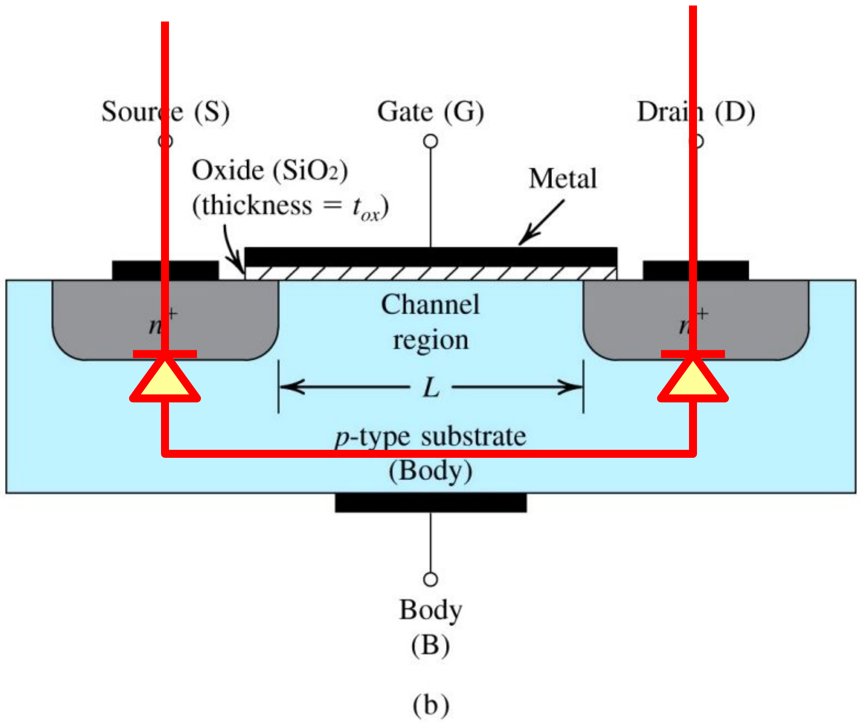


**Figure 5.1:** Physical structure…

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# 5.1.3. Creating a Channel for Current Flow

- **Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.
  - **step #1:** $v_{GS}$ is applied to the gate terminal, causing a positive build up of positive charge along metal electrode.
  - **step #2:** This "build up" causes free holes to be repelled from region of *p*-type substrate under gate.

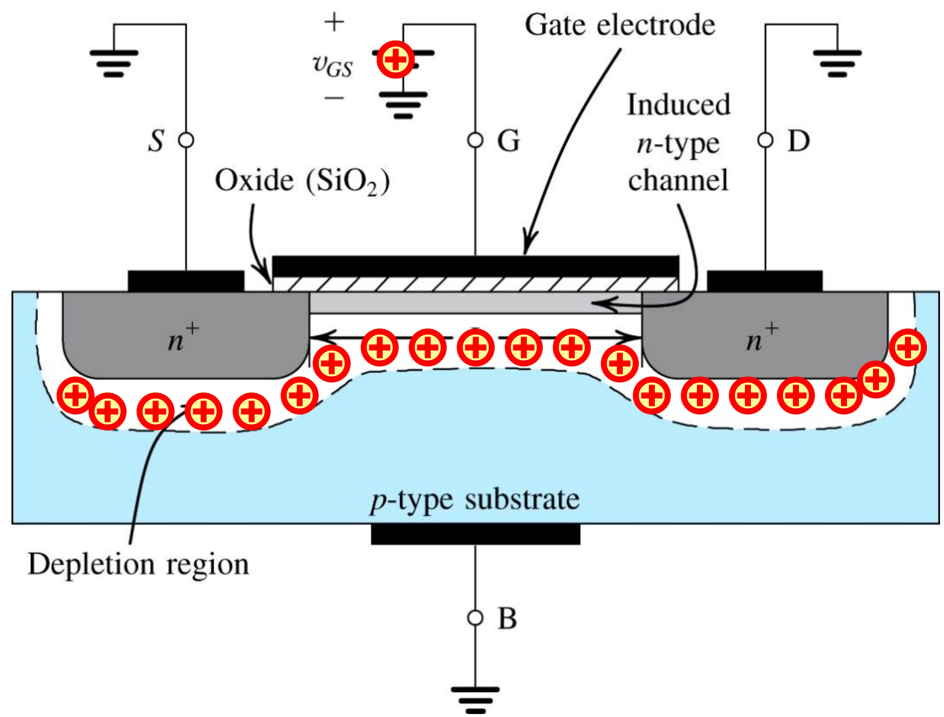


**Figure 5.2:** The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate

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**Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.

- **step #3:** This "migration" results in the uncovering of negative **bound charges**, originally neutralized by the free holes
- **step #4:** The positive gate voltage also attracts electrons from the $n^+$ source and drain regions into the channel.

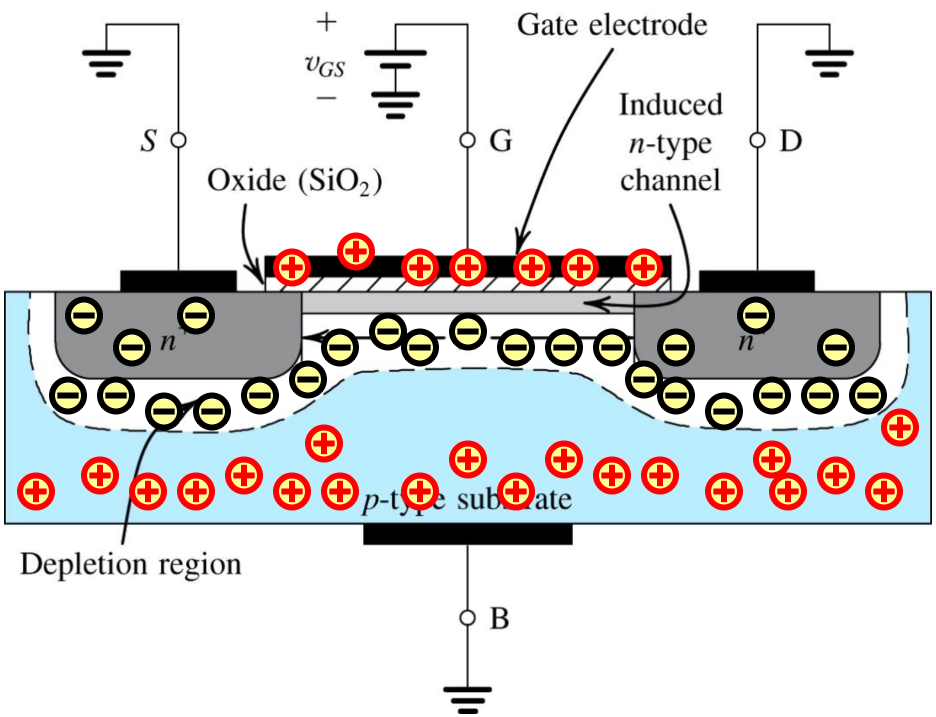


**Figure 5.2:** The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate

**Q:** What happens if (1) source and drain are grounded and (2) positive voltage is applied to gate? Refer to figure to right.

this induced channel is also known as an **inversion layer**

- **step #5:** Once a sufficient number of "these" electrons accumulate, an *n*-region is created…
  - …connecting the source and drain regions
- **step #6:** This provides path for current flow between *D* and *S*.

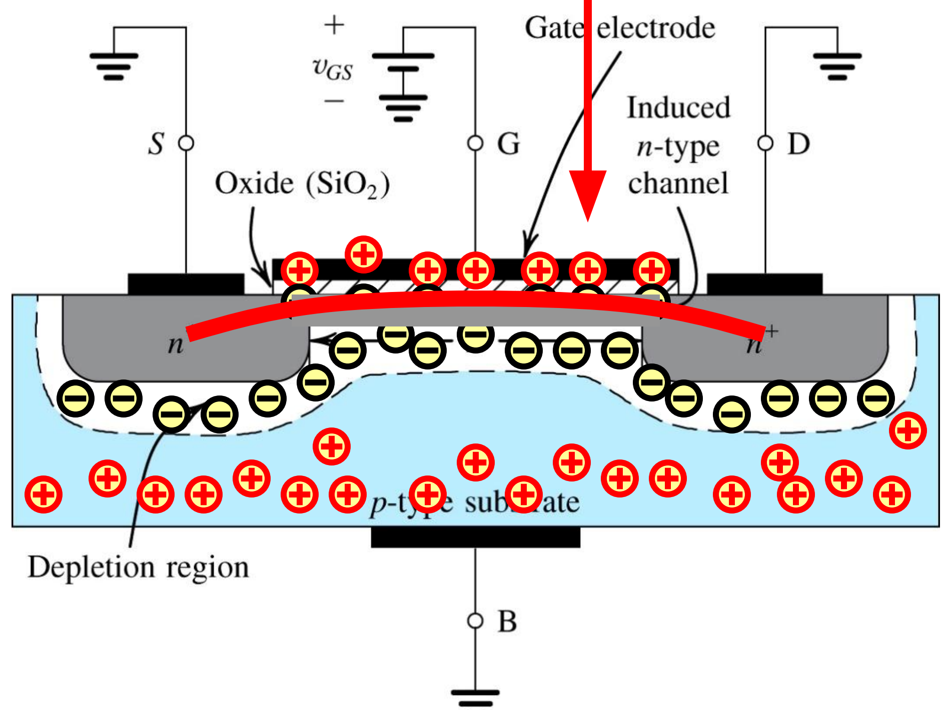


**Figure 5.2:** The enhancement-type NMOS transistor with a positive voltage applied to the gate. An *n* channel is induced at the top of the substrate beneath the gate

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# 5.1.3. **Creating a Channel for Current Flow**

$V_{tn}$ is used for *n*-type MOSFET, $V_{tp}$ is used for *p*-channel

- **threshold voltage** ($V_t$) – is the minimum value of $v_{GS}$ required to form a conducting channel between drain and source
  - typically between 0.3 and $0.6V_{dc}$
- **field-effect** – when positive $v_{GS}$ is applied, an electric field develops between the gate electrode and induced *n*-channel – the conductivity of this channel is affected by the strength of field
  - $SiO_2$ layer acts as dielectric
- **effective / overdrive voltage** – is the difference between $v_{GS}$ applied and $V_t$.

$$\text{(eq5.1)} \quad v_{OV} \equiv v_{GS} - V_t$$

- **oxide capacitance** ($C_{ox}$) – is the capacitance of the parallel plate capacitor per unit gate area ($F/m^2$)

$\varepsilon_{ox}$ is permittivity of $SiO_2 = 3.45\mathbf{E}{-11}(F/m)$

$t_{ox}$ is thickness of $SiO_2$ layer

$$\text{(eq5.3)} \quad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \text{ in } F/m^2$$

# 5.1.3. **Creating a Channel for Current Flow**

- **Q:** What is main requirement for *n*-channel to form?
  - **A:** The voltage across the "oxide" layer must exceed $V_t$.
- For example, when $v_{DS} = 0$…
  - the voltage at every point along channel is zero
  - the voltage across the oxide layer is uniform and equal to $v_{GS}$
- **Q:** How can one express the magnitude of electron charge contained in the channel?
  - **A:** See below…

*W* and *L* represent width and length of channel respectively

$$\text{(eq5.2)} \quad |Q| = C_{ox}(WL)v_{OV} \text{ in } C$$

- **Q:** What is effect of $v_{OV}$ on *n*-channel?
  - **A:** As $v_{OV}$ grows, so does the depth of the *n*-channel as well as its conductivity.

# 5.1.4. Applying a Small $v_{DS}$

- **Q:** For small values of $v_{DS}$, how does one calculate $i_{DS}$ (aka. $i_D$)? **A:** Equation (5.7)…
- **Q:** What is the origin of this equation?
  - **A:** Current is defined in terms of **charge per unit length of *n*-channel** as well as **electron drift velocity.**

$\mu_n$ represents mobility of electrons at surface of the n-channel in $m^2/Vs$

$$\text{(eq5.7) } i_D = \underbrace{\left(C_{ox} W v_{OV}\right)}_{\substack{\text{charge per unit}\\ \text{length of}\\ n\text{-channel}\\ \text{in } C/m}} \underbrace{\left(\frac{\mu_n v_{DS}}{L}\right)}_{\substack{\text{electron}\\ \text{drift velocity}\\ \text{in } m^2/Vs}} \text{ in } A$$

# 5.1.4. **Applying a Small $v_{DS}$**

- **Q:** How does one calculate charge per unit length of *n*-channel ($Q/uL$)?
  - **A:** For small values of $v_{DS}$, one can still assume that voltage between gate and *n*-channel is constant (along its length) – and equal to $v_{GS}$.
  - **A:** Therefore, effective voltage between gate and *n*-channel remains equal to $v_{OV}$.
  - **A:** Therefore, (5.2) from two slides back applies.

# 5.1.4. **Applying a Small $v_{DS}$**

- **Q:** How does one calculate charge per unit length of *n*-channel ($Q/uL$)?
  - **A:** Use (5.2) to calculate charge per unit $L$ of channel.
- **Q:** How does one calculate electron drift velocity?
  - **A:** Note that $v_{DS}$ establishes an electric field $E$ across length of *n*-channel, this may calculate e-drift velocity.

action: divide both sides by L

(eq5.2) $|Q| = C_{ox}(WL)v_{OV}$ in $C$

(eq5.4) $\dfrac{|Q|}{L} = C_{ox}Wv_{OV}$ in $C/m$

(eq5.5) $|E| = \dfrac{v_{DS}}{L}$ in $V/m$

(eq5.6) e-drift velocity $= \ldots$

$$\ldots = \mu_n|E| \text{ in } \frac{V}{m}\frac{m^2}{Vs} = \frac{m}{s}$$

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# 5.1.4. **Applying a Small $v_{DS}$**

- **Q:** How does one calculate charge per unit length of

el.

Note that these two values may be employed to define current in amperes (aka. *C/s*).

  - **A:** Note that $v_{DS}$ establishes an electric field *E* across length of *n*-channel, this may calculate e-drift velocity.

action: divide both sides by L

$$\text{(eq5.2)}\quad |Q| = C_{ox}(WL)v_{OV} \ \text{ in } C$$

$$\text{(eq5.4)}\quad \frac{|Q|}{L} = C_{ox}Wv_{OV} \ \text{ in } C/m$$

$$\text{(eq5.5)}\quad |E| = \frac{v_{DS}}{L} \ \text{ in } V/m$$

$$\text{(eq5.6)}\quad \text{e-drift velocity} = \ldots$$

$$\ldots = \mu_n |E| \ \text{ in } \frac{V}{m}\frac{m^2}{Vs} = \frac{m}{s}$$

# 5.1.4. **Applying a Small $v_{DS}$**

- **Q:** What is observed from equation (5.7)?
  - **A:** For small values of $v_{DS}$, the *n*-channel acts like a variable resistance whose value is controlled by $v_{OV}$.

(eq5.7) $$i_D = \left[ (\mu_n C_{ox}) \frac{W}{L} v_{OV} \right] v_{DS} \text{ in } A$$

(eq5.8a) $$r_{DS} = \frac{v_{DS}}{i_D} = \frac{1}{\underbrace{(\mu_n C_{ox})}_{\text{process transconductance parameter}} \underbrace{\left(\frac{W}{L}\right)}_{\text{aspect ratio}} v_{OV}} \text{ in } \Omega$$

# 5.1.4. **Applying a Small** $v_{DS}$

Note that this $v_{OV}$ represents the depth of the *n*-channel - what if it is not assumed to be constant? How does this equation change?

Note that this is one **VERY IMPORTANT** equation in Chapter 5.

n equation (5.7)?

$v_{DS}$, the *n*-channel acts like a

ose value is controlled by $v_{OV}$.

$$\text{(eq5.7)}\quad i_D = \left[(\mu_n C_{ox})\frac{W}{L}v_{OV}\right]v_{DS} \text{ in } A$$

$$\text{(eq5.8a)}\quad r_{DS} = \frac{v_{DS}}{i_D} = \frac{1}{\underbrace{(\mu_n C_{ox})}_{\text{process transconductance parameter}}\underbrace{\left(\frac{W}{L}\right)}_{\text{aspect ratio}}v_{OV}} \text{ in } \Omega$$

# 5.1.4. **Applying a Small $v_{DS}$**

- **Q:** What three factors is $r_{DS}$ dependent on?
  - **A: process transconductance parameter for NMOS** ($\mu_n C_{ox}$) – which is determined by the manufacturing process
  - **A: aspect ratio** ($W/L$) – which is dependent on size requirements / allocations
  - **A: overdrive voltage** ($v_{OV}$) – which is applied by the user

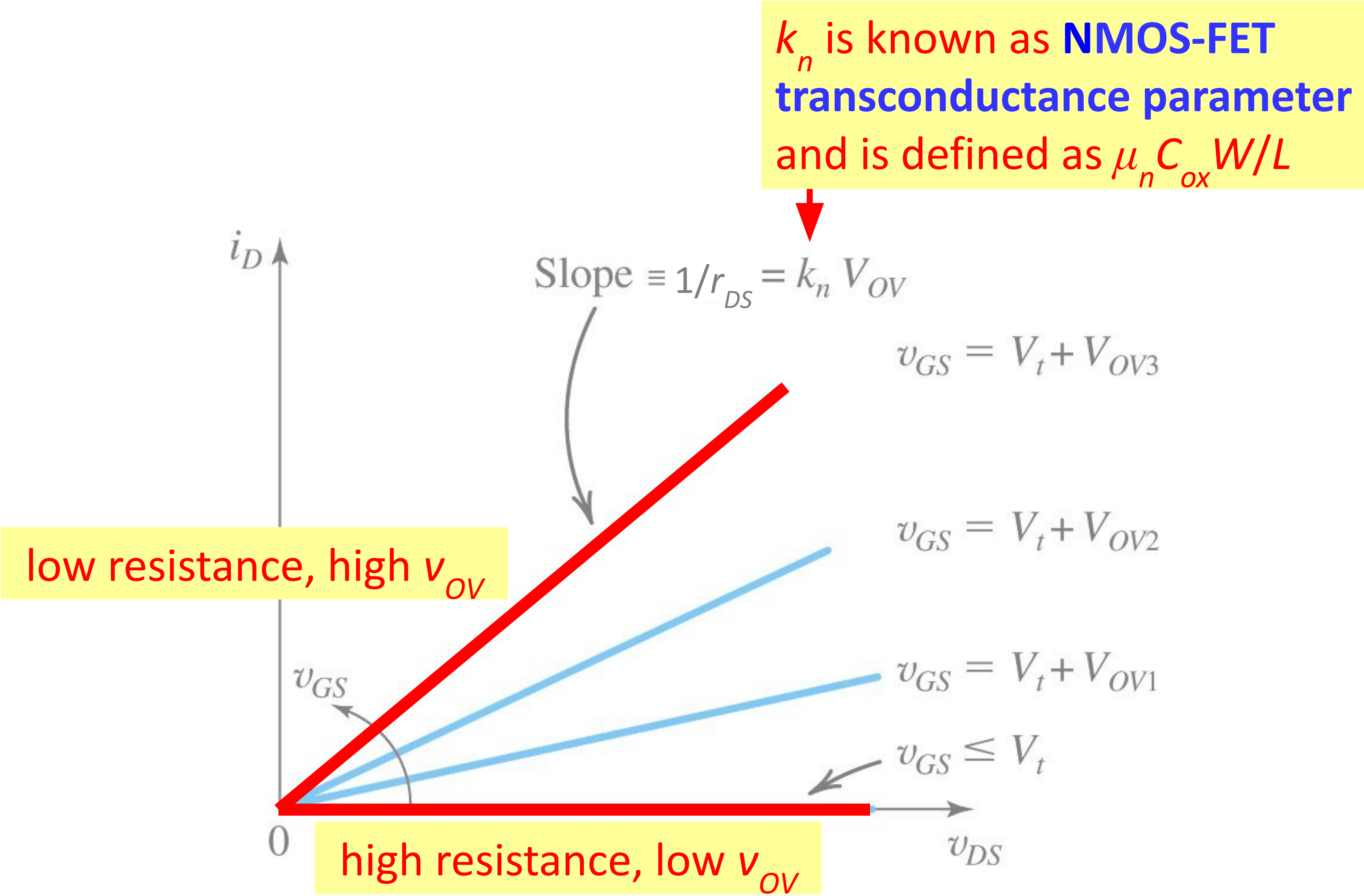



**Figure 5.4:** The $i_D$-$v_{DS}$ characteristics of the MOSFET in Figure 5.3. when the voltage applied between drain and source $V_{DS}$ is kept small.

# 5.1.5. **Operation as $v_{DS}$ is Increased**

- **Q:** What happens to $i_D$ when $v_{DS}$ increases beyond "small values"?
  - **A:** The relationship between them ceases to be linear.
- **Q:** How can this non-linearity be explained?
  - **step #1:** Assume that $v_{GS}$ is held constant at value greater than $V_t$.
  - **step #2:** Also assume that $v_{DS}$ is applied and appears as voltage drop across *n*-channel.
  - **step #3:** Note that voltage decreases from $v_{GS}$ at the source end of channel to $v_{GD}$ at drain end, where…
    - $v_{GD} = v_{GS} - v_{DS}$
    - $v_{GD} = V_t + v_{OV} - v_{DS}$

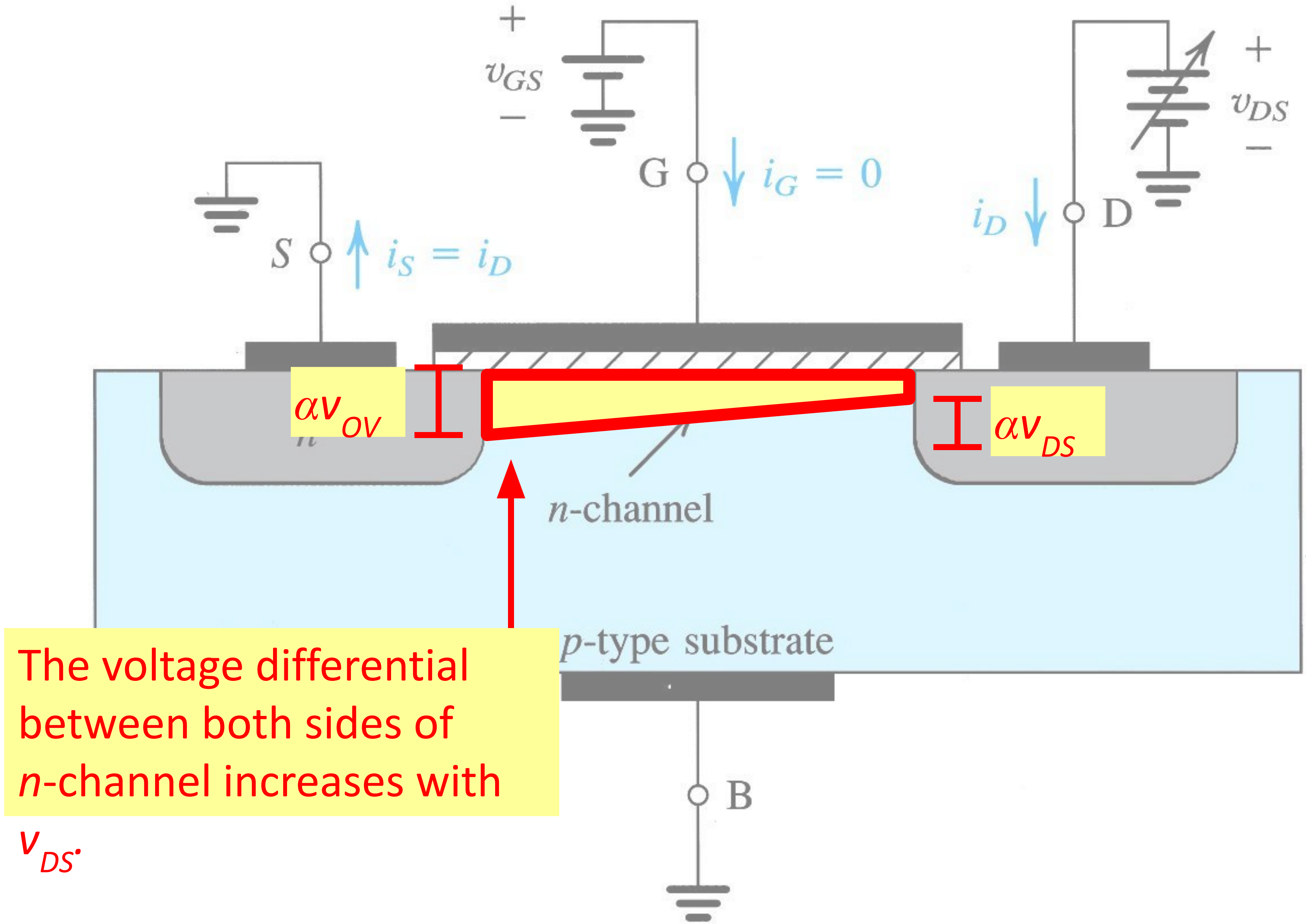
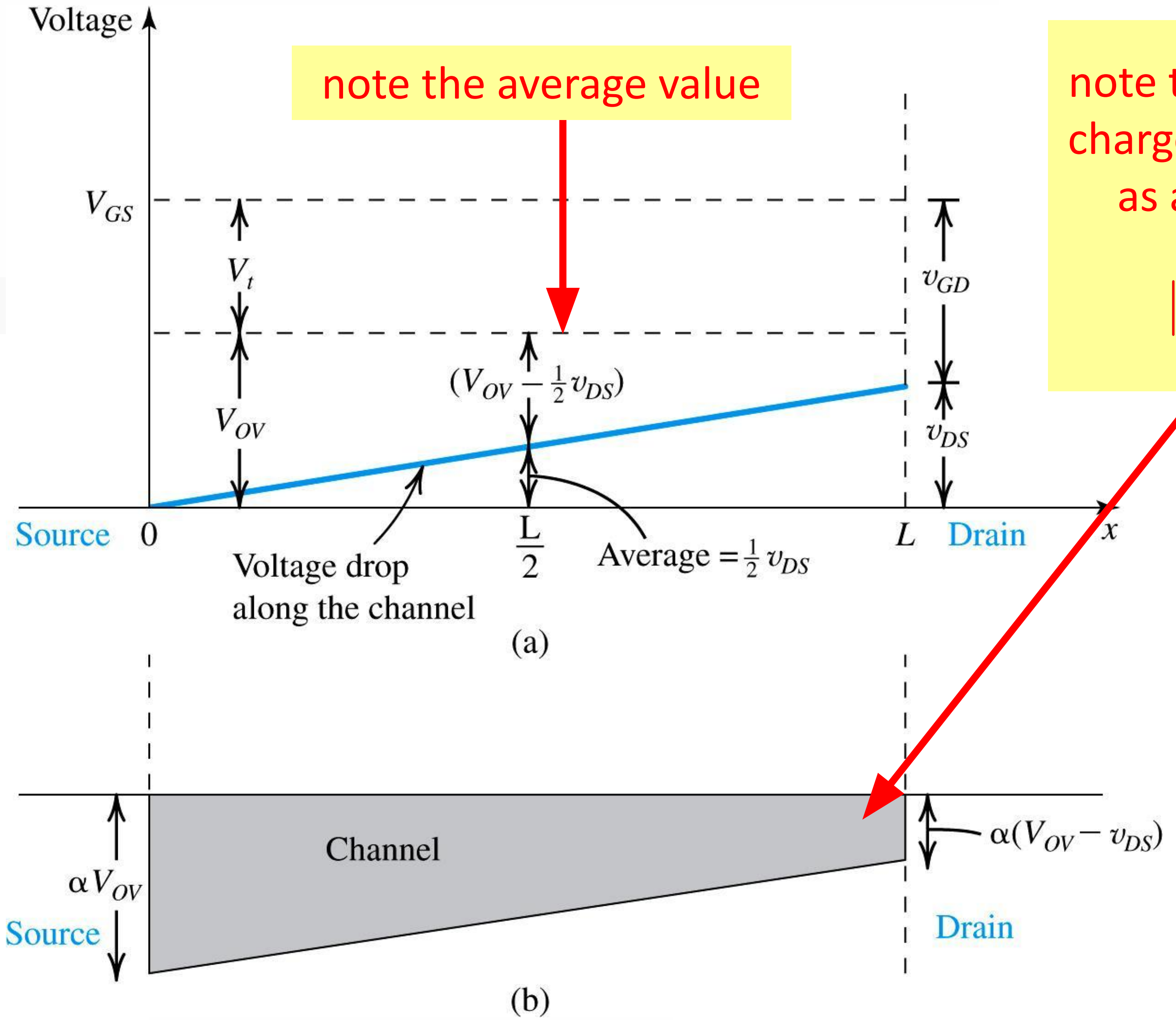



**Figure 5.5:** Operation of the e-NMOS transistor as $v_{DS}$ is increased.




note that we can define total charge stored in channel |Q| as area of this trapezoid

$$|Q| = \left(v_{OV} - \tfrac{1}{2}v_{DS}\right)L$$

**Figure 5.6(a):** For a MOSFET with $v_{GS} = V_t + v_{OV}$ application of vDS causes the voltage drop along the channel to vary linearly, with an average value of vDS at the midpoint. Since $v_{GD} > V_t$, the channel still exists at the drain end. (b) The channel shape corresponding to the situation in (a). While the depth of the channel at the source is still proportional to $v_{OV}$, the drain end is not.

# Q: How can this non-linearity be explained?

- **step #4:** Define $i_{DS}$ in terms of $v_{DS}$ and $v_{OV}$.

action: replace $v_{OV}$ with $v_{OV} - \frac{1}{2}v_{DS}$

(eq5.7) $$i_D = \left[ (\mu_n C_{ox}) \frac{W}{L} \left( v_{OV} - \tfrac{1}{2} v_{DS} \right) \right] v_{DS}$$

$$i_D = \begin{cases} (\mu_n C_{ox}) \frac{W}{L} \left( v_{OV} - \tfrac{1}{2} v_{DS} \right) v_{DS} & \text{if } v_{DS} < v_{OV} \\ (\mu_n C_{ox}) \frac{W}{L} \left( v_{OV} - \tfrac{1}{2} v_{DS} \right) v_{DS} & \text{otherwise} \end{cases}$$

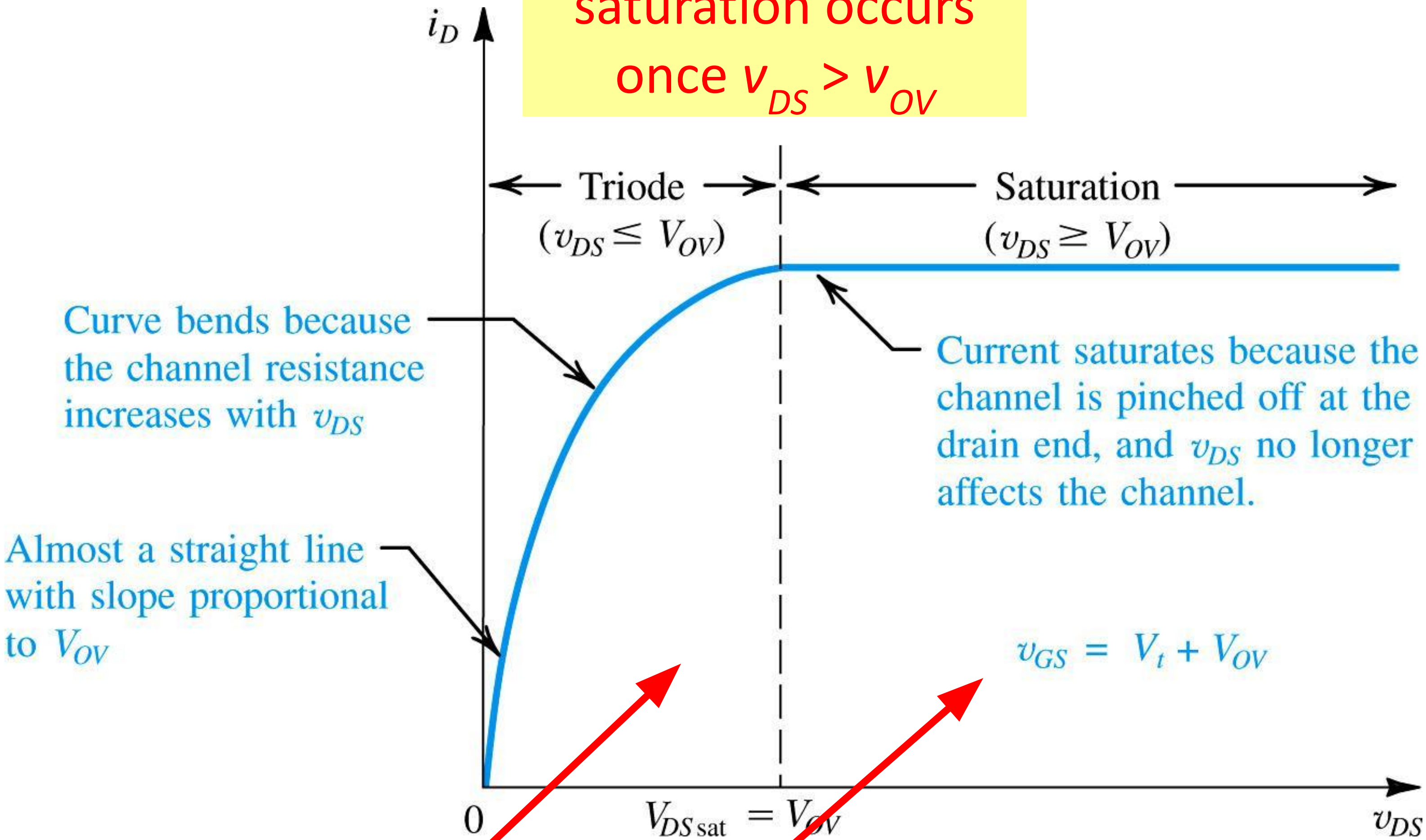
if $v_{DS} > v_{OV}$ then $v_{DS} = v_{OV}$

$i_D$ is dependent on the apparent $v_{OV}$ (not $v_{DS}$ inherently) which does not change after $v_{DS} > v_{OV}$

(eq5.14) $$i_D = \begin{cases} (\mu_n C_{ox}) \frac{W}{L} \left( v_{OV} - \tfrac{1}{2} v_{DS} \right) v_{DS} & \text{if } v_{DS} < v_{OV} \\ \frac{1}{2} (\mu_n C_{ox}) \frac{W}{L} v_{OV}^2 & \text{otherwise} \end{cases}$$ in $A$

triode vs. saturation region

saturation occurs once $v_{DS} > v_{OV}$




$$\text{(eq5.14) } i_D = \begin{cases} \text{triode: } (\mu_n C_{ox})\dfrac{W}{L}\left(v_{OV} - \tfrac{1}{2}v_{DS}\right)v_{DS} & \text{if } v_{DS} < v_{OV} \\ \text{saturation: } \dfrac{1}{2}(\mu_n C_{ox})\dfrac{W}{L}v_{OV}^2 & \text{otherwise} \end{cases} \text{ in } A$$

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# 5.1.6. **Operation for** $v_{DS} >> v_{OV}$

- In section 5.1.5, we assume that *n*-channel is tapered but **channel pinch-off** does not occur.
  - Trapezoid doesn't become triangle for $v_{GD} > V_t$
- **Q:** What happens if $v_{DS} > v_{OV}$?
  - **A:** MOSFET enters **saturation region**. Any further increase in $v_{DS}$ has no effect on $i_D$.

pinch-off does not mean blockage of current

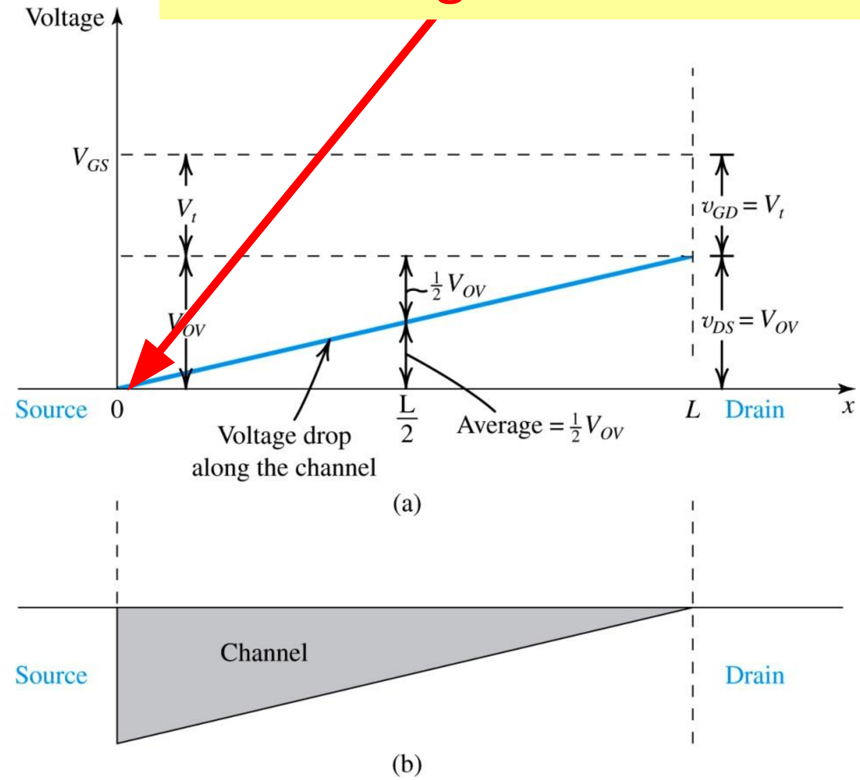



**Figure 5.8:** Operation of MOSFET with $v_{GS} = V_t + v_{OV}$ as $v_{DS}$ is increased to $v_{OV}$. At the drain end, $v_{GD}$ decreases to $V_t$ and the channel depth at the drain-end reduces to zero (pinch-off). At this point, the MOSFET enters saturation more of operation. Further increasing $v_{DS}$ (beyond $v_{OV}$) has no effect on the channel shape and $i_D$ remains constant.

# **Example 5.1:** NMOS MOSFET

- **Example 5.1. Problem Statement:** Consider an NMOS process technology for which $L_{min} = 0.4\mu m$, $t_{ox} = 8nm$, $\mu_n = 450cm^2/Vs$, $V_t = 0.7V$.
- **Q(a):** Find $C_{ox}$ and $k'_n$.
- **Q(b):** For a MOSFET with $W/L = 8\mu m/0.8\mu m$, calculate the values of $v_{OV}$, $v_{GS}$, and $v_{DSmin}$ needed to operate the transistor in the saturation region with dc current $I_D = 100\mu A$.
- **Q(c):** For the device in (b), find the values of $v_{OV}$ and $v_{GS}$ required to cause the device to operate as a 1000*ohm* resistor for very small $v_{DS}$.

# 5.1.7. The *p*-Channel MOSFET

- Figure 5.9(a) shows cross-sectional view of a *p*-channel enhancement-type MOSFET.
  - structure is similar but "opposite" to *n*-channel
- **complementary devices** – two devices such as the *p*-channel and *n*-channel MOSFET's.

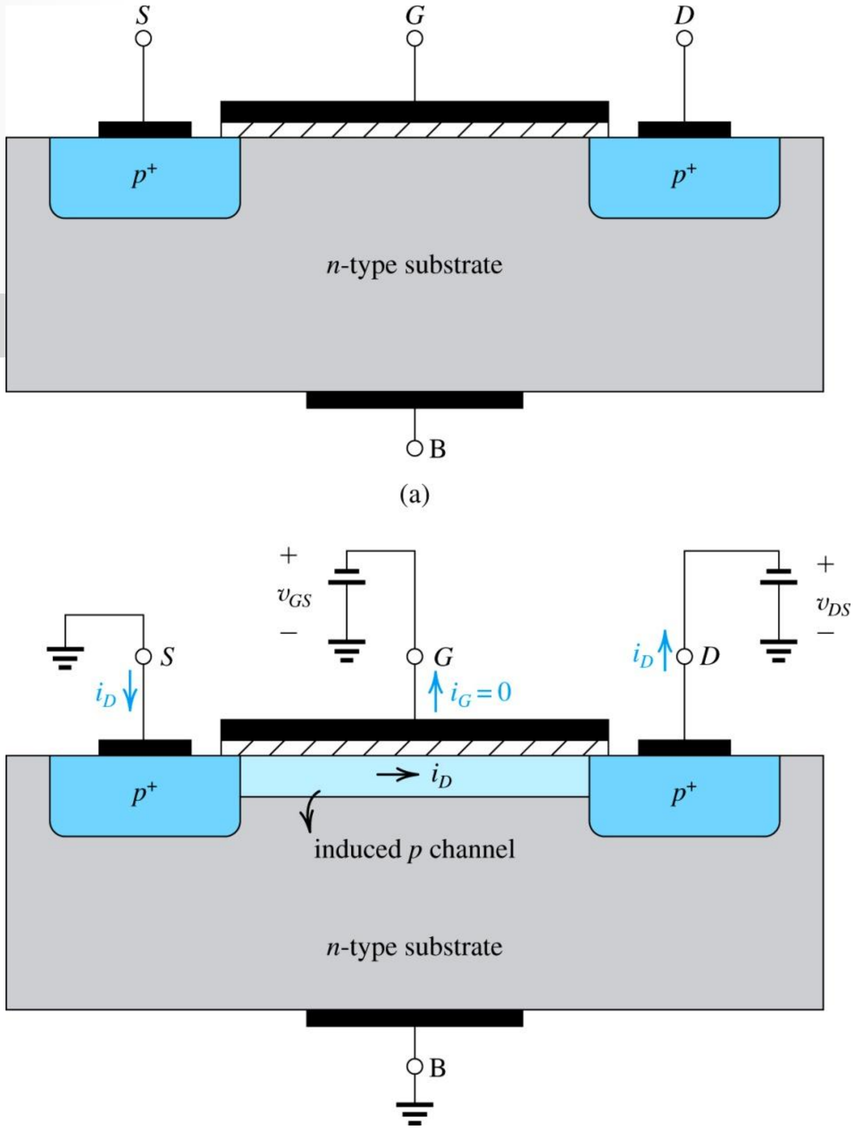
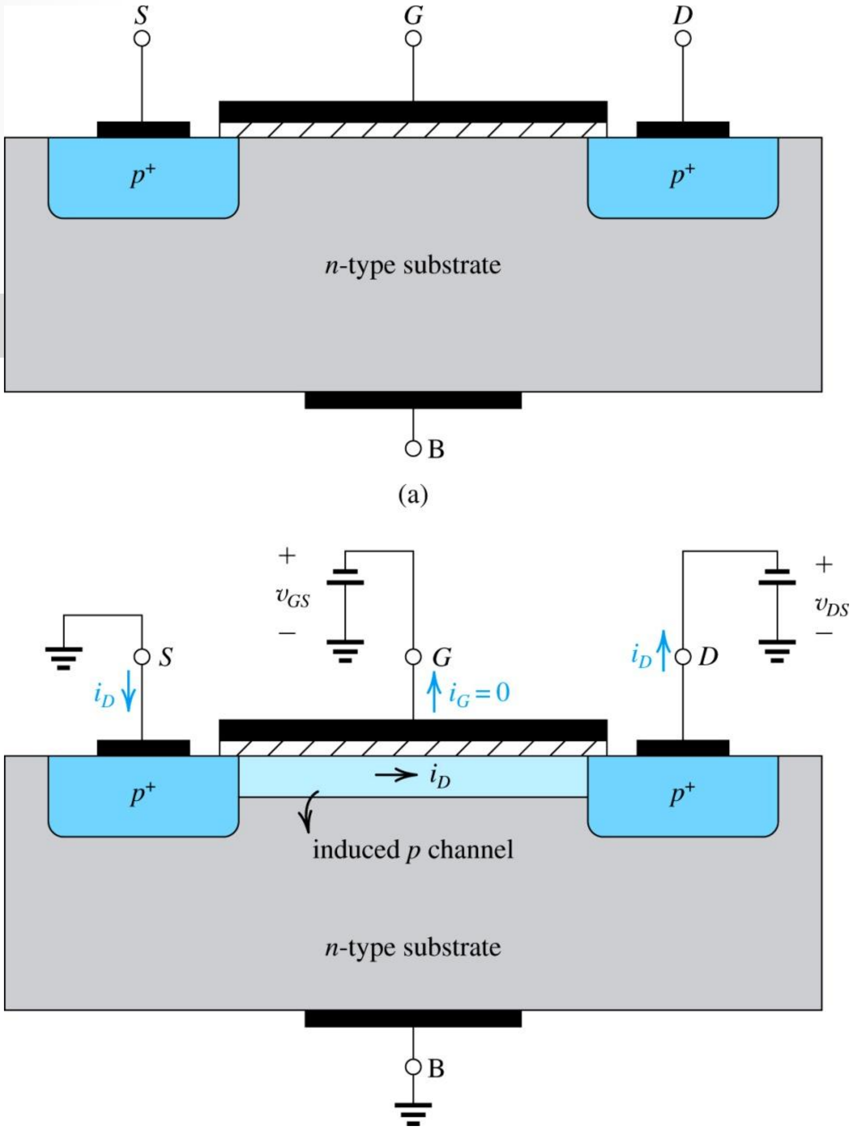



**Figure 5.9(a):** Physical structure of the PMOS transistor. Note that it is similar to the NMOS transistor shown in Figure 5.1(b), except that all semiconductor regions are reversed in polarity. **(b)** A negative voltage $v_{GS}$ of magnitude greater than $|V_{tp}|$ induces a p-channel, and a negative $v_{DS}$ causes a current $i_D$ to flow from source to drain.

# 5.1.7. The *p*-Channel MOSFET

- **Q:** What are main differences between *n*-channel and *p*-channel?
  - **A:** Negative (not positive) voltage applied to gate "closes" the channel
    - allowing path for current flow
  - **A:** Threshold voltage (previously represented as $V_t$) is represented as $V_{tp}$
    - $|v_{GS}| > |V_{tp}|$ to close channel

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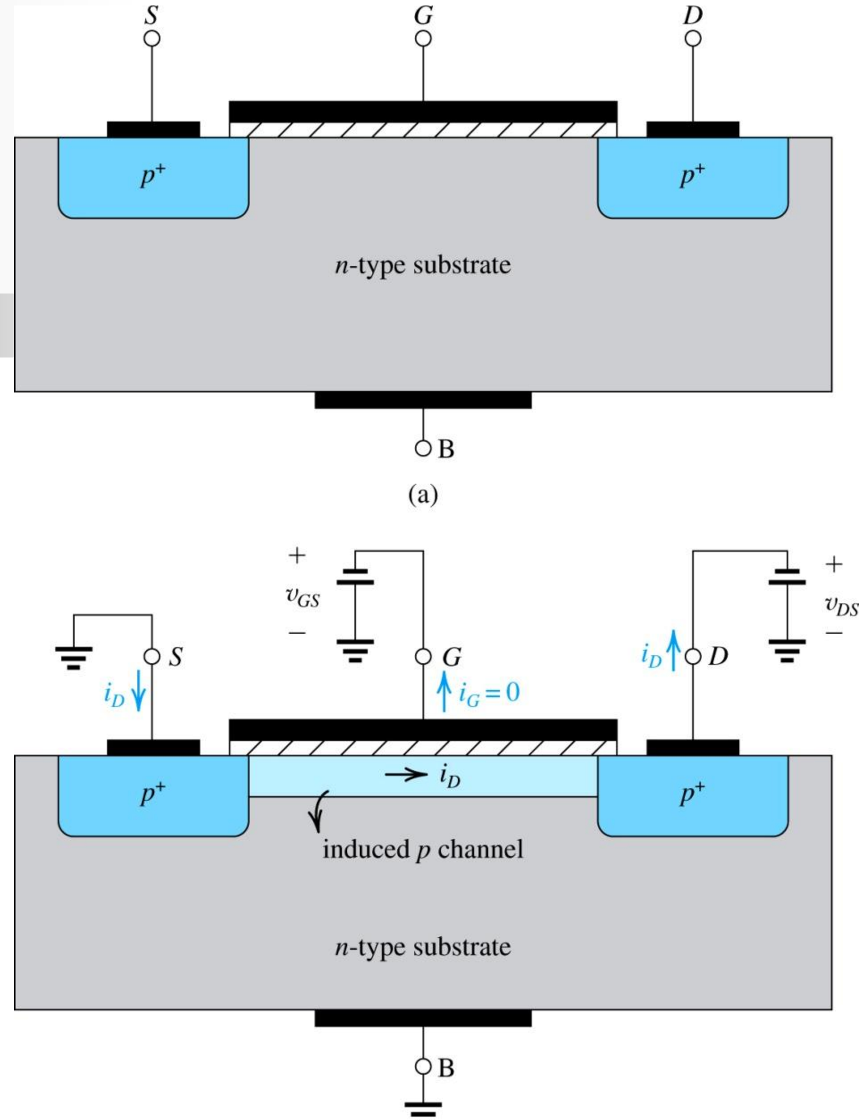
# 5.1.7. The *p*-Channel MOSFET

- **Q:** What are main differences between *n*-channel and *p*-channel?
  - **A:** Process transconductance parameters are defined differently
    - $k'_p = \mu_p C_{ox}$
    - $k_p = \mu_p C_{ox}(W/L)$
  - **A:** The rest, essentially, is the same, but with reverse polarity...

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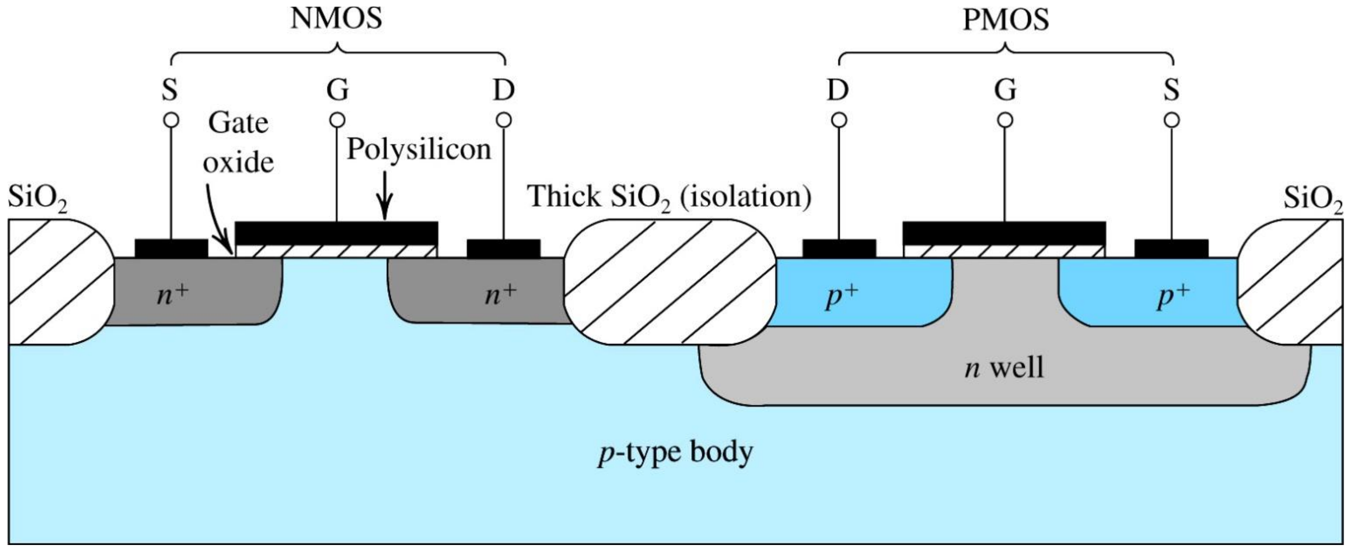
# 5.1.7. The *p*-Channel MOSFET

- PMOS technology originally dominated the MOS field (over NMOS). However, as manufacturing difficulties associated with NMOS were solved, "they" took over
- **Q:** Why is NMOS advantageous over PMOS?
  - **A:** Because electron mobility $\mu_n$ is 2 – 4 times greater than hole mobility $\mu_p$.
- **complementary MOS** (CMOS) **technology** – is technology which allows fabrication of both N and PMOS transistors on a single chip.

# 5.1.8. Complementary MOS or CMOS

- CMOS employs MOS transistors of both polarities.
  - more difficult to fabricate
  - more powerful and flexible
  - now more prevalent than NMOS or PMOS



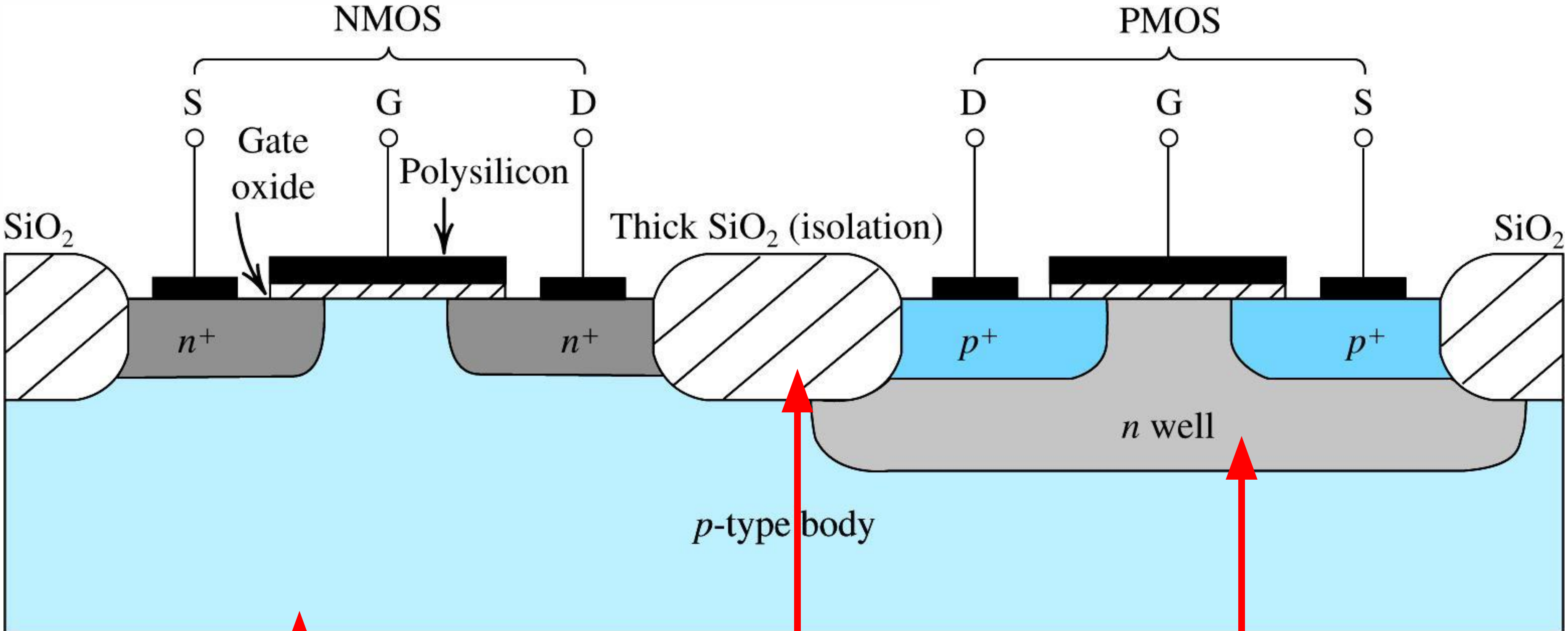



**Figure 5.10:** Cross-section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate *n*-type region, known as an *n* well. Another arrangement is also possible in which an *n*-type body is used and the *n* device is formed in a *p* well. Not shown are the connections made to the *p*-type body and to the *n* well; the latter functions as the body terminal for the *p*-channel device.

*p*-type semiconductor provides the MOS body (and allows generation of n-channel)

$SiO_2$ is used to isolate NMOS from PMOS

*n*-well is added to allow generation of *p*-channel

# Quick Recap!

- The equation used to define $i_D$ depends on relationship btw $v_{DS}$ and $v_{OV}$.
  - $v_{DS} << v_{OV}$
  - $v_{DS} < v_{OV}$
  - $v_{DS} => v_{OV}$
  - $v_{DS} >> v_{OV}$

$\mu_n$ represents mobility of electrons at surface of the n-channel in $m^2/Vs$

$$\text{(eq5.7)}\quad i_D = \underbrace{\left(C_{ox} W v_{OV}\right)}_{\text{charge per unit length of } n\text{-channel in } C/m} \underbrace{\left(\frac{\mu_n v_{DS}}{L}\right)}_{\text{electron drift velocity in } m^2/Vs} \text{ in } A$$

$$\text{(eq5.14)}\quad i_D = \left(\mu_n C_{ox}\right)\frac{W}{L}\left(v_{OV} - \tfrac{1}{2}v_{DS}\right)v_{DS} \text{ in } A$$

$$\text{(eq5.17)}\quad i_D = \frac{1}{2}\left(\mu_n C_{ox}\right)\frac{W}{L}v_{OV}^2 \text{ in } A$$

(e This has not been covered yet! A

# 5.2. Current-Voltage Characteristics

- Figure 5.11. shows an *n*-channel enhancement MOSFET.
- There are four terminals:
  - drain (D), gate (G), body (B), and source (S).
- Although, it is assumed that body and source are connected.

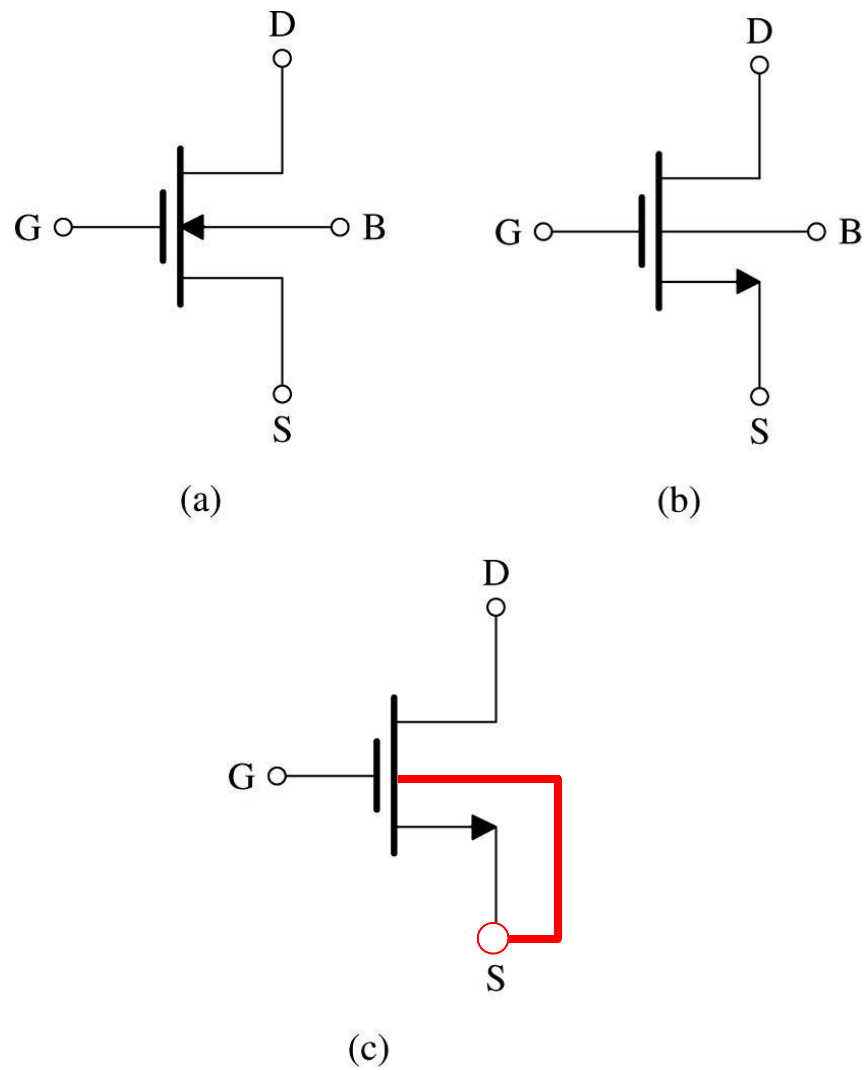
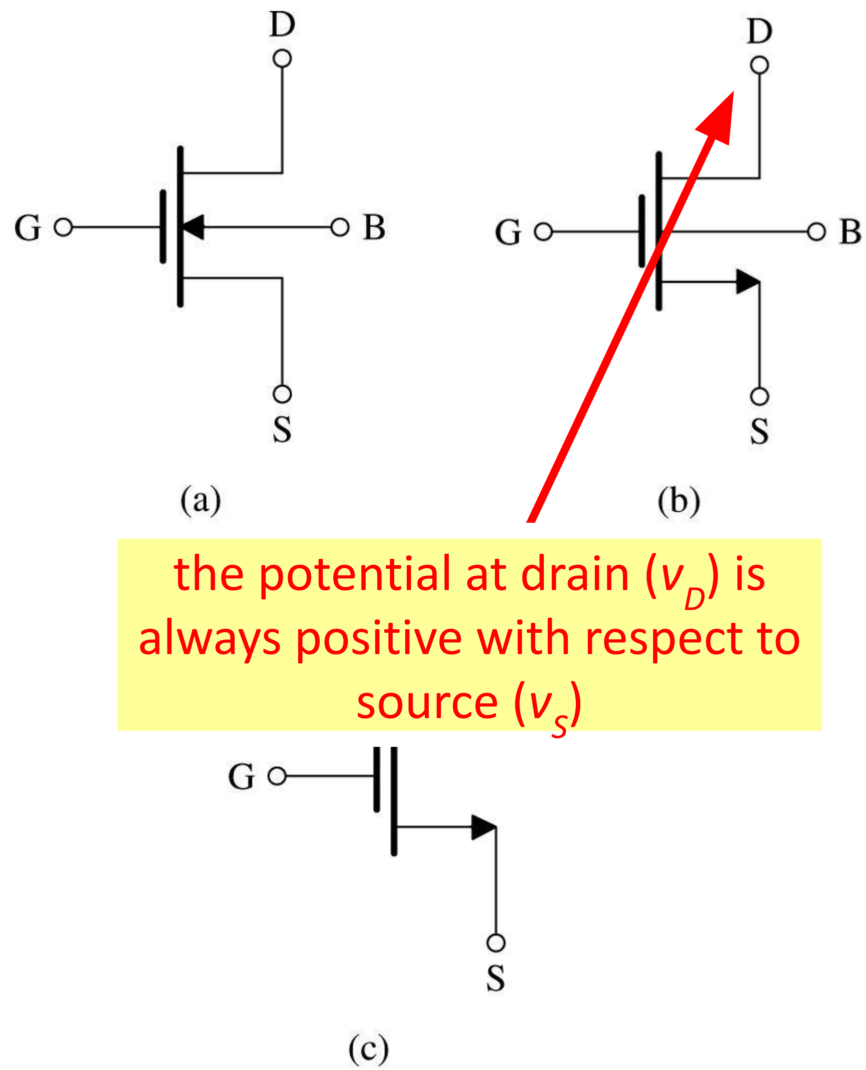



**Figure 5.11 (a):** Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., *n* channel). **(c)** Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

# 5.2. Current-Voltage Characteristics

- Although MOSFET is symmetrical device, one often designates terminals as source and drain.
- **Q:** How does one make this designation?
  - **A:** By polarity of voltage applied.
- Arrowheads designate "normal" direction of current flow
  - Note that, in part (b), we designate current as D□S.
  - No need to place arrow with B.



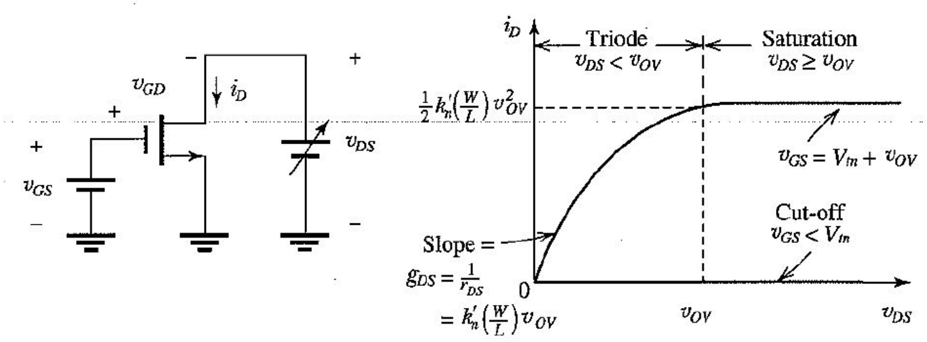

the potential at drain ($v_D$) is always positive with respect to source ($v_S$)

**Figure 5.11 (a):** Circuit symbol for the *n*-channel enhancement-type MOSFET. **(b)** Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., *n* channel). **(c)** Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

# 5.2.2. The $i_D$-$v_{DS}$ Characteristics

- Table 5.1. provides a compilation of the conditions and formulas for operation of NMOS transistor in three regions.
  - cutoff
  - triode
  - saturation

**Table 5.1** Regions of Operation of the Enhancement NMOS Transistor




- $v_{GS} < V_{tn}$ : no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$ : a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

*Triode Region*

Continuous channel, obtained by:

$$v_{GD} > V_{tn}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

*Saturation Region*

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

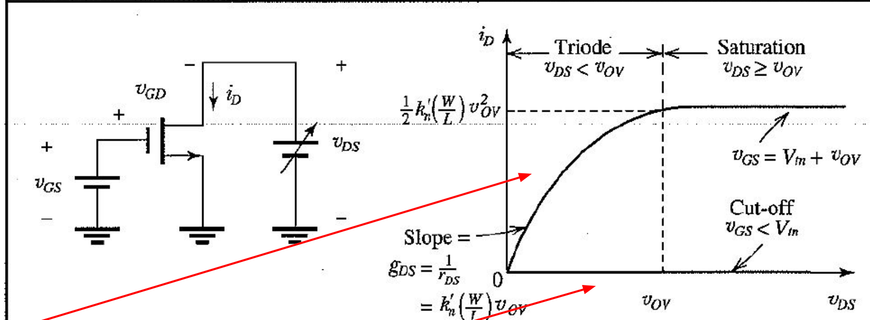
or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

# 5.2.2. The $i_D$-$v_{DS}$ Characteristics

- At top of table, it shows circuit consisting of NMOS transistor and two dc supplies ($v_{DS}$, $v_{GS}$)
- This circuit is used to demonstrate $i_D$-$v_{DS}$ characteristic
  - 1st set $v_{GS}$ to desired constant
  - 2nd vary $v_{DS}$
- Two curves are shown…
  - $v_{GS} < V_{tn}$
  - $v_{GS} = V_{tn} + v_{OV}$

**Table 5.1** Regions of Operation of the Enhancement NMOS Transistor




- $v_{GS} < V_{tn}$ : no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$ : a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

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or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

*Saturation Region*

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

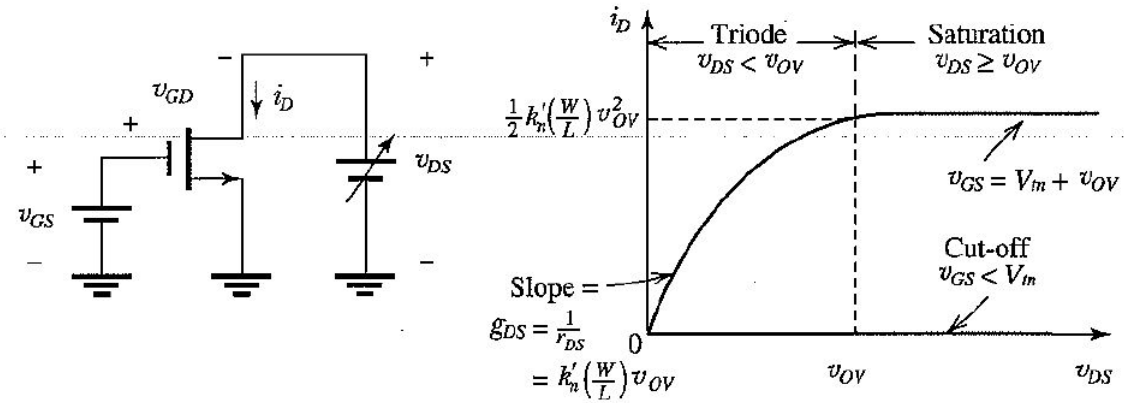
Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

## Table 5.1 Regions of Operation of the Enhancement NMOS Transistor




- $v_{GS} < V_{tn}$ : no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$ : a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

### *Triode Region*

Continuous channel, obtained by:

$$v_{GD} > V_{tn}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

### *Saturation Region*

Pinched-off channel, obtained by:

$$v_{GD} \leqslant V_{tn}$$

or equivalently:

$$v_{DS} \geqslant v_{OV}$$

Then

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

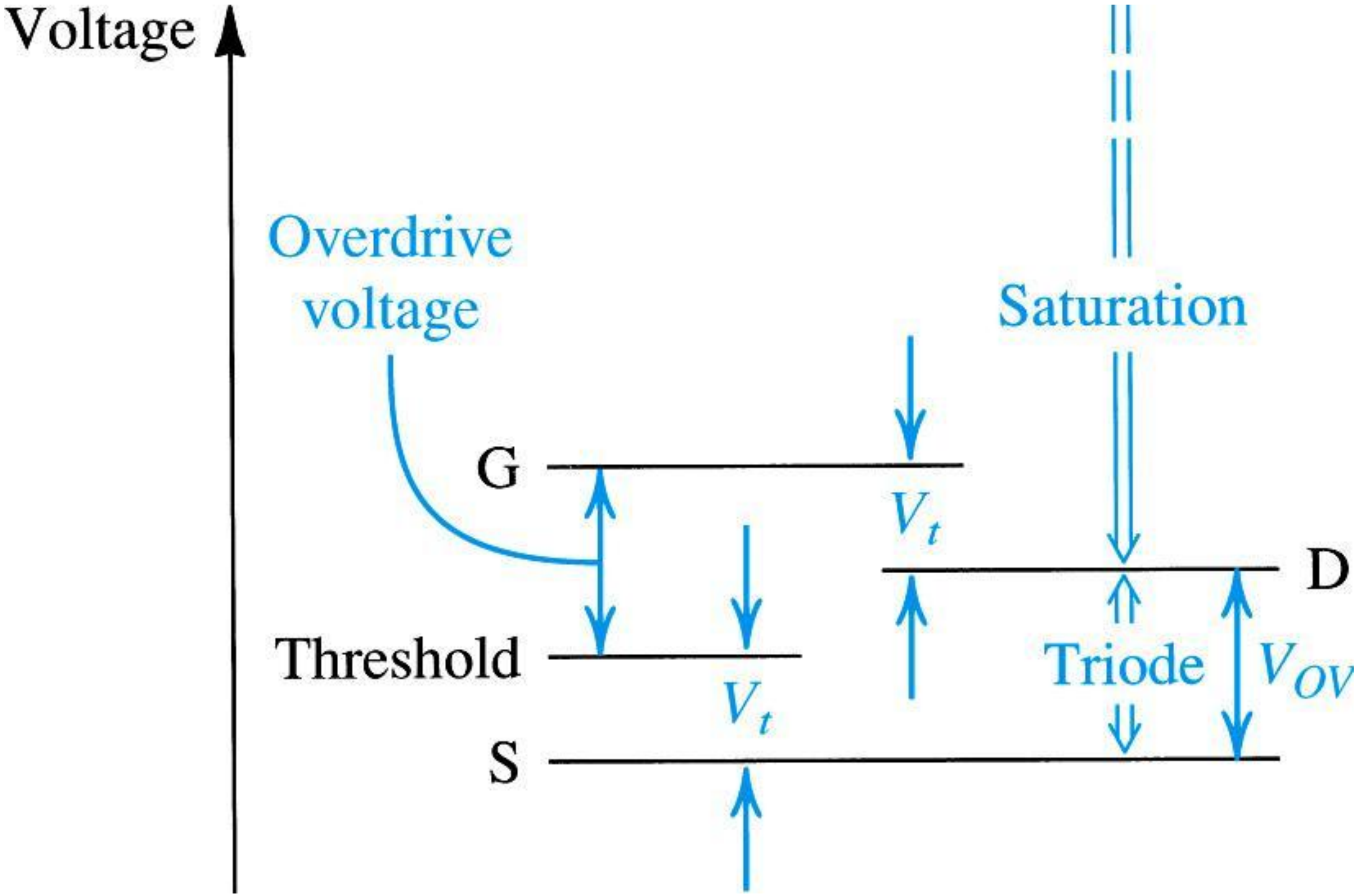



**Figure 5.12:** The relative levels of the terminal voltages of the enhancement NMOS transistor for operation in the triode region and in the saturation region.

equation (5.14)

as $v_{GS}$ increases, so do the **(1)** saturation current and **(2)** beginning of the saturation region

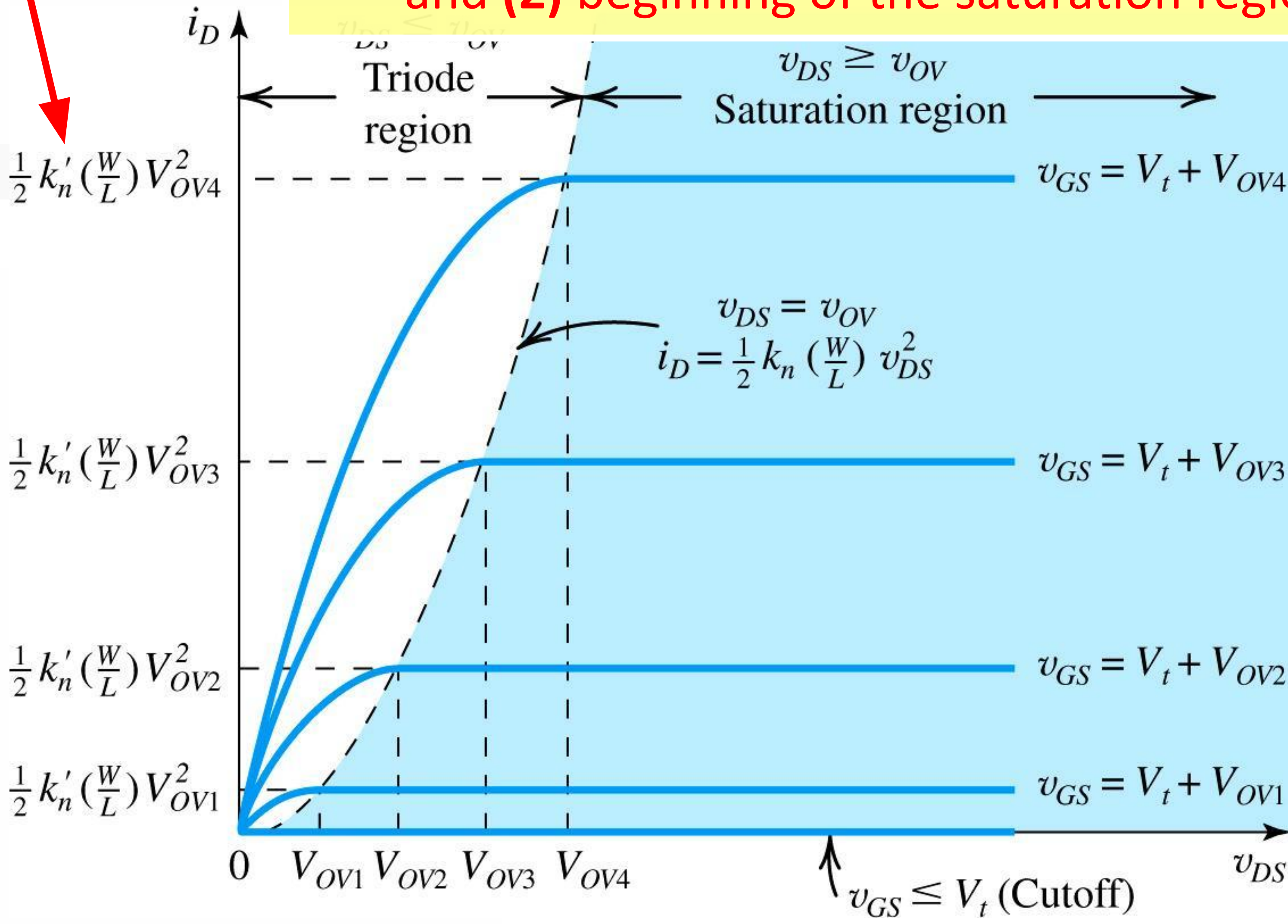



**Figure 5.13:** The $i_D - v_{DS}$ characteristics for an enhancement-type NMOS transistor

# 5.2.2. The $i_D$-$v_{GS}$ Characteristic

- **Q:** When MOSFET's are employed to design amplifier, in what range will they be operated?
  - **A:** saturation
- In saturation, the drain current ($i_D$) is…
  - dependent on $v_{GS}$
  - independent of $v_{DS}$
- In effect, it becomes a voltage-controlled current source.
  - This is key for amplification.

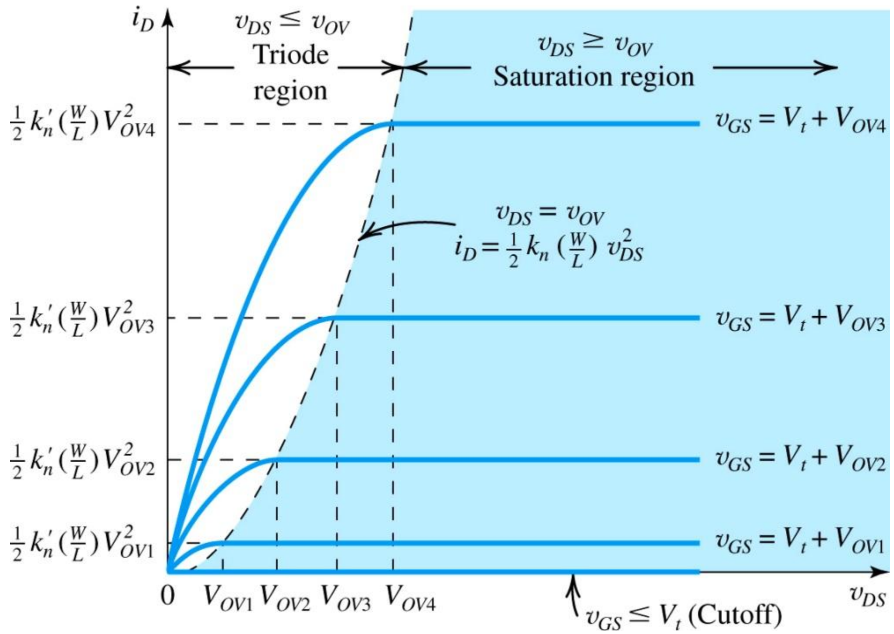



**Figure 5.13:** The $i_D - v_{DS}$ characteristics for an enhancement-type NMOS transistor

# 5.2.2. The $i_D$-$v_{GS}$ Characteristic

- **Q:** What is one problem with (5.21)?
  - **A:** It is nonlinear w/ respect to $v_{OV}$ … however, this is not of concern now.

- In effect, it becomes a voltage-controlled current source.
  - This is key for amplification.
  - Refer to (5.21).

(eq5.21) $$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\overbrace{(v_{GS} - V_{tn})^2}^{v_{OV}^2}$$

this relationship provides basis for application of MOSFET as amplifier

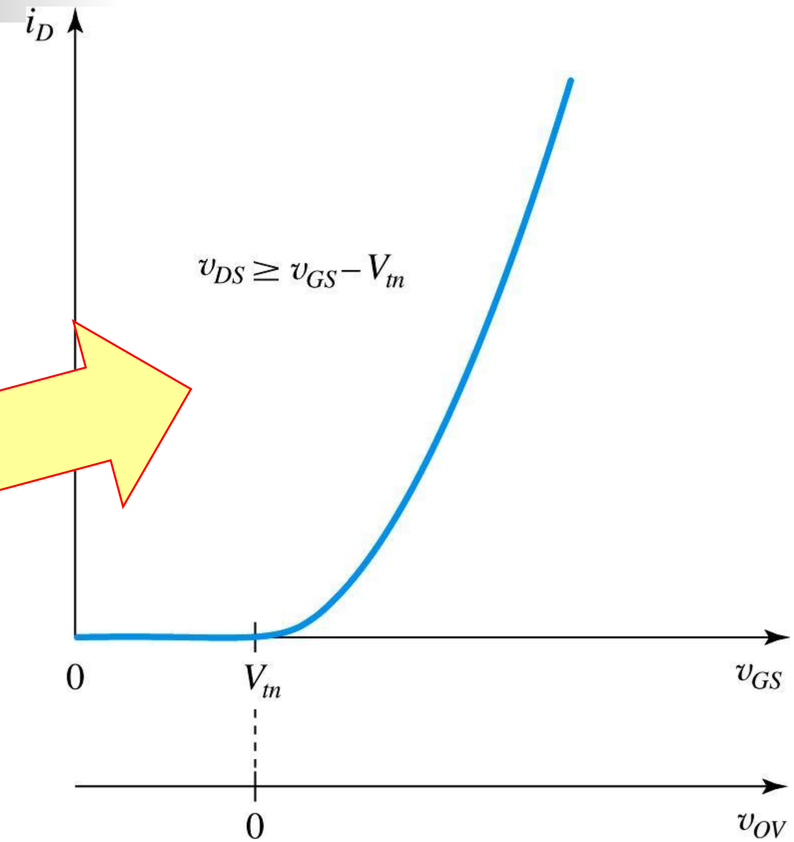



**Figure 5.14:** The $i_D$-$v_{GS}$ characteristic of an NMOS transistor operating in the saturation region. The $i_D$-$v_{OV}$ characteristic can be obtained by simply re-labeling the horizontal axis, that is, shifting the origin to the point $v_{GS} = V_{tn}$.

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# 5.2.2. The $i_D$-$v_{GS}$ Characteristic

- The view of transistor as CVCS is exemplified in figure 5.15.
  - This circuit is known as the **large-signal equivalent circuit.**
  - Current source is ideal.
  - Infinite output resistance represents independent, in saturation, of $i_D$ from $v_{DS}$.

note that, in this circuit, $i_D$ is completely independent of $v_{DS}$ (because no shunt resistor exists)

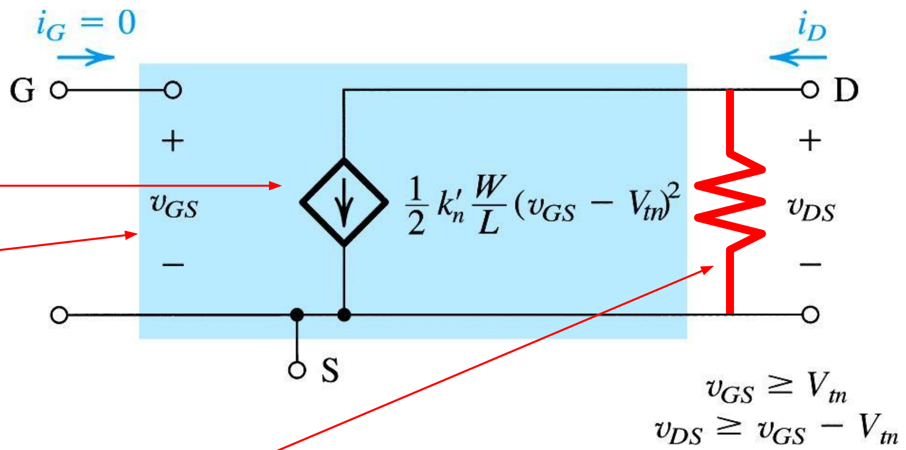


**Figure 5.15:** Large-signal equivalent-circuit model of an *n*-channel MOSFET operating in the saturation

# **Example 5.2:** NMOS Transistor

- **Example 5.2. Problem Statement:** Consider an NMOS transistor fabricated in an 0.18-$\mu m$ process with L = 0.18$\mu m$ and W = 2$\mu m$. The process technology is specified to have $C_{ox}$ = 8.6$fF/\mu m^2$, $\mu_n$ = 450$cm^2/Vs$, and $V_{tn}$ = 0.5$V$.
- **Q(a):** Find $V_{GS}$ and $V_{DS}$ that result in the MOSFET operating at the edge of saturation with ID = 100$\mu A$.
- **Q(b):** If VGS is kept constant, find VDS that results in $I_D$ = 50$\mu A$.
- **Q(c):** To investigate the use of the MOSFET as a linear amplifier, let it be operating in saturation with $V_{DS}$ = 0.3$V$. Find the change in $i_D$ resulting from $v_{GS}$ changing from 0.7$V$ by +0.01$V$ and -0.01$V$.

# 5.2.4. **Finite Output Resistance in Saturation**

- In previous section, we assume (in saturation) $i_D$ is independent of $v_{DS}$.
- Therefore, a change $\Delta v_{DS}$ causes no change in $i_D$.
  - This implies that the incremental resistance $R_S$ is infinite.
  - It is based on the idealization that, once the *n*-channel is pinched off, changes in $v_{DS}$ will have no effect on $i_D$.
  - The problem is that, in practice, this is not completely true.

# 5.2.4. Finite Output Resistance in Saturation

- **Q:** What effect will increased $v_{DS}$ have on *n*-channel once pinch-off has occurred?
  - **A:** It will cause the pinch-off point to move slightly away from the drain & create new depletion region.
  - **A:** Voltage across the (now shorter) channel will remain at ($v_{OV}$).
  - **A:** However, the additional voltage applied at $v_{DS}$ will be seen across the "new" depletion region.

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# 5.2.4. **Finite Output Resistance in Saturation**

this is the most important point here

- **Q:** What effect will increased $v_{DS}$ have on *n*-channel once pinch-off has occurred?
  - **A:** This voltage accelerates electrons as they reach the drain end, and sweep them across the "new" depletion region.
  - **A:** However, at the same time, the length of the *n*-channel will decrease.
    - Known as **channel length modulation**.

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# 5.2.4. Finite Output Resistance in Saturation

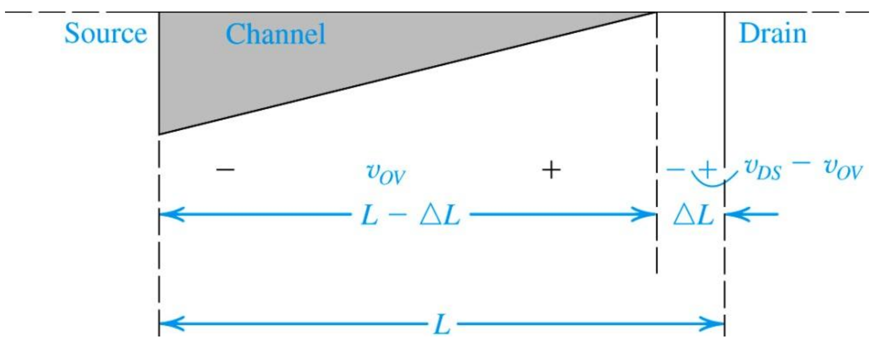



**Figure 5.16:** Increasing $v_{DS}$ beyond $v_{DSsat}$ causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length by $\Delta L$

- **Q:** How do we account for "this effect" in $i_D$?
  - **A:** Refer to (5.23).

valid when $v_{DS} > v_{OV}$

(eq5.17) $i_D = \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2$ in $A$

(eq5.23) $i_D = \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2(1+\lambda v_{DS})$ in $A$

valid when $v_{DS} >> v_{OV}$

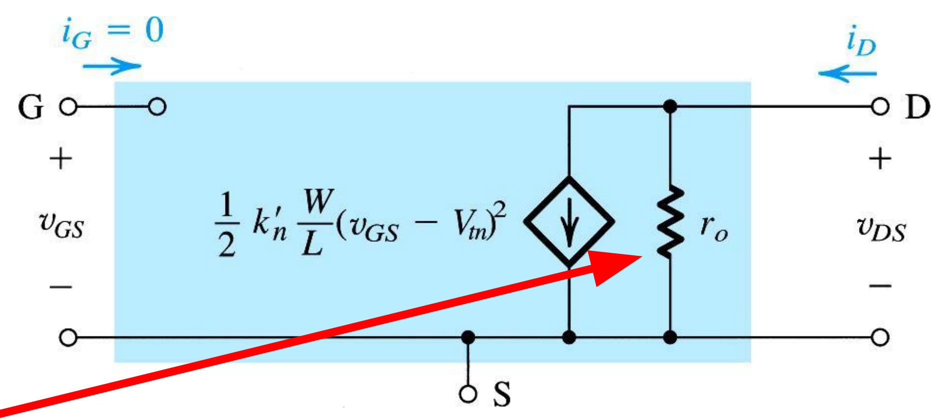



**Figure 5.18:** Large-Signal Equivalent Model of the $n$-channel MOSFET in saturation, incorporating the output resistance $r_o$. The output resistance models the linear dependence of $i_D$ on $v_{DS}$ and is given by (5.23)

  - **A:** Addition of finite output resistance ($r_o$).

# 5.2.4. Finite Output Resistance in Saturation

- **Q:** How is $r_o$ defined?
  - **step #1:** Note that $r_o$ is the 1/slope of $i_D$-$v_{DS}$ characteristic.
  - **step #2:** Define relationship between $i_D$ and $v_{DS}$ using (5.23).
  - **step #3:** Take derivative of this function.
  - **step #4:** Use above to define $r_o$.
- Note that $r_o$ may be defined in terms of $i_D$, where $i_D$ does not take in to account channel length modulation…

$$\text{(eq5.24)}\quad r_o = \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1}_{v_{GS}=constant}$$

$$\text{(eq5.23)}\quad \frac{\partial i_D}{\partial v_{DS}} = \frac{\partial}{\partial v_{DS}}\left(\underbrace{\frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2(1+\lambda v_{DS})}_{(5.23)}\right)$$

$$\text{(eq5.23)}\quad \frac{\partial i_D}{\partial v_{DS}} = \frac{\partial}{\partial v_{DS}}\left(\underbrace{\frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2(1+\lambda v_{DS})}_{(5.23)}\right)$$

$$\text{(eq5.23)}\quad \frac{\partial i_D}{\partial v_{DS}} = \frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2\lambda$$

$$\boxed{\begin{aligned}&\text{(eq5.25)}\quad r_o = \left[\frac{1}{2}(\mu_n C_{ox})\frac{W}{L}v_{OV}^2\lambda\right]^{-1}_{v_{GS}=constant}\\ &\text{(eq5.24)}\quad r_o = \frac{1}{\lambda i_D} = \frac{V_A}{i_D}\end{aligned}}$$

# 5.2.4. Finite Output Resistance in Saturation

- **Q:** What is $\lambda$?
  - **A:** A device parameter with the units of $V^{-1}$, the value of which depends on manufacturer's design and manufacturing process.
    - much larger for newer tech's
- Figure 5.17 demonstrates the effect of channel length modulation on $v_{DS}$-$i_D$ curves
  - In short, we can draw a straight line between $V_A$ and saturation.

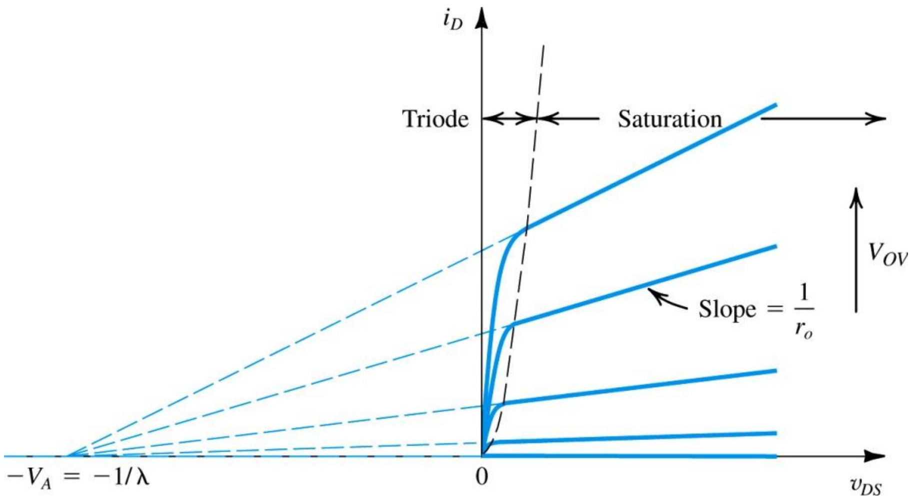



**Figure 5.17:** Effect of $v_{DS}$ on $i_D$ in the saturation region. The MOSFET parameter $V_A$ depends on the process technology and, for a given process, is proportional to the channel length $L$.

# 5.2.5. **Characteristics of the *p*-channel MOSFET**

- Characteristics of the *p*-channel MOSFET are similar to the *n*-channel, however with many signs reversed.
- Please review section 5.2.5 from the text, with focus on table 5.2.

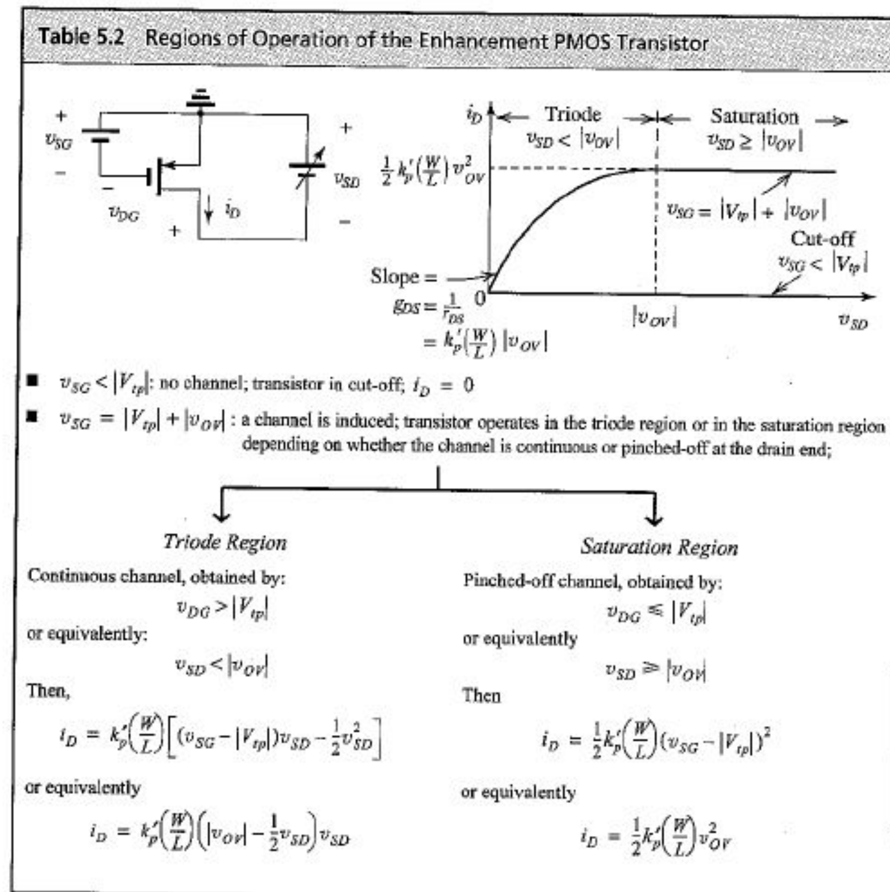


**Table 5.2** Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cut-off; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$ : a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

*Triode Region*

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently:

$$v_{SD} < |v_{OV}|$$

Then,

$$i_D = k'_p\left(\frac{W}{L}\right)\left[(v_{SG} - |V_{tp}|)v_{SD} - \frac{1}{2}v_{SD}^2\right]$$

or equivalently

$$i_D = k'_p\left(\frac{W}{L}\right)\left(|v_{OV}| - \frac{1}{2}v_{SD}\right)v_{SD}$$

*Saturation Region*

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

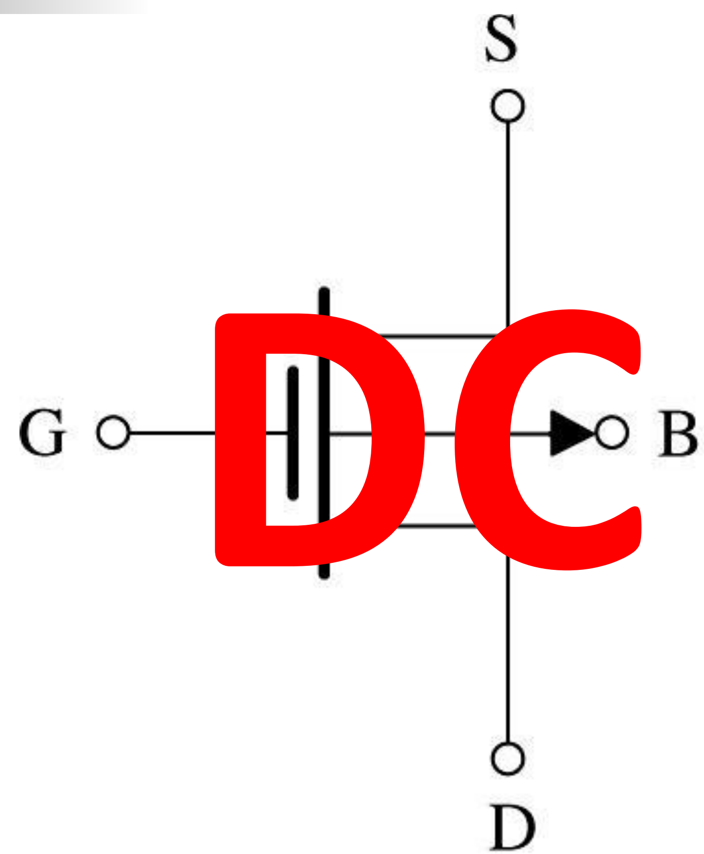
$$i_D = \frac{1}{2}k'_p\left(\frac{W}{L}\right)(v_{SG} - |V_{tp}|)^2$$

or equivalently

$$i_D = \frac{1}{2}k'_p\left(\frac{W}{L}\right)v_{OV}^2$$

# 5.3. MOSFET Circuits at DC

- We move on to discuss how MOSFET's behave in dc circuits.
- We will neglect the effects of channel length modulation (assuming $\lambda = 0$).
- We will work in terms of overdrive voltage ($v_{OV}$), which reduces need to distinguish between PMOS and NMOS.



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# **Example 5.3:** NMOS Transistor

- **Problem Statement:** Design the circuit of Figure 5.21, that is, determine the values of $R_D$ and $R_S$ – so that the transistor operates at $I_D = 0.4mA$ and $V_D = +0.5V$. The NMOS transistor has $V_t = 0.7V$, $\mu_n C_{ox} = 100\mu A/V^2$, L = 1$\mu m$, and W = 32$\mu m$. Neglect the channel-length modulation effect (i. e. assume that $\lambda = 0$).

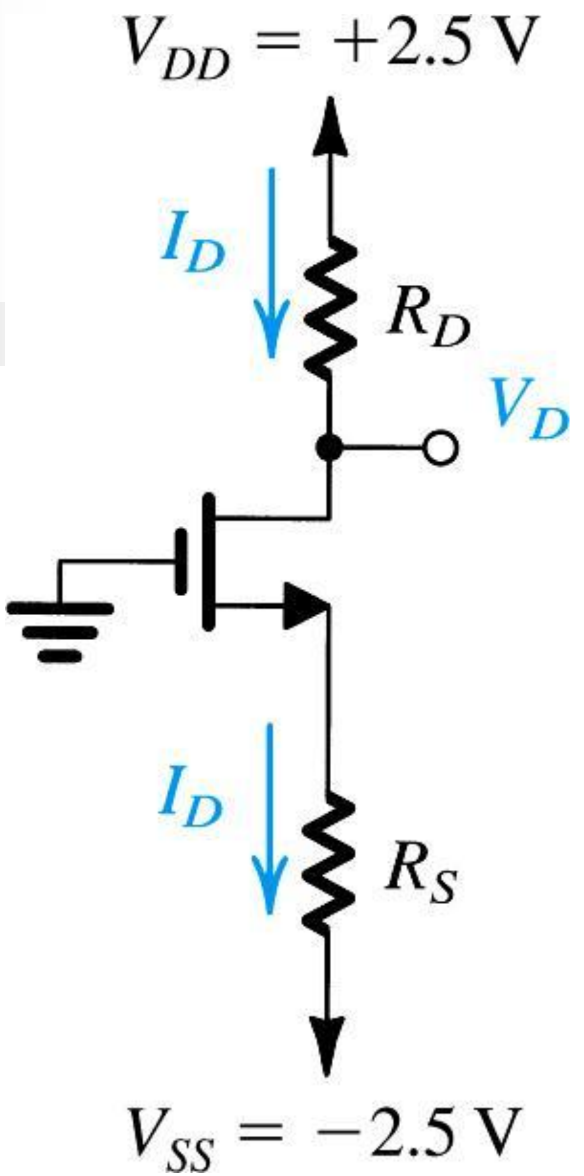
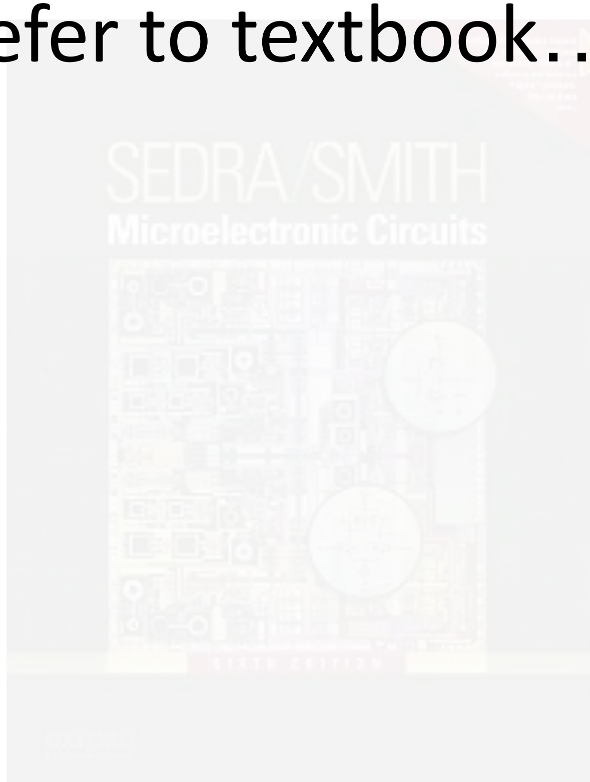



**Figure 5.21:** Circuit for Example 5.3.

# Example 5.4:

- Refer to textbook…



# Example 5.5: MOSFET

- **Problem Statement:** Design the circuit in Figure 5.23 to establish a drain voltage of 0.1$V$. What is the effective resistance between drain and source at this operating point? Let $V_{tn} = 1V$ and $k'_n(W/L) = 1mA/V^2$.

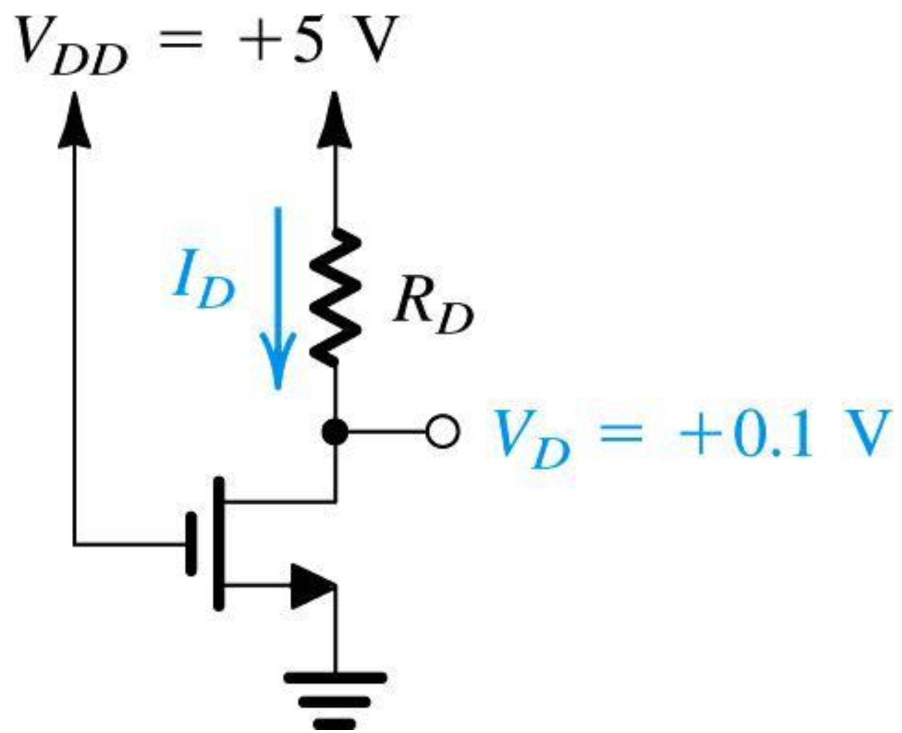
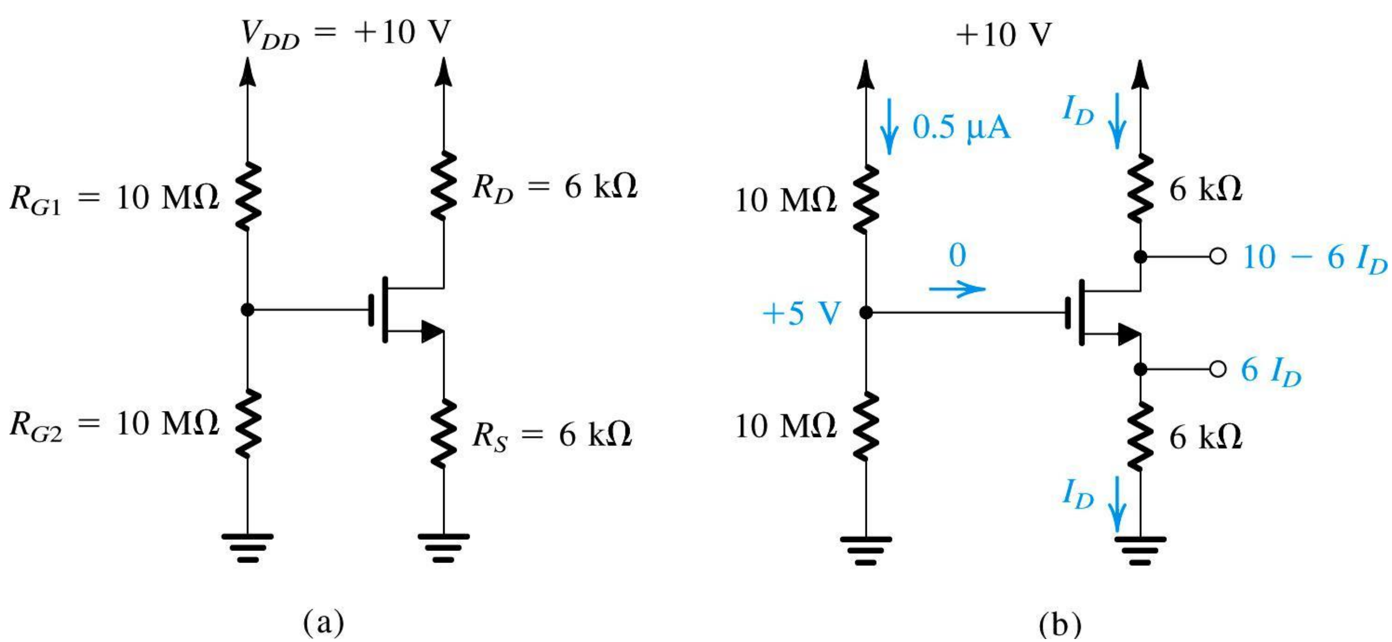



**Figure 5.23:** Circuit for Example 5.5.

# **Example 5.6:** MOSFET

- **Problem Statement:** Analyze the circuit shown in Figure 5.24(a) to determine the voltages at all nodes and the current through all branches. Let $V_{tn} = 1V$ and $k'_n(W/L) = 1mA/V^2$. Neglect the channel-length modulation effect (i.e. assume $\lambda = 0$).

**Figure 5.24: (a)** Circuit for Example 5.6. **(b)** The circuit with some of the analysis details shown.



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# **Example 5.7:** PMOS Transistor

- **Problem Statement:** Design the circuit of Figure 5.25 so that transistor operates in saturation with $I_D = 0.5mA$ and $V_D = +3V$. Let the enhancement-type PMOS transistor have $V_{tp} = -1V$ and $k'_p(W/L) = 1mA/V^2$. Assume $\lambda = 0$.
- **Q:** What is the largest value that $R_D$ can have while maintaining saturation-region operation?

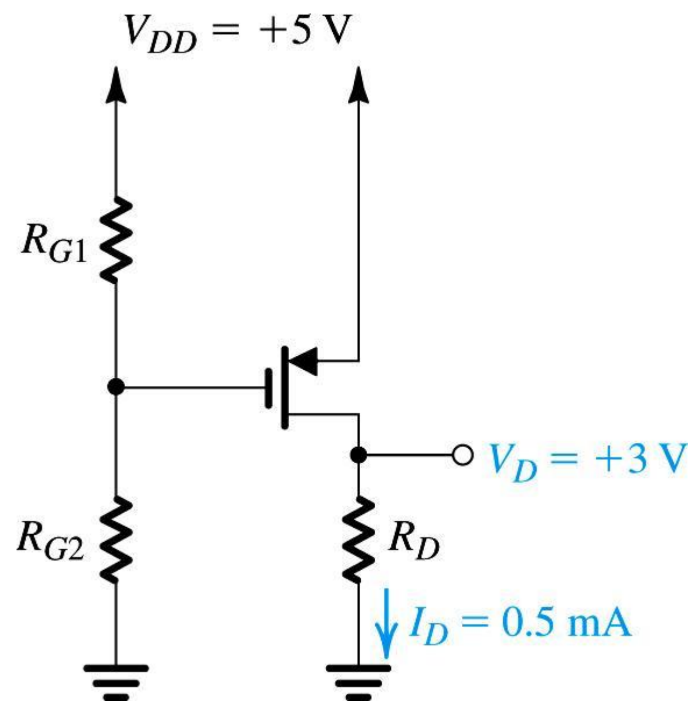



**Figure 5.25:** Circuit for Example 5.7.

# **Exercise 5.8:** CMOS Transistor

- **Problem Statement:** The NMOS and PMOS transistors in the circuit of Figure 5.26(a) are matched, with $k'_n(W_n/L_n) = k'_p(W_p/L_p) = 1mA/V^2$ and $V_{tn} = -V_{tp} = 1V$. Assuming $\lambda = 0$ for both devices.
- **Q:** Find the drain currents $i_{DN}$ and $i_{DP}$, as well as voltage $v_O$ for $v_I = 0V$, $+2.5V$, and $-2.5V$.

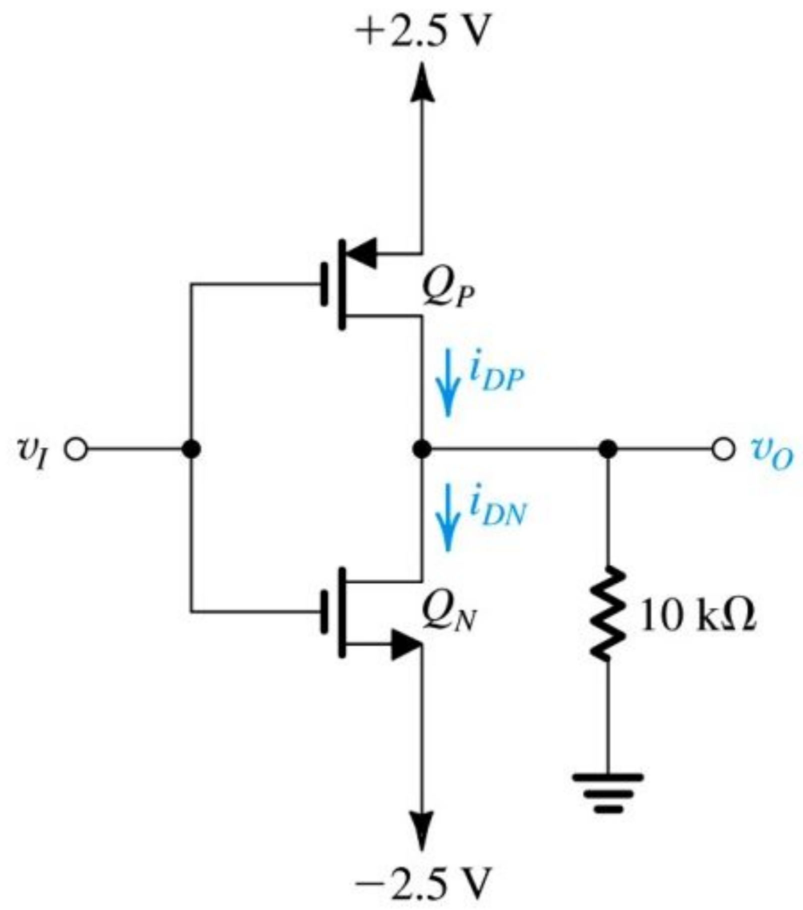
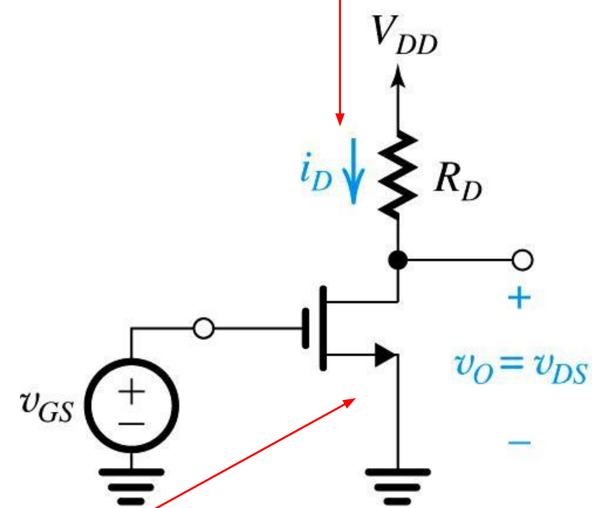



**Figure 5.26:** Circuits for Example 5.8.

# 5.4.1. **Obtaining a Voltage Amplifier**

example of transconductance amplifier

- In section 1.5 of text, we learned that voltage controlled current source (VCCS) can serve as **transconductance amplifier**.
  - the following slides (with blue tint) are a review
- **Q:** How can we translate current output to voltage?
  - **A:** Measure voltage drop across load resistor.




(a)

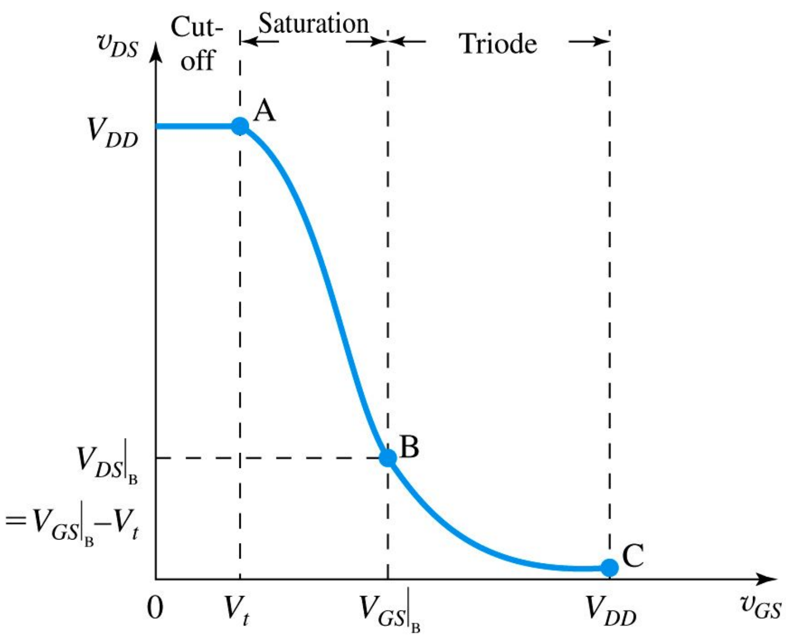
$v_{out}$ supply function of input $v_{GS}$

(eq5.30) $v_{DS} = V_{DD} - i_D R_D$

**Figure 5.27: (a)** simple MOSFET amplifier with input $v_{GS}$ and output $v_{DS}$

# 5.4.2. **Voltage Transfer Characteristic**

- **voltage transfer characteristics (VTC)** – plot of out voltage vs. input
- **three regions** exist in VTC
  - $v_{GS} < V_t$ □ **cut off FET**
    - $v_{OV} = v_{GS} - V_t < 0$
    - $I_D = 0$
    - $v_{DS}$ ??? $v_{OV}$
    - $v_{out} = v_{DD}$
  - $V_t < v_{GS} < v_{DS} + V_t$ □ **saturation**
    - $v_{OV} = v_{GS} - V_t > 0$
    - $I_D = ½\, k_n(v_{GS} - V_t)^2$
    - $v_{DS} >> v_{OV}$
    - $v_{out} = V_{DD} - I_D R_D$
  - $v_{DS} + V_t < v_{GS} < V_{DD}$ □ **triode**
    - $v_{OV} = v_{GS} - V_t > 0$
    - $I_D = k_n(v_{GS} - V_t - v_{DS})v_{DS}$
    - $v_{DS} > v_{OV}$
    - $v_{out} = V_{DD} - I_D R_D$



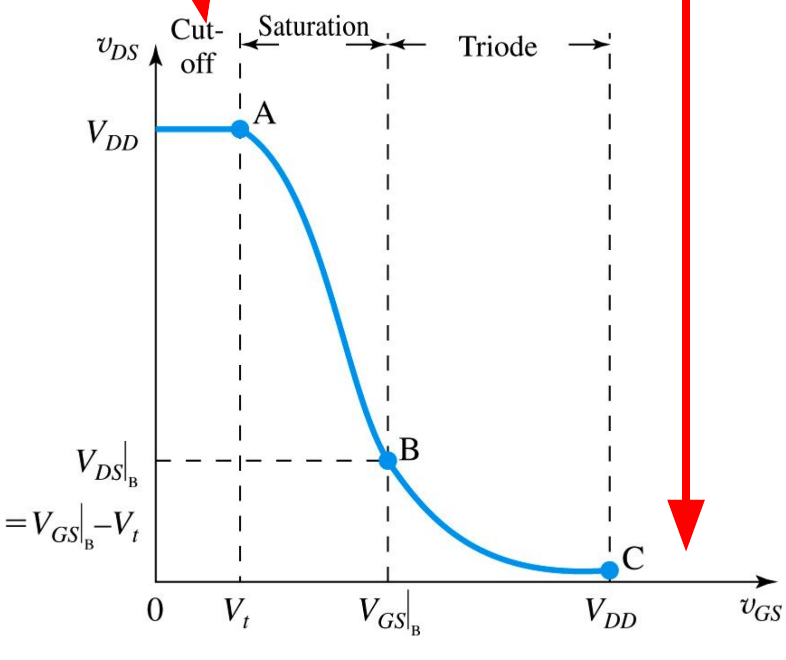

(b)

**Figure 5.27: (b)** the voltage transfer characteristic (VTC) of the amplifier from previous slide

cutoff FET

cutoff AMP

# Voltag Characteristic




(b)

**Figure 5.27: (b)** the voltage transfer characteristic (VTC) of the amplifier from previous slide

- **Q:** What observations may be drawn?
  - **A:** Cutoff FET represents transistor blocking, cutoff AMP represents $v_{out} = 0$
  - **A:** As $v_{GS}$ increases…
    - $v_{DS}$ (effectively) decreases
    - $i_D$ increases
    - $v_{out}$ decreases nonlinearly
    - gain ($G$) decreases
  - **A:** Once $v_{DS} > v_{DD}$, all power is dissipated by resistor $R_D$

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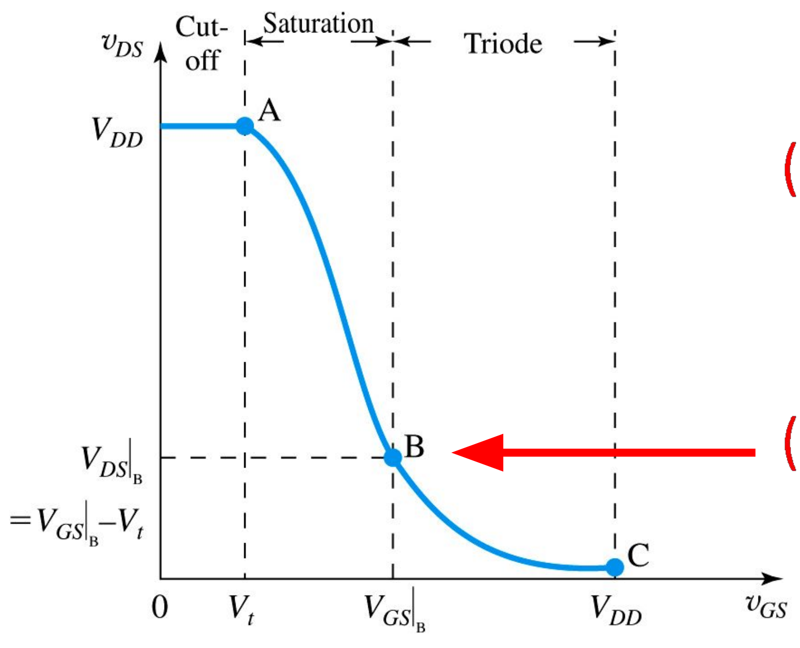
# 5.4.2. **Voltage Transfer Characteristic**

**Q:** How do we define $v_{DS}$ in terms of $v_{GS}$ for saturation?




(b)

**Figure 5.27: (b)** the voltage transfer characteristic (VTC) of the amplifier from previous slide

this is equation is simply ohm's law / KVL

(eq5.32) $$v_{DS} = V_{DD} - \underbrace{\left[\frac{1}{2}k_n\left(v_{GS} - V_t\right)^2\right]}_{i_D} R_D$$

(eq5.33) $$V_{GS}\big|_B = V_t + \frac{\sqrt{2k_nR_DV_{DD} + 1} - 1}{k_nR_D}$$

**Q:** How do we define point *B* – boundary between saturation and triode regions?

# 5.4.3. Biasing the MC to Obtain Linear Amplification

This equation differs from (5.32) because it considers dc component only.

this equation is simply ohm's law

(eq5.34) $V_{DS} = V_{DD} - \left[ \frac{1}{2} k_n \left( V_{GS} - V_t \right)^2 \right] R_D$

$V_{source} - I_D R_D$

- **Q:** How can we linearize VTC?
  - **A:** Appropriate biasing technique
  - **A:** Dc voltage $v_{GS}$ is selected to obtain operation at point *Q* on segment *AB*
- **Q:** How do we choose $v_{GS}$?
  - **A:** Will discuss shortly…

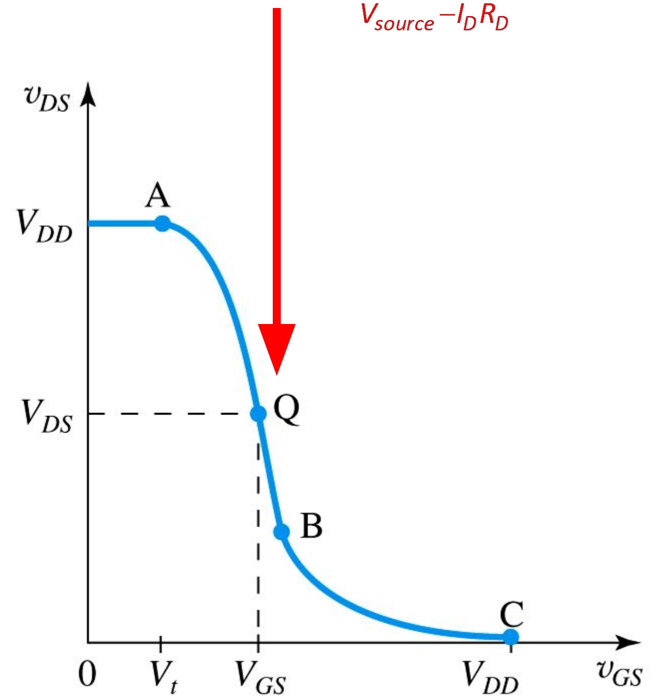



**Figure 5.28:** biasing the MOSFET amplifier at point *Q* located on segment *AB* of VTC

# 5.4.3. Biasing the MOSFET to Obtain Linear Amplification

this equation is simply ohm's law

(eq5.34) $V_{DS} = V_{DD} - \left[\frac{1}{2}k_n\left(V_{GS} - V_t\right)^2\right]R_D$

$V_{source} - I_D R_D$

- **bias point / dc operating pt. (*Q*)** – point of linearization for MOSFET
  - Also known as **quiescent point**.
- **Q:** How will *Q* help us?
  - **A:** Because VTC is linear near *Q*, we may perform linear amplification of signal << Q

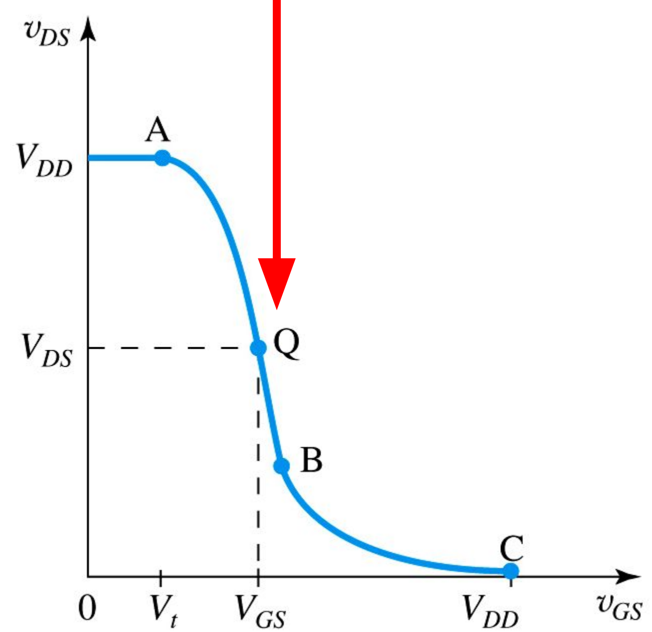
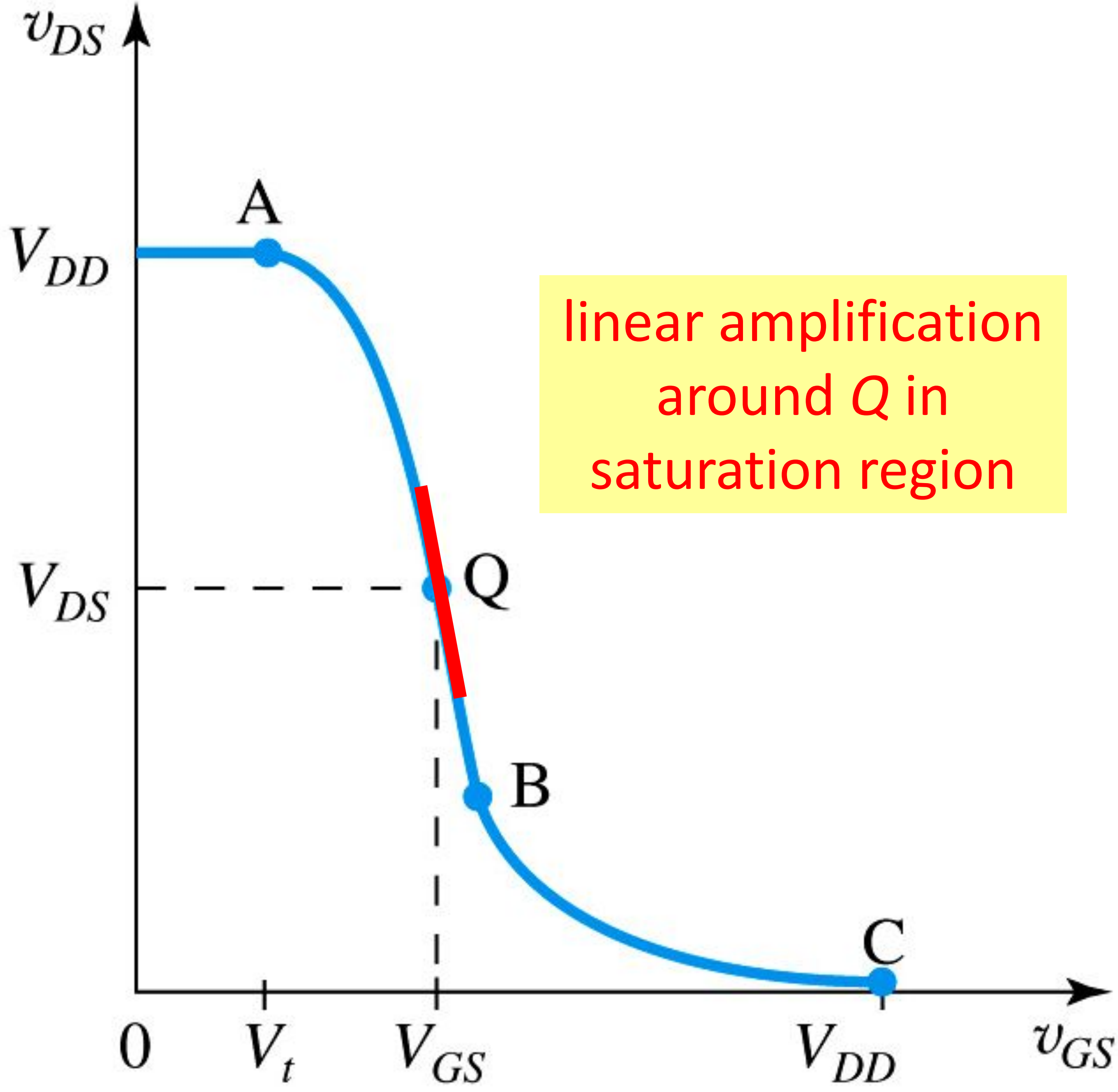



**Figure 5.28:** biasing the MOSFET amplifier at point *Q* located on segment *AB* of VTC



$v_{DS}$
A
$V_{DD}$
linear amplification around $Q$ in saturation region
$V_{DS}$
Q
B
C
0
$V_t$
$V_{GS}$
$V_{DD}$
$v_{GS}$

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# 5.4.3. Biasing the MOSFET to Obtain Linear Amplification

- **Q:** How is linear gain achieved?
  - **step #1:** Bias MOSFET with dc voltage $V_{GS}$ as defined by (5.34)
  - **step #2:** Superimpose amplifier input ($v_{gs}$) upon $V_{GS}$.
  - **step #3:** Resultant $v_{ds}$ should be linearly proportional to small-signal component $v_{gs}$.

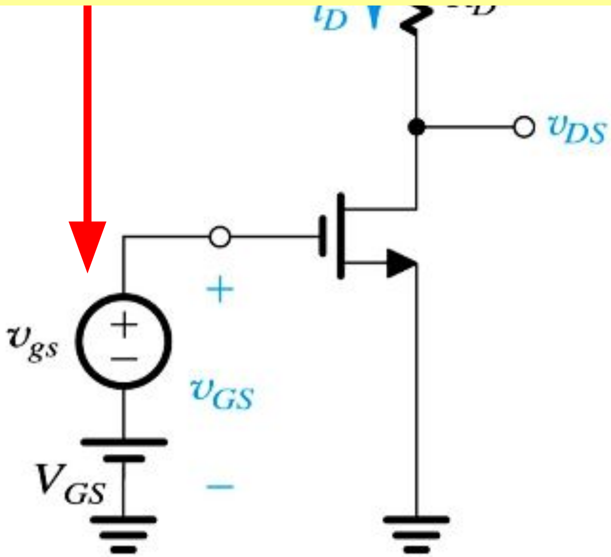
$$\mathbf{v}_{GS}(t) = V_{GS} + \mathbf{v}_{gs}(t)$$

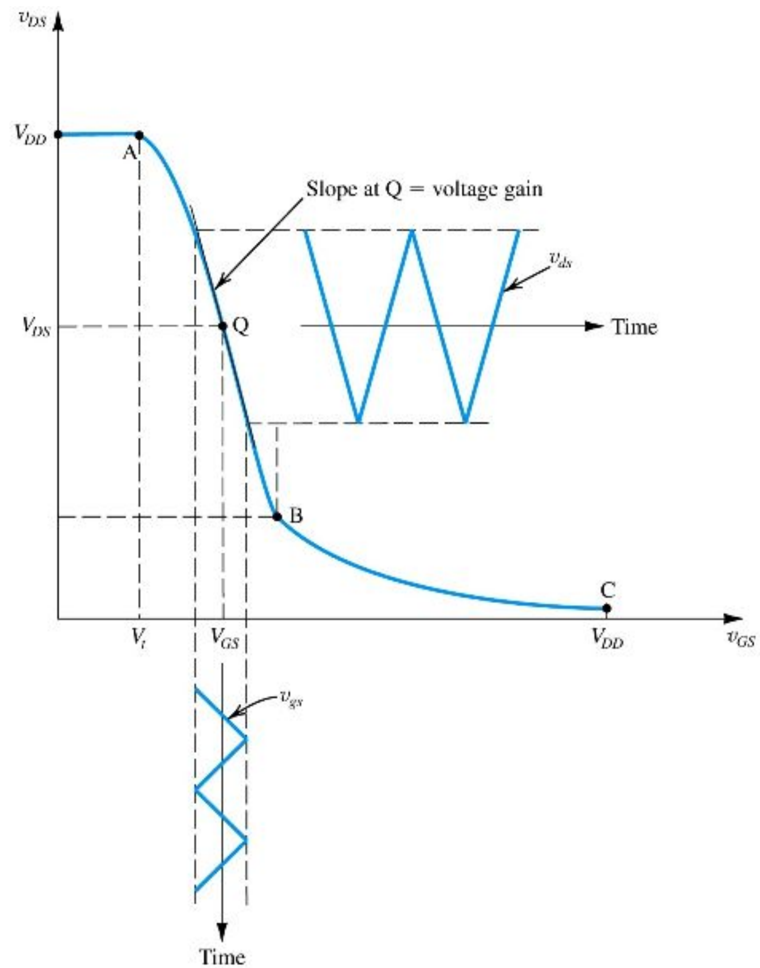
$$\mathbf{v}_{ds}(t) \propto \mathbf{v}_{gs}(t)$$

# **Q:** How is linear gain achieved?

**Figure 5.29:** The MOSFET amplifier with a small time-varying signal $v_{gs}(t)$ superimposed on the dc bias voltage $V_{GS}$. The MOSFET operates on a short almost-linear segment of the VTC around the bias point $Q$ and provides an output voltage $v_{ds} = A_v v_{gs}$

As long as $\mathbf{v}_{gs}(t)$ is small, its effect on $v_{DS}(t)$ will be linear – *facilitating linear amplification.*




(b)

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# Q: How is linear gain achieved?

- **step #4:** Note if $v_{gs}$ is small, output $v_{ds}$ will be nearly linearly proportional to it.
  - Slope will be constant.

(eq5.35) $$A_v = \left.\frac{dv_{DS}}{dv_{GS}}\right|_{v_{GS} \approx V_{GS}}$$

means that $v_{gs}$ is small

**action:** replace $v_{DS}$ with (5.32)

(eq5.35) $$A_v = \left.\frac{d\left(V_{DD} - \frac{1}{2}k_n\left(v_{GS} - V_t\right)^2 R_D\right)}{dv_{GS}}\right|_{v_{GS} \approx V_{GS}}$$

**action:** simplify

(eq5.36) $$A_v = -k_n\left(V_{GS} - V_t\right)R_D$$

**action:** replace with $V_{OV}$

(eq5.37) $$A_v = -k_n V_{OV} R_D$$

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# 5.4.4. **Small-Signal Voltage Gain**

- **Q:** What observations can be made about voltage gain?
  - **A:** Gain is negative.
  - **A:** Gain is proportional to:
    - load resistance ($R_D$)
    - transistor conductance parameter ($k_n$)
    - overdrive voltage ($v_{OV}$)

(eq5.35) $$A_v = \left.\frac{dv_{DS}}{dv_{GS}}\right|_{v_{GS} \approx V_{GS}}$$

means that $v_{gs}$ is small

**action:** replace $v_{DS}$ with (5.32)

(eq5.35) $$A_v = \left.\frac{d\left(V_{DD} - \frac{1}{2}k_n\left(v_{GS} - V_t\right)^2 R_D\right)}{dv_{GS}}\right|_{v_{GS} \approx V_{GS}}$$

**action:** simplify

(eq5.36) $$A_v = -k_n\left(V_{GS} - V_t\right)R_D$$

**action:** replace with $V_{OV}$

(eq5.37) $$A_v = -k_n V_{OV} R_D$$

# 5.4.4. **Small-Signal Gain**

- Equation (5.38) is another version of (5.37) which incorporates (5.17).
- It demonstrates that gain is ratio of:
  - voltage drop across $R_D$
  - half of over voltage

(eq5.37) $A_v = -k_n V_{OV} R_D$

**action:** incorporate (5.17) $i_D = \frac{1}{2} k_n v_{OV}^2$

(eq5.38) $A_v = -\left( \frac{I_D R_D}{V_{OV}/2} \right)$

# 5.4.4. **Small-Signal Gain**

This does not mean that output may be 10x supply ($V_{DD}$).

For example, 0.13*mm* CMOS technology with *VDD* = 1.3*V* yields maximum gain of 13*V/V*.

- **Q:** How does (5.38) relate to ph
  - **A:** For modern CMOS techn than 0.2*V*.
  - **A:** This means that max achievable gain is approximately $10V_{DD}$.

$$\max(A_v) = -\left(\frac{\overbrace{\max(I_D)R_D}^{V_{DD}}}{\underbrace{V_{OV}}_{0.1V}/2}\right) = -10V_{DD}$$

# Example 5.9:
## MOSFET Amplifier

- **Problem Statement:** Consider the amplifier circuit shown in Figure 5.29(a). The transistor is specified to have $V_t = 0.4V$, $k'_n = 0.4mA/V^2$, $W/L = 10$, and $\lambda = 0$. Also, let $V_{DD} = 1.8V$, $R_D = 17.5k\Omega hms$, and $V_{GS} = 0.6V$.
- **Q(a):** For $v_{gs} = 0$ (and hence $v_{ds} = 0$), find $V_{OV}$, $I_D$, $V_{DS}$, and $A_v$.
- **Q(b):** What is the maximum symmetrical signal swing allowed at the drain? Hence, find the maximum allowable amplitude of a sinusoidal $v_{gs}$.

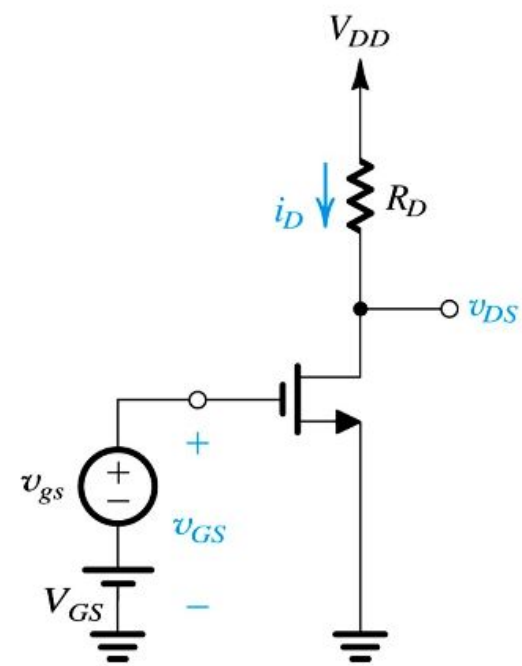


**Figure 5.29:**

# 5.4.5. **Determining the VTC via Graphical Analysis**

(eq5.39) $$i_D = \frac{V_{DD}}{R_D} - \frac{v_{DS}}{R_D}$$

- Graphical method for determining VTC is shown in Figure 5.31
- Rarely used in practice, b/c difficult to draw *vi*-relationship.
- Based on observation that, for each value of $v_{GS}$, circuit will operate at intersection of $i_D$ and $v_{DS}$.

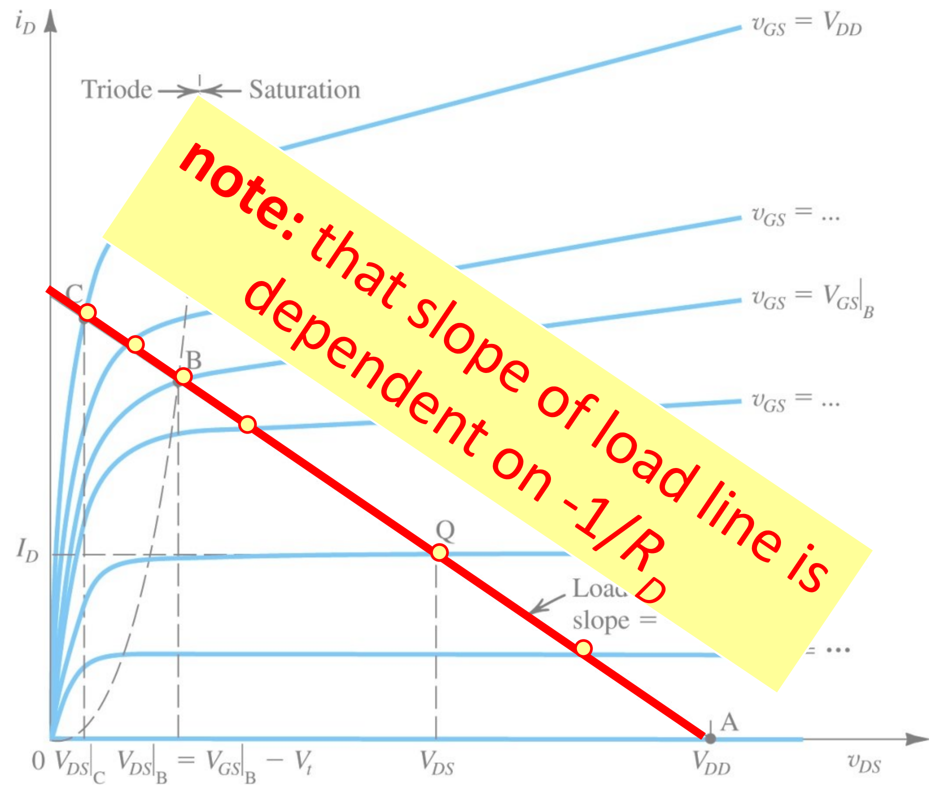


**Figure 5.31:** Graphical construction to determine the voltage transfer characteristic of the amplifier in Fig. 5.29(a).

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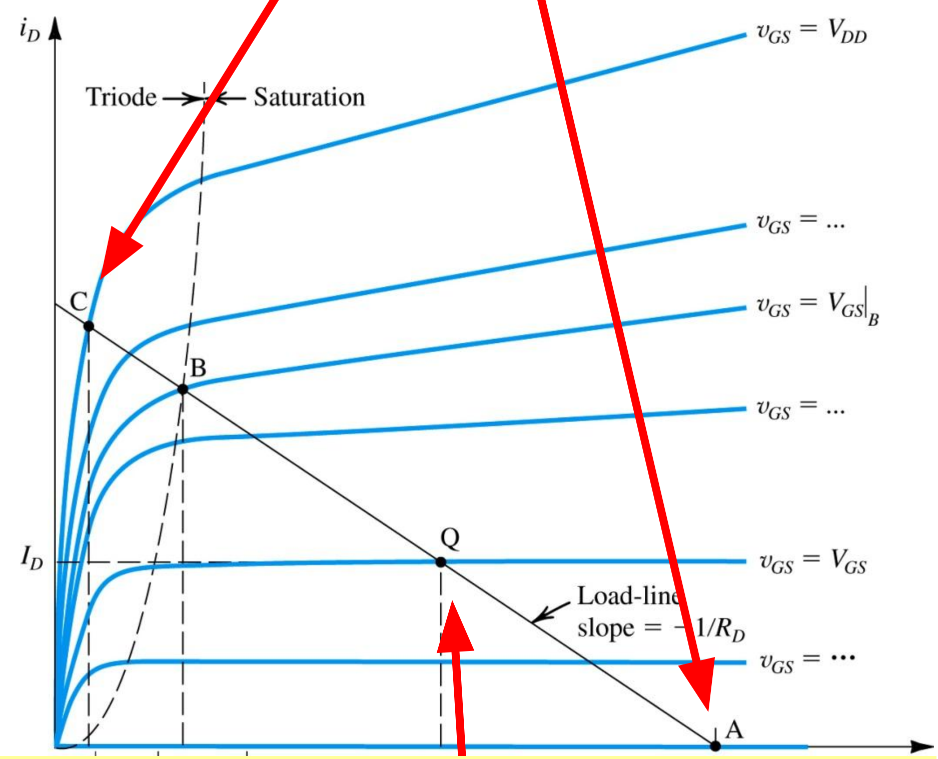
# 5.4.5. **Determining the VTC via Graphical Analysis**

- point A – where $v_{GS} = V_t$
- point Q – where MOSFET may be biased for amplifier operation
  - $v_{GS} = V_{GS}$, $v_{DS} = V_{DS}$
- point B – where MOSFET leaves saturation / enters triode
- point C – where MOSFET is deep in triode region and $v_{GS} = V_{DD}$

Points *A* (open) and *C* (closed) are suitable for switch applications




Point *Q* is suitable for amplifier applications

# 5.4.5. Determining the VTC via Graphical Analysis

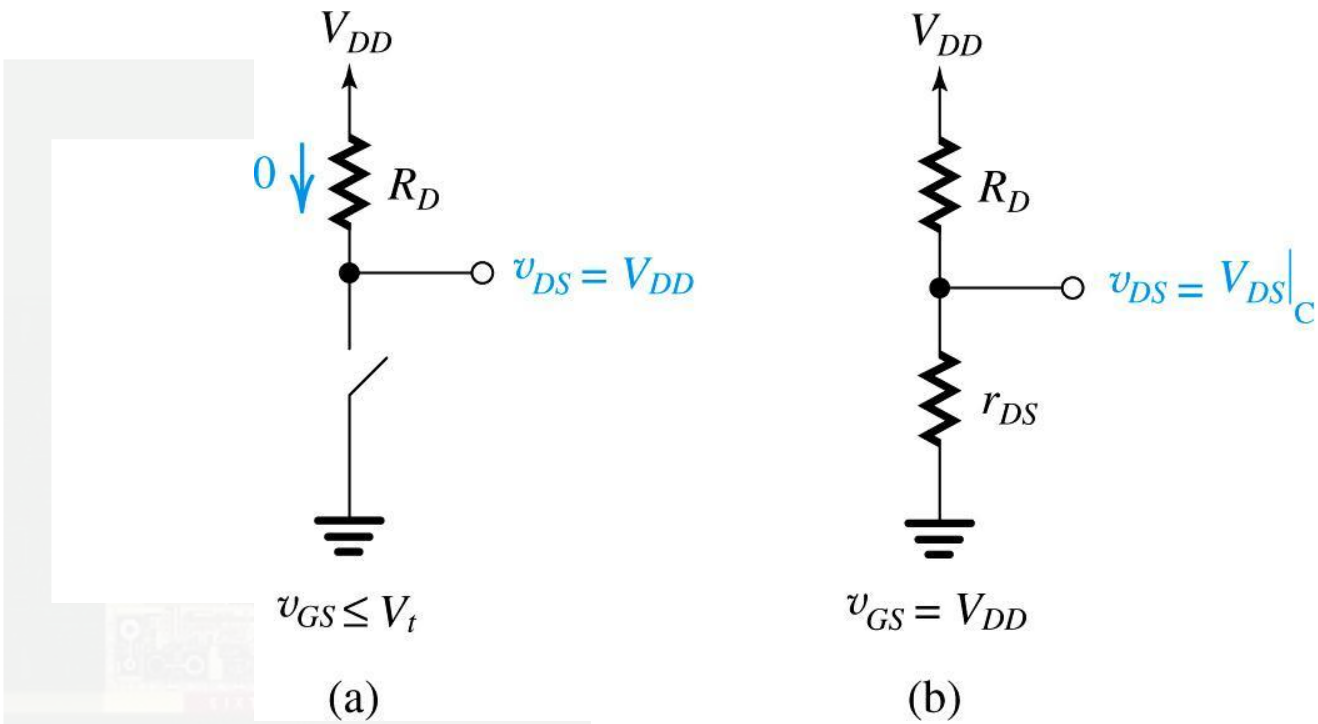


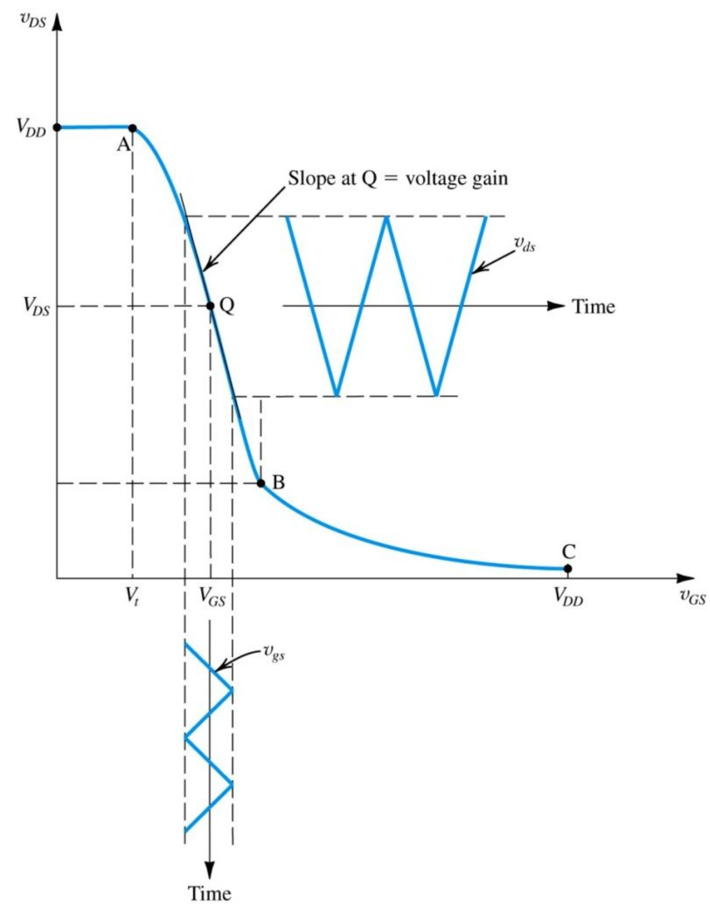

**Figure 5.32:** Operation of the MOSFET in Figure 5.29(a) as a switch: **(a)** Open, corresponding to point A in Figure 5.31; **(b)** Closed, corresponding to point C in Figure 5.31. The closure resistance is approximately equal to $r_{DS}$ because $V_{DS}$ is usually very small.

# 5.4.6. **Locating the Bias Point Q**

- **bias point** ($Q$) – is determined by value of $v_{GS}$ and load resistance $R_D$.
- Two considerations in deciding Q:
  - Required gain.
  - Allowable signal swing at output.

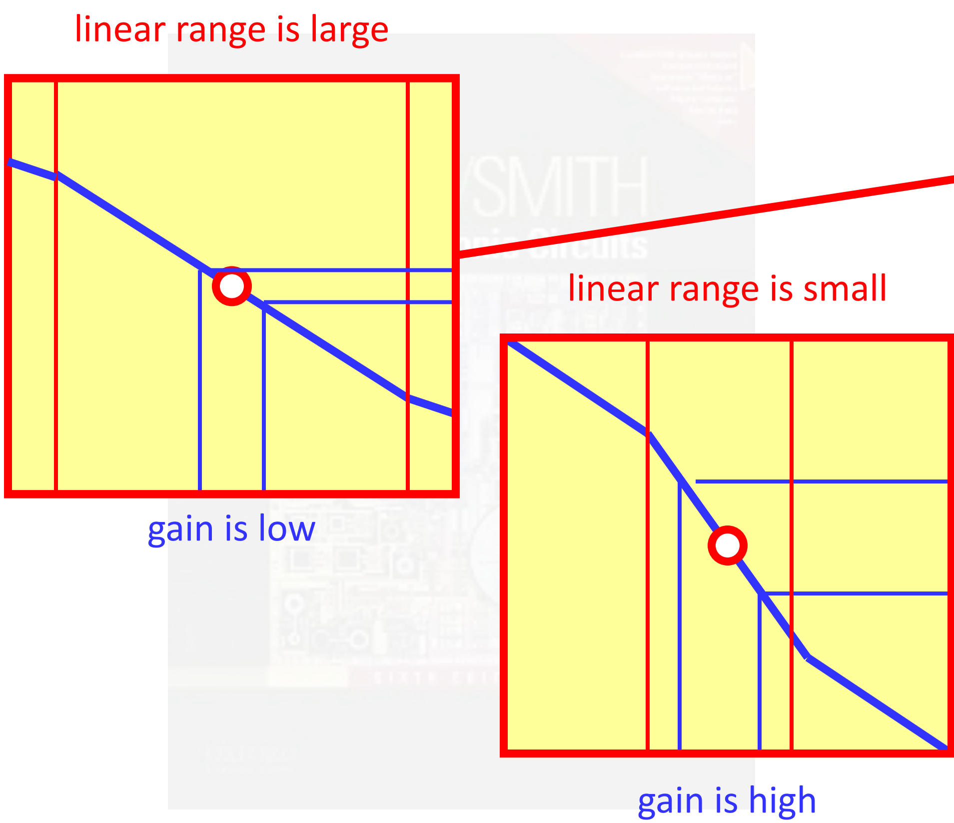
# 5.4.6. **Locating the Bias Point Q**

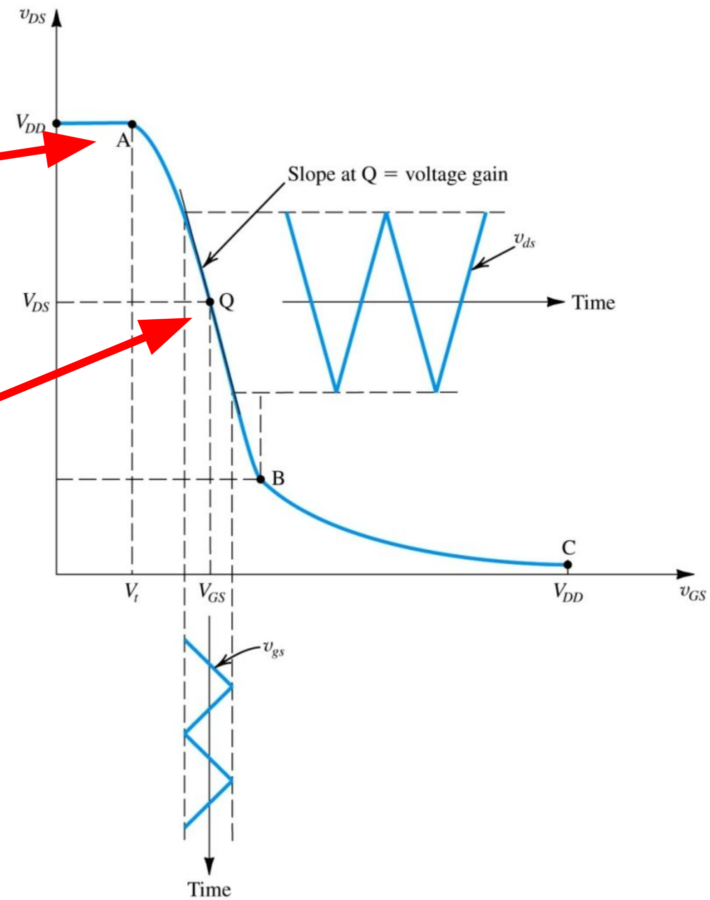
- **Q:** How is *Q* for VTC defined (assuming $R_D$ is fixed)?
  - A: As point *Q* approaches *B*:
    - gain increases
    - maximum $v_{gs}$ swing decreases



# 5.4.6. Locating the Bias Point Q

Note that a trade-off between gain and linear range exists.

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# 5.4.6. Locating the Bias Point Q

The objective is to prevent $v_{DS}$ from "clipping" or entering triode region

- To define **load resistance $R_D$**, one should refer to the $i_D$ - $v_{DS}$ plane.
- Two examples of $R_D$ are shown to right for illustration:
  - **Q2:** too close to triode
    - not enough **legroom**
  - **Q1:** too close to $V_{DD}$
    - not enough **headroom**
- Ideally, we want to be somewhere in the middle.

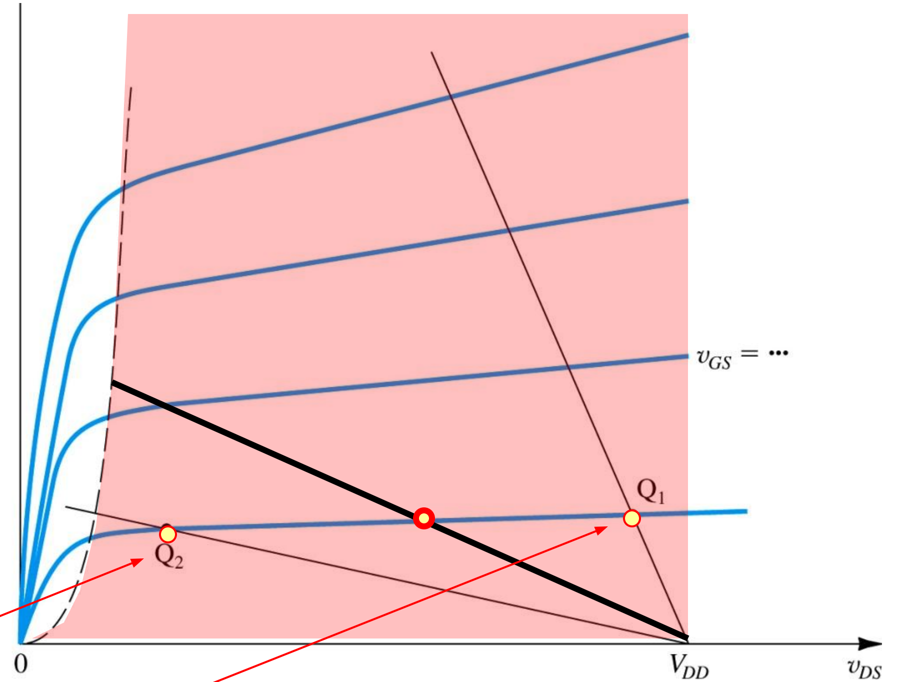


**Figure 5.33:** Two load lines and corresponding bias points. Bias point $Q_1$ does not leave sufficient room for positive signal swing at the drain (too close to $V_{DD}$). Bias point $Q_2$ is too close to the boundary of the triode region and might not allow for sufficient negative signal swing.

# 5.5. S… Ope… Models

input voltage to be amplified

dc bias voltage

output voltage

- Previously it was stated that linear amplification may be obtained from MOSFET via…
  - Operation in saturation region
  - Utilization of small-input
- This section will explore small-signal operation in detail
  - Note the conceptual amplifier circuit to right

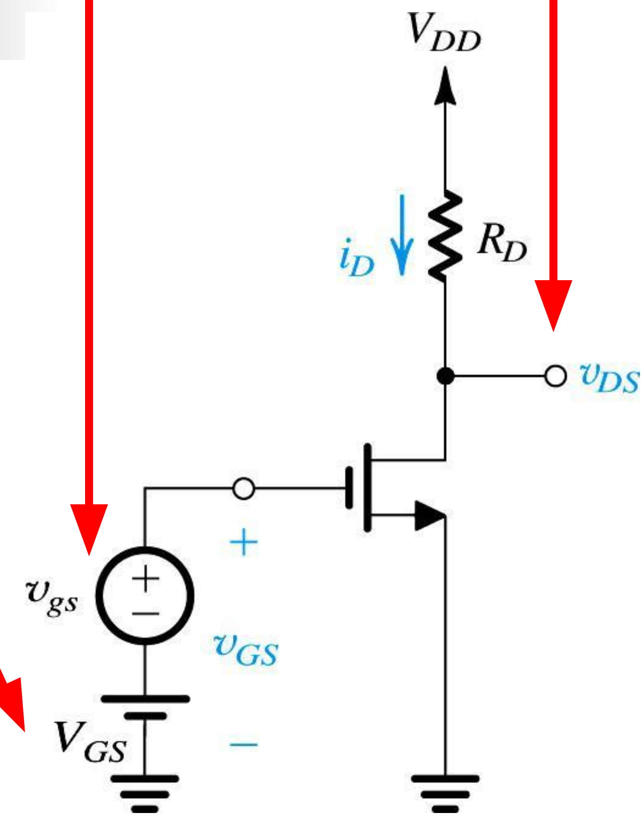


**Figure 5.34:** Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

# 5.5.1. The DC Bias Point

- **Q:** How is dc bias current $I_D$ defined?

only applies in saturation where $V_{DS} \geq V_{OV}$

(eq5.40) $$I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2 = \frac{1}{2}k_n V_{OV}^2$$

(eq5.41) $$V_{DS} = V_{DD} - R_D I_D$$

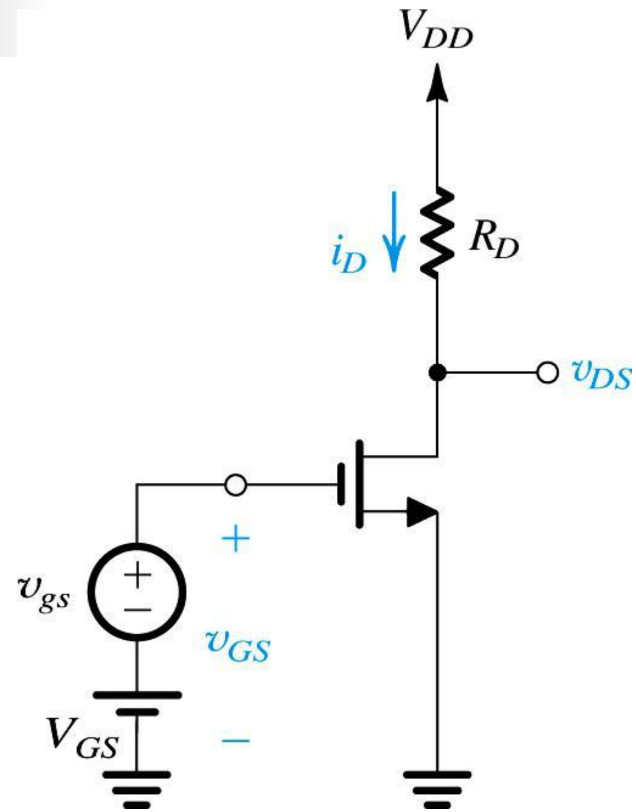


**Figure 5.34:** Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

# 5.5.2. The Signal Current in the Drain Terminal

- **Q:** What is effect of $v_{gs}$ on $i_D$?
  - **step #1:** Define $v_{GS}$ as in (5.42).
  - **step #2:** Define $i_D$, separate terms as function of $V_{GS}$ and $v_{gs}$

(eq5.42) $v_{GS} = V_{GS} + v_{gs}$

action: state (5.17)

(eq5.17) $i_D = \frac{1}{2}k_n\left(\underbrace{V_{GS} + v_{gs} - V_t}_{v_{OV}}\right)^2$

action: expand the squared term via $V_{GS} - V_t$ and $v_{gs}$

(eq5.43) $i_D = \frac{1}{2}k_n\left[(V_{GS} - V_t)^2 + \ldots \underbrace{\ldots + 2(V_{GS} - V_t)v_{gs} + v_{gs}^2}_{(V_{GS}+v_{gs}-V_t)}\right]$

action: simplify

3) $i_D = \frac{1}{2}k_n(V_{GS} - V_t)^2 + \ldots$

$\ldots + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$

Note that this differs from previous analyses - because of attempt to isolate the effect of $v_{gs}$ from $V_{GS}$.

# Q: What is effect of $v_{gs}$ on $i_D$?

Note that to minimize nonlinear distortion, $v_{gs}$ should be kept small.

$$\tfrac{1}{2}k_n v_{gs}^2 << k_n(V_{GS}-V_t)v_{gs}$$

$$v_{gs} << 2(V_{GS}-V_t)$$

$$v_{gs} << 2v_{OV}$$

- **step #3:** Classify terms.
  - dc bias current ($I_D$).
  - linear gain – is desirable.
  - nonlinear distortion – is undesirable, because rep. distortion.

(eq5.43) $$i_D = \underbrace{\frac{1}{2}k_n\left(V_{GS}-V_t\right)^2}_{\text{dc bias current } (I_D)} + \underbrace{k_n\left(V_{GS}-V_t\right)v_{gs}}_{\text{linear gain term}} + \underbrace{\frac{1}{2}k_n v_{gs}^2}_{\text{nonlinear distortion term}}$$

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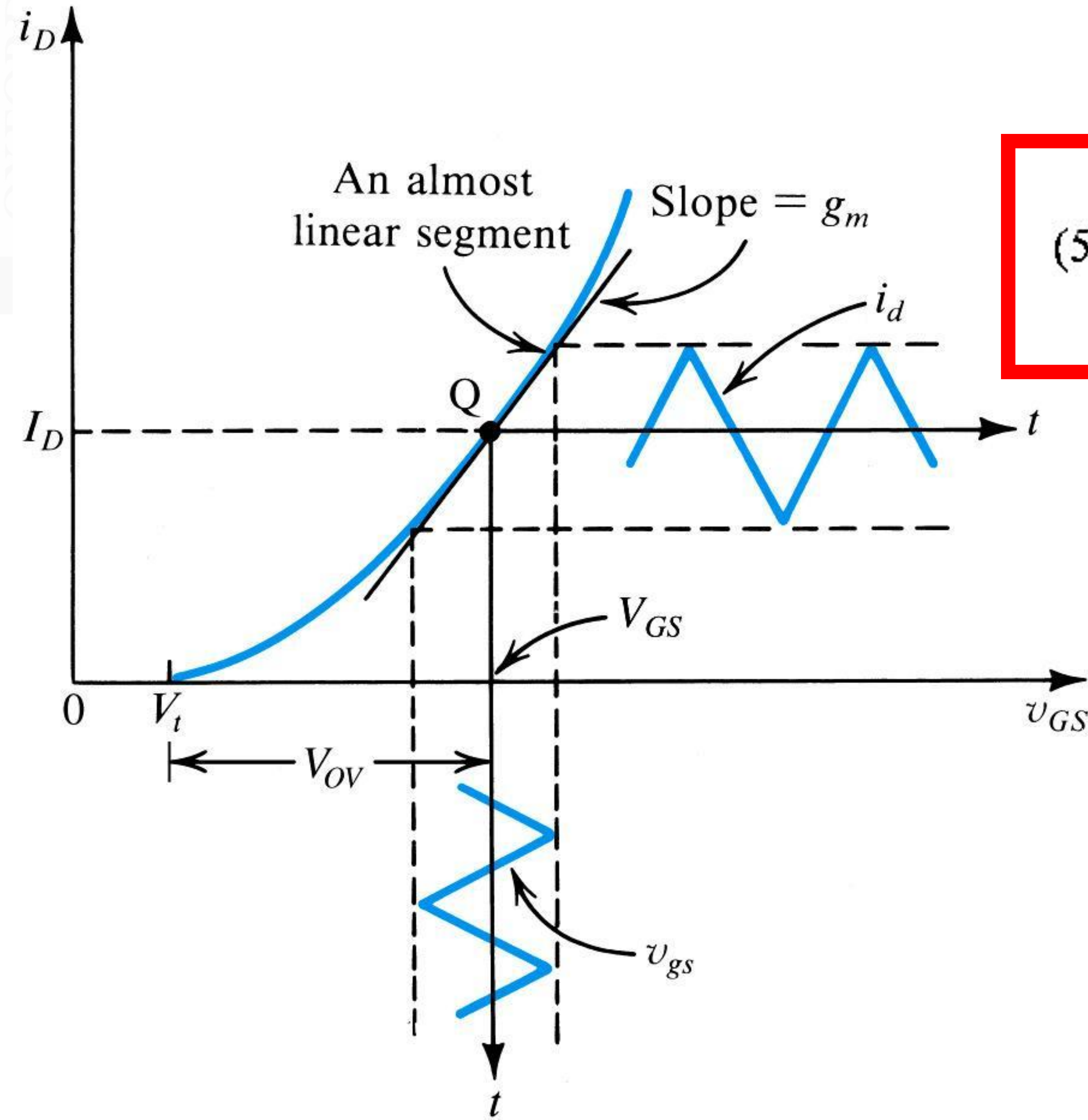
# Q: What is effect of $v_{gs}$ on $i_D$?

- **step #4:** Adapt (5.43) for small-signal condition.
  - If $v_{gs} << 2v_{OV}$, neglect distortion.

(eq5.43) $$i_D = \underbrace{\frac{1}{2}k_n\left(V_{GS} - V_t\right)^2}_{\text{dc bias current } (I_D)} + \underbrace{\boxed{k_n\left(V_{GS} - V_t\right)v_{gs}}}_{\text{linear gain term}} + \underbrace{\cancel{\frac{1}{2}k_n v_{gs}^2}}_{\text{nonlinear distortion term}}$$

(eq5.47) **MOSFET transconductance** $$g_m = \frac{v_{gs}}{i_d} = k_n\left(V_{GS} - V_t\right)$$




$$g_m \equiv \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GS}} \tag{5.49}$$

**Figure 5.35:** Small-signal operation of the MOSFET amplifier.

# 5.5.3. The Voltage Gain

- **Q:** How is voltage gain ($A_v$) defined?
  - **step #1:** Define $v_{DS}$ for circuit of Figure 5.34 using KVL.

**action:** apply small-signal condition

$$v_{DS} = V_{DD} - R_D i_D = V_{DD} - R_D \left( I_D + i_d \right)$$

**action:** regroup terms **action:** simplify

$$v_{DS} = \underbrace{V_{DD} - R_D I_D}_{\text{dc component } (V_{DS})} - R_D i_d = V_{DS} - \underbrace{R_D i_d}_{v_{ds}}$$

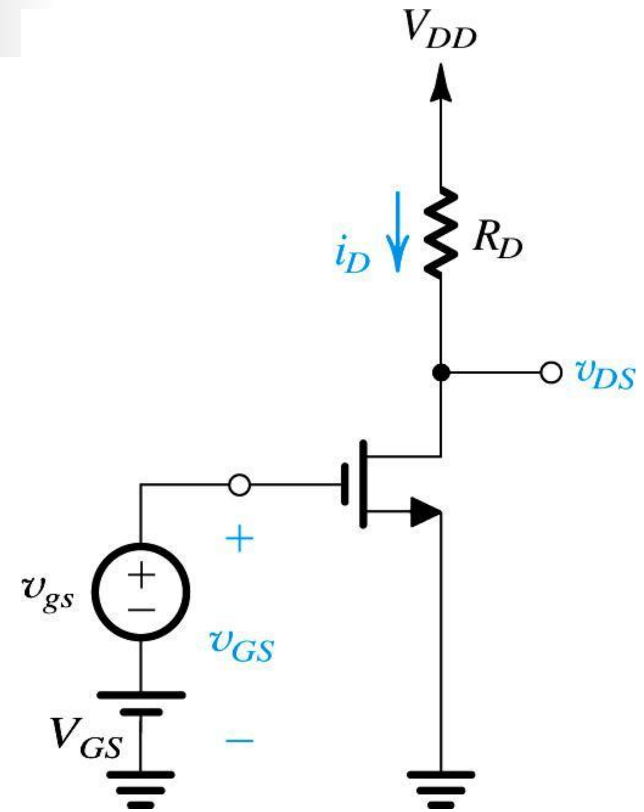



**Figure 5.34:** Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

# Q: How is voltage gain ($A_v$) defined?

- **step #2:** Isolate $v_{ds}$ component of $v_{DS}$.
- **step #3:** Solve for gain ($A_v$).

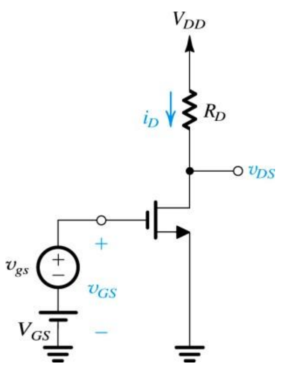



**Figure 5.34:** Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

action: isolate $v_{ds}$

(eq5.50) $v_{ds} = -R_D i_d$

action: insert (5.47)

(eq5.50) $v_{ds} = -R_D \left( g_m v_{gs} \right)$

(5.47)

action: solve for gain

(eq5.51) $A_v = \dfrac{v_{ds}}{v_{gs}} = -g_m R_D$

# 5.5.3. **The Voltage Gain**

- Output signal is shifted from input by 180°.
- Input signal $v_{gs} << 2(V_{GS} - V_t)$.
- Operation should remain in MOSFET saturation region
  - $v_{DS} > v_{GS} - V_t$ (legroom)
  - $v_{DS} < V_{DD}$ (headroom)

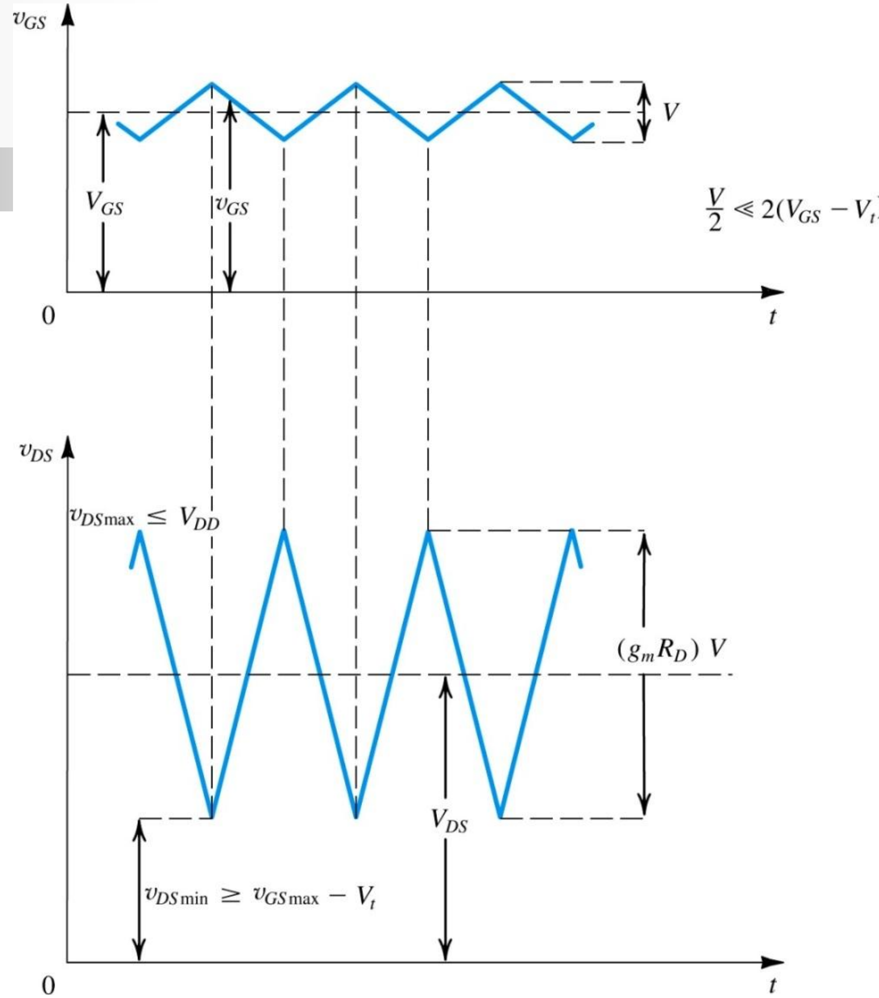



**Figure 5.36:** Total instantaneous voltage $v_{GS}$ and $v_{DS}$ for the circuit in Figure 5.34.

# 5.5.5. **Small-Signal Equivalent Models**

- From signal POV, FET behaves as VCCS.
  - Accepts $v_{gs}$ between gate and source
  - Provides current ($i_D$) at drain
  - Input resistance is high
    - b/c gate terminal draws $i_G = 0$
  - Output resistance is high

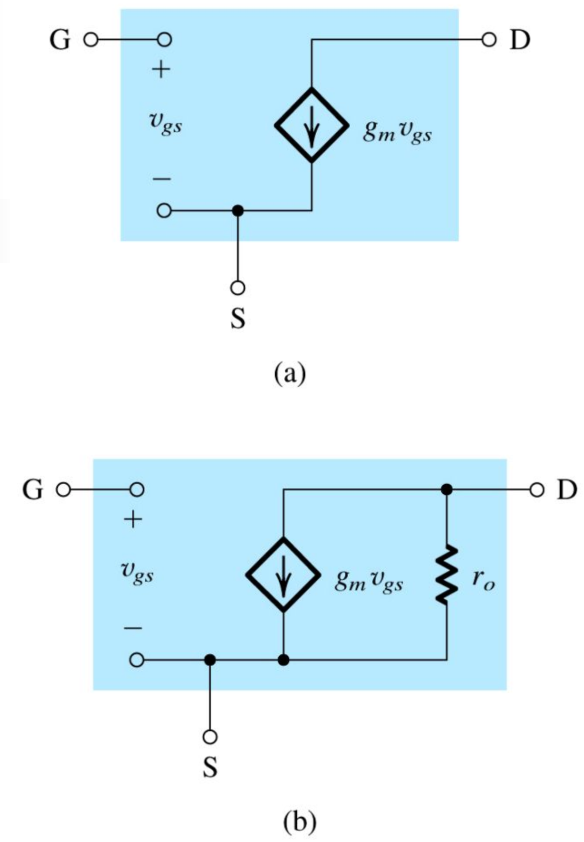



**Figure 5.37:** Small-signal models for the MOSFET: **(a)** neglecting the dependence of $i_D$ on $v_{DS}$ in saturation (the channel-length modulation effect) and **(b)** including the effect of channel length modulation

# 5.5.5. **Small-Signal Equivalent Models**

Note that this resistor ($r_o$) takes on value 10*kOhm* to 1*MOhm* and represents channel-length modulation.

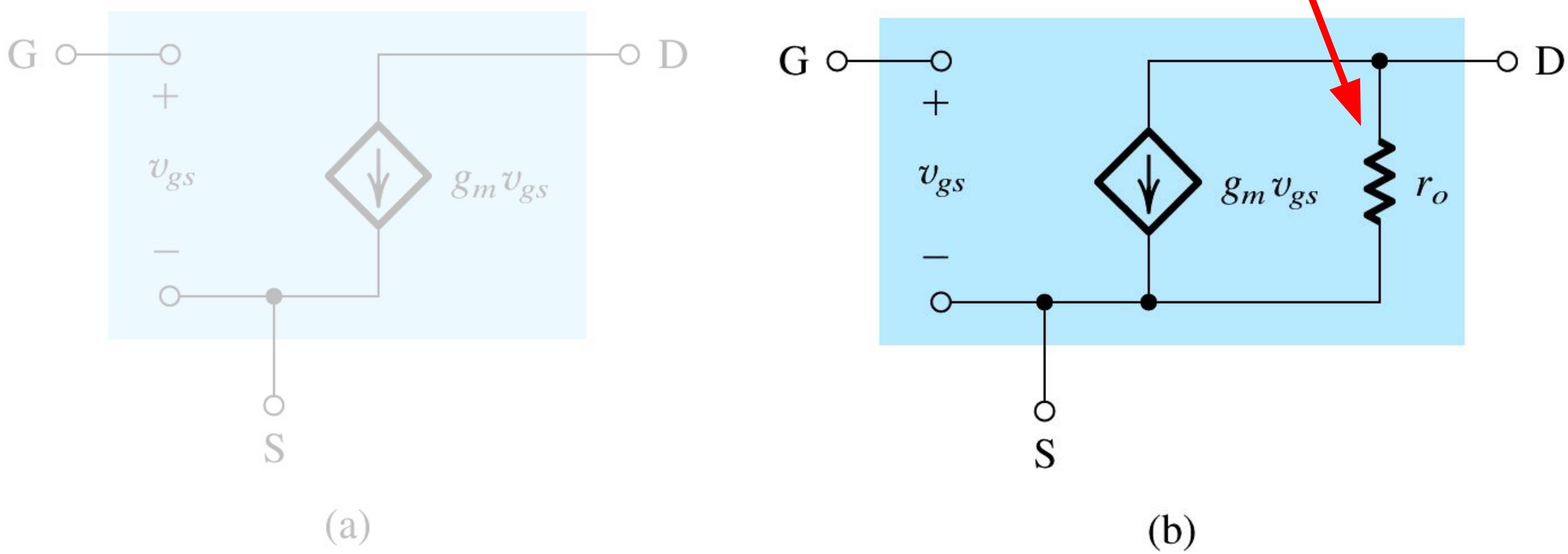



**Figure 5.37:** Small-signal models for the MOSFET: **(a)** neglecting the dependence of $i_D$ on $v_{DS}$ in saturation (the channel-length modulation effect) and **(b)** including the effect of channel length modulation

# More Observations

- Model (b) is more accurate than model (a)
- $r_o = V_A / I_D$
- Small signal parameters ($g_m$, $r_o$) both depend on dc bias point
- If channel-length modulation is considered, (5.51) becomes (5.54).

less accurate, b/c does not consider channel length modulation

(eq5.51) $$A_v = \frac{v_{ds}}{v_{gs}} = -g_m R_D$$

(eq5.54) $$A_v = \frac{v_{ds}}{v_{gs}} = -g_m \left( R_D \,||\, r_o \right)$$

more accurate, b/c does consider channel length modulation

# 5.5.6. **The Transconductance** $g_m$

- Observations from (5.47)
  - $g_m$ is proportional to $\mu_n$, $C_{ox}$, ratio $W/L$, dc component $V_{OV}$.
  - MOSFET with short / wide channel provides maximum gain.
  - Gain may be increased via $V_{GS}$, but not without reducing allowable swing of $v_{gs}$.

(eq5.47) $$g_m = \frac{v_{gs}}{i_d} = k_n\left(V_{GS} - V_t\right)$$

**action:** make some substitutions

(eq5.47) $$g_m = \underbrace{k_n' \frac{W}{L}}_{k_n}\left(V_{GS} - V_t\right)$$

**action:** simplify

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

# 5.5.6: **The Transconductance $g_m$**

- Observations from (5.47)
  - $g_m$ is proportional to square root of dc bias current ($I_D$)
  - For given $I_D$, $g_m$ is proportional to $(W/L)^{1/2}$
- This behavior is sharp contrast to the bipolar junction transistor (BJT).
  - For which, $g_m$ is proportional to $g_m$ alone (not size or geometry).

(eq5.40) $$I_D = \frac{1}{2} k'_n \frac{W}{L} V_{OV}^2$$

**action:** solve (5.40) for $V_{OV}$

(eq5.40) $$V_{OV} = \frac{\sqrt{2I_D}}{\sqrt{k'_n}\sqrt{W/L}}$$

---

(eq5.55) $$g_m = k'_n \frac{W}{L} V_{OV}$$

**action:** substitute for $V_{OV}$ as defined above

(eq5.56) $$g_m = k'_n \frac{W}{L} \sqrt{\frac{2I_D}{k'_n W/L}}$$

**action:** simplify

(eq5.56) $$g_m = \sqrt{2k'_n}\sqrt{W/L}\sqrt{I_D}$$

# 5.5.6. **The Transconductance** $g_m$

- **Q:** How does MOSFET compare to BJT? Assume $I_D$ = 0.5$mA$, $k'_n$ = 120$mA/V^2$.
  - **A:** MOSFET $g_m$ = 0.35$mA/V$
    - $W/L = 1$
  - **A:** MOSFET $g_m$ = 3.5$mA/V$
    - $W/L = 100$
  - **A:** BJT $g_m$ = 20$mA/V$

# 5.5.6: **The Transconductance $g_m$**

- Figure 5.38 illustrates the relationship defined in (5.57).

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

**action:** replace $k_n' \frac{W}{L}$

(eq5.56) $$g_m = \left( \frac{2I_D}{(V_{GS} - V_t)^2} \right) V_{OV}$$

**action:** simplify

(eq5.57) $$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}}$$

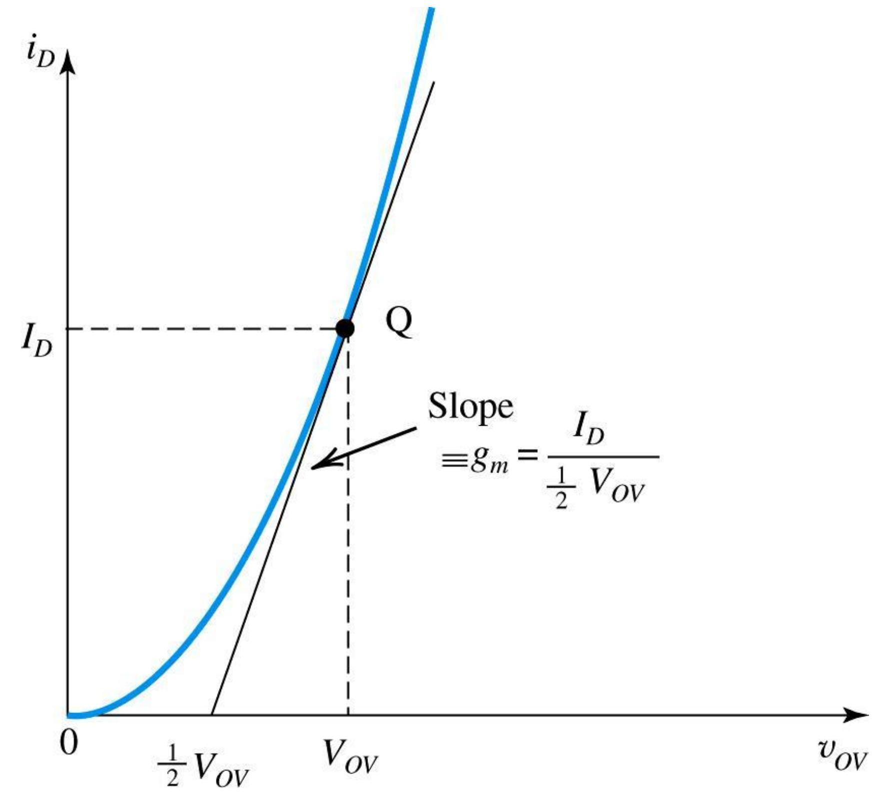



**Figure 5.38:** The slope of the tangent at the bias point $Q$ intersects the $v_{OV}$ axis at $1/2V_{OV}$. Thus $g_m = I_D/(1/2V_{OV})$.

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# 5.5.6: **The Transconductance** $g_m$

- In summary, there are three relationships for determining $g_m$:
  - (5.55), (5.56), and (5.57)
- These relationships are dependent on three design parameters:
  - $W/L$, $V_{OV}$, $I_D$

(eq5.55) $$g_m = k_n' \frac{W}{L} V_{OV}$$

(eq5.56) $$g_m = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D}$$

(eq5.57) $$g_m = \frac{2I_D}{V_{OV}}$$

# **Example 5.10:** MOSFET Amplifier

- **Example 5.10 Problem Statement:** Figure 5.39(a) shows a discrete common-source MOSFET amplifier utilizing a drain-to-gate resistance $R_G$ for biasing purposes. Such a biasing arrangement will be studied in Section 5.7. The input signal $v_I$ is coupled to the gate via a large capacitor, and the output signal at the drain is coupled to the load resistance $R_L$ via another large capacitor. The transistor has $V_t = 1.5V$, $k'_n(W/L) = 0.25mA/V^2$, and $V_A = 50V$. Assume the coupling capacitors to be sufficiently large so as to act as short circuits at the signal-frequencies of interest.
- **Q:** We wish to analyze this amplifier circuit to determine its **(a)** small-signal voltage gain, its **(b)** input resistance, and the largest allowable input signal.

**note:** capacitors block dc signals completely, but have no effect on small-signal

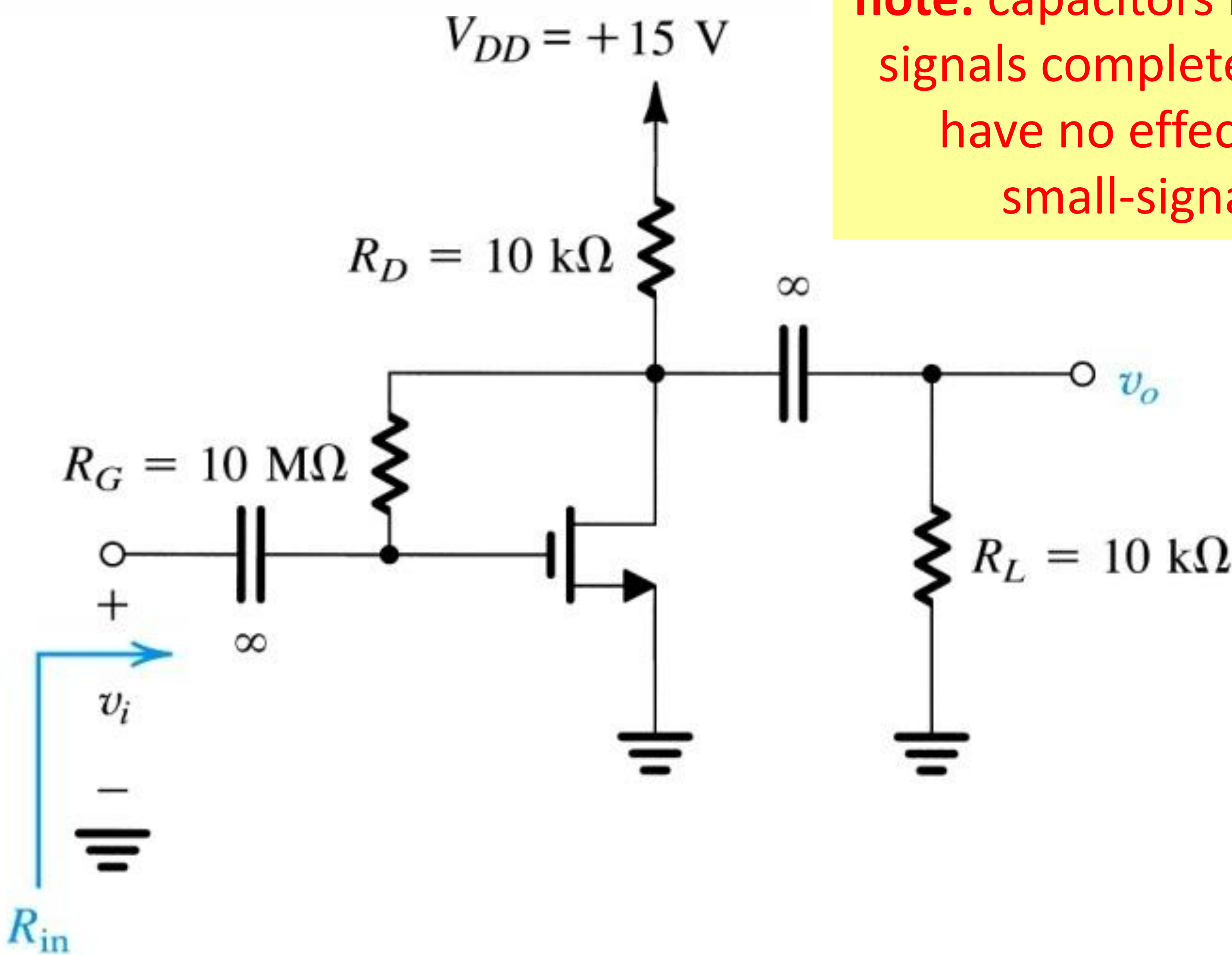


**Figure 5.39:** Example 5.10 amplifier circuit.

# 5.5.7. **The T Equivalent-Circuit Model**

- Through circuit transformation, it is possible to develop alternative circuit models
- *T*-Equivalent-Ckt Model is shown to right.

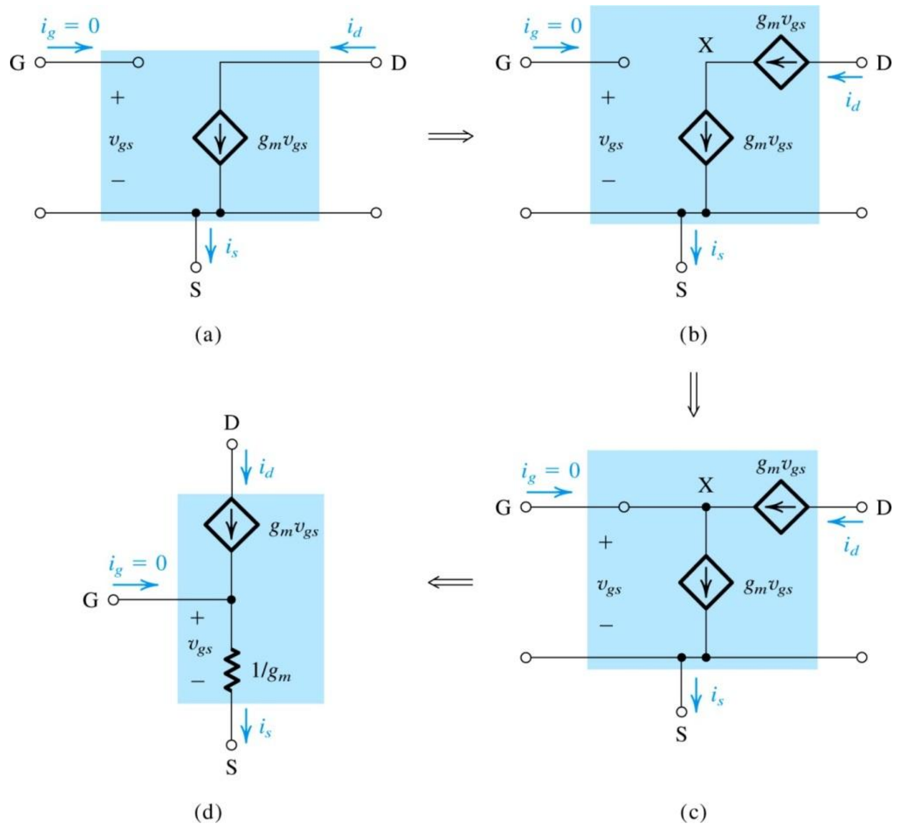


**Figure 5.40:** Development of the T equivalent-circuit model for the MOSFET. For simplicity, $r_o$ has been omitted; however, it may be added between D and S in the T model of **(d)**.

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# 5.5.7. **The T Equivalent-Circuit Model**

- **Q:** How is this model developed?
  - **step #1:** Begin with **small signal model** (assume $R_o$=0).
  - **step #2:** Place second current source in series with the first.
    - Has no effect on circuit operation.

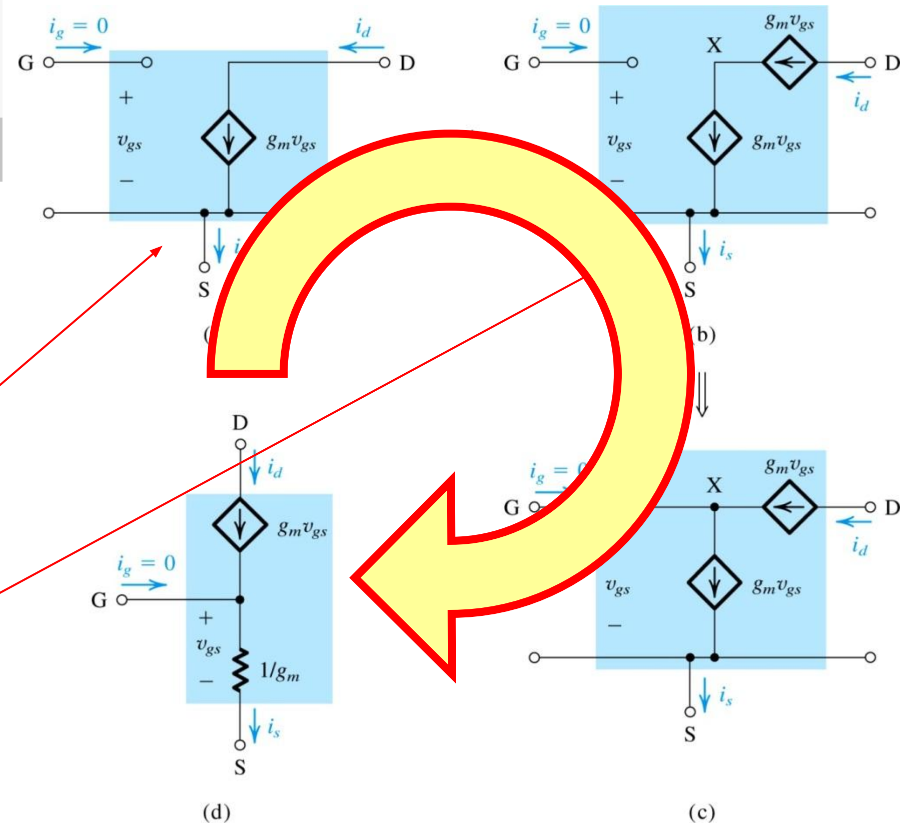


**Figure 5.40:** Development of the T equivalent-circuit model for the MOSFET. For simplicity, $r_o$ has been omitted; however, it may be added between D and S in the T model of **(d)**.

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# Q: How is T Equivalent-Circuit Model developed?

- **step #3:** Create new node $X$, which connects gate and drain terminals
  - b/c the two current sources are equal, $i_g = 0$
- **step #4:** replace initial current source with equivalent resistance.
  - $i_{DS} = g_m v_{gs} = v_{gs}/R_{gs}$

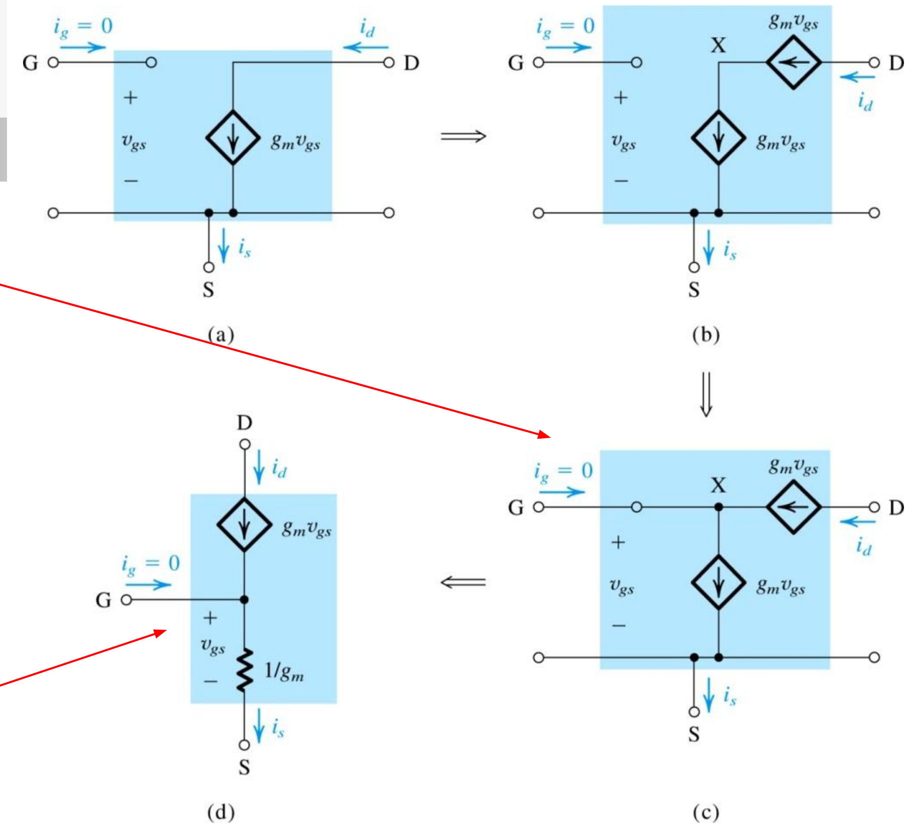



**Figure 5.40:** Development of the T equivalent-circuit model for the MOSFET. For simplicity, $r_o$ has been omitted; however, it may be added between D and S in the T model of **(d)**.

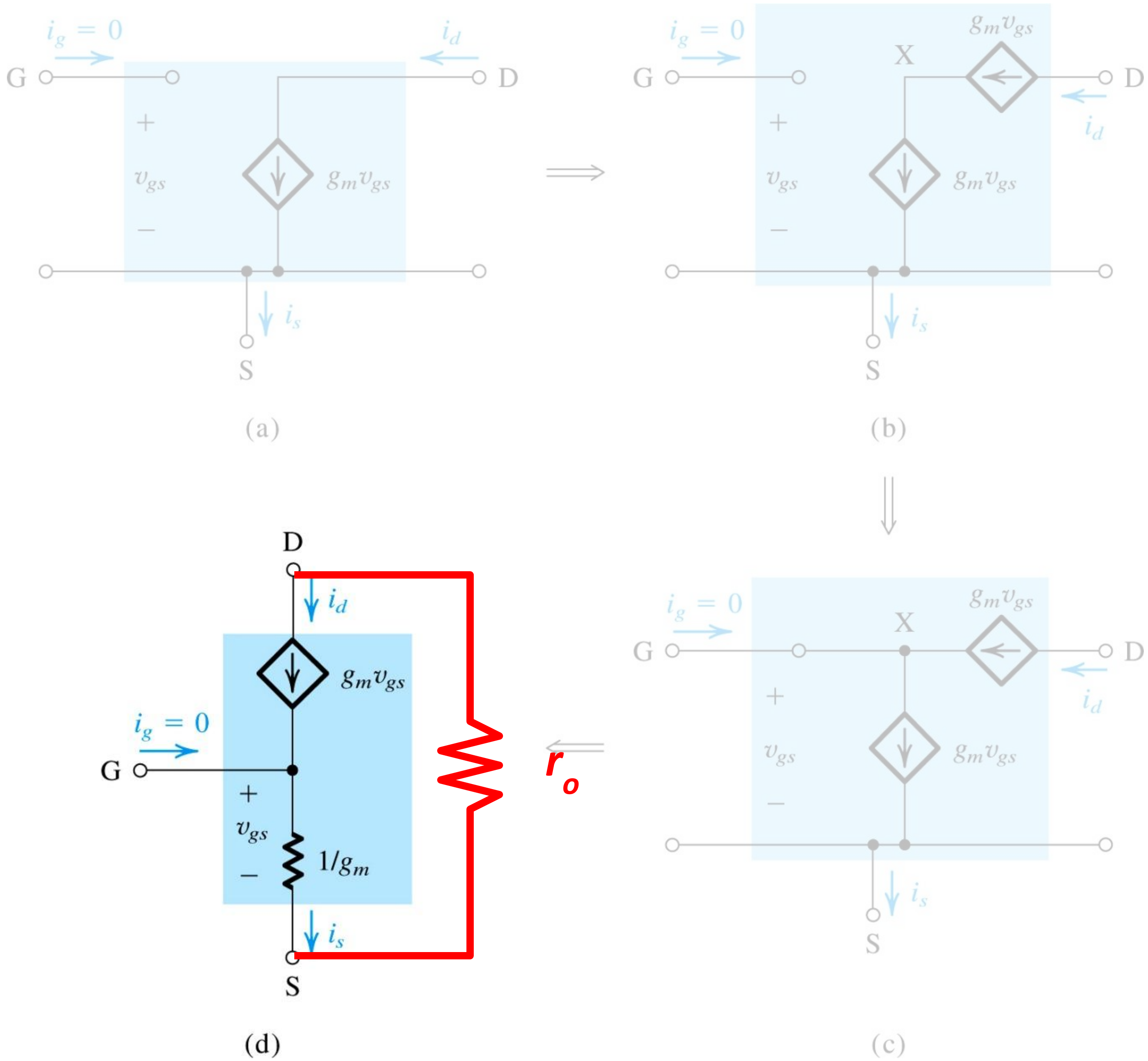



**Figure 5.40:** Development of the T equivalent-circuit model for the MOSFET. For simplicity, $r_o$ has been omitted; however, it may be added.

## Table 5.3 Small-Signal Equivalent-Circuit Models for the MOSFET

***Small-Signal Parameters***

**NMOS transistors**

■ Transconductance:

$$g_m = \mu_n C_{ox} \frac{W}{L} V_{OV} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{V_{OV}}$$

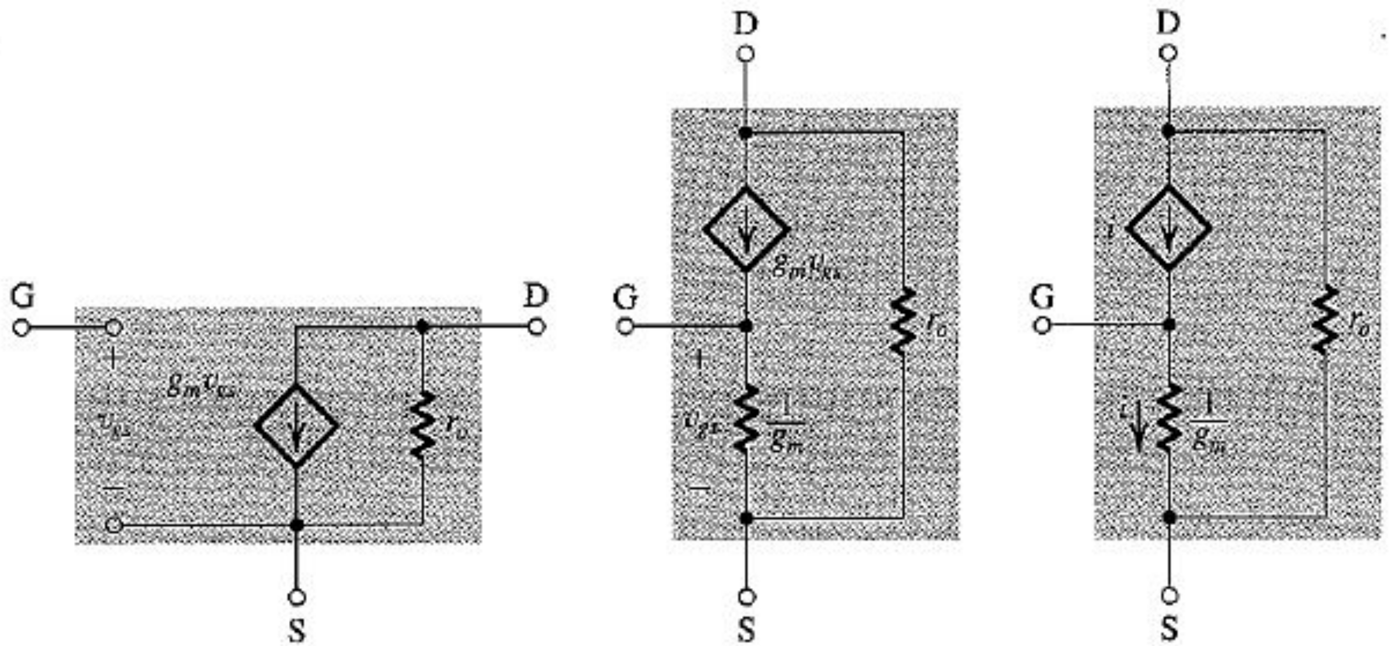
■ Output resistance:

$$r_o = V_A / I_D = 1/\lambda I_D$$

**PMOS transistors**

Same formulas as for NMOS *except* using $|V_{OV}|$, $|V_A|$, and replacing $\mu_n$ with $\mu_p$.

***Small-Signal Equivalent Circuit Models***




Hybrid-$\pi$ model

T models

## Summary

- The enhancement-type MOSFET is current the modt widely used semiconductor device. It is the basis of CMOS technology, which is the most popular IC fabrication technology at this time. CMOS provides both n-channel (NMOS) and p-channel (PMOS) transistors, which increases design flexibility. The minimum MOSFET channel length achievable with a given CMOS process is used to characterize the process
- The overdrive voltage $|V_{OV}| = |V_{GS}| - |V_t|$ is the key quantity that governs the operation of the MOSFET. For amplifier applications, the MOSFET must operate in the saturation region.

# Summary

- In saturation, $i_D$ shows some linear dependence on $v_{DS}$ as a result of the change in channel length. This channel-length modulation phenomenon becomes more pronounced as $L$ decreases. It is modeled by ascribing an output resistance $r_o = |V_A|/I_D$ to the MOSFET model. Although the effect of ro on the operation of discrete-circuit MOS amplifiers is small, that is not the case in IC amplifiers.
- The essence of the use of MOSFET as an amplifier is that in saturation $v_{GS}$ controls $i_D$ in the manner of a voltage-controller current source. When the device is dc biased in the saturation region, a small-signal input ($v_{gs}$) may be amplified linearly.

# Summary

- In cases where a resistance is connected in series with the source lead of the MOSFET, the T model is the most conveinant to use.
- The three basic configurations of the MOS amplifiers are shown in Figure 5.43.
- The CS amplifier has an ideally infinite input resistance and reasonably high gain – but a rather high output resistance and limited frequency response. It is used to obtain most of the gain in a cascade amplifier.
- Adding a resistance $R_s$ in the source lead of the CS amplifier can lead to beneficial results.

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# Summary

- The CG amplifier has a low input resistance and thus it alone has limited and specialized applications. However, its excellent high-frequency response makes it attractive in combination with the CS amplifier.
- The source follow has (ideally) infinite input resistance, a voltage gain lower than but close to unity, and a low output resistance. It is employed as a voltage buffer and as the output stage of a multistage amplifier.
- A key step in the design of transistor amplifiers is to bias the transistor to operate at an appropriate point in the saturation region.